

A RECIPROCITY LAW FOR QUANTUM DILOGARITHM VALUES AT REAL MULTIPLICATION POINTS, WITH APPLICATIONS TO HILBERT’S TWELFTH PROBLEM, STARK UNITS AND SICS

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ABSTRACT. We prove a reciprocity law for the values of Faddeev’s quantum dilogarithm at real multiplication points, which Radchenko and Wheeler showed to be algebraic. For a real quadratic field K , each value is attached to a lattice I in K , a totally positive unit ε with $\varepsilon I = I$, and a torsion point of K/I . Assuming one of Radchenko and Wheeler’s identities, the Fourier proportionality relation, we show that the squared absolute values lie in the maximal abelian extension K^{ab} , and that each idele of K acts on them by moving the lattice and the torsion point, with an inversion when the idele is negative at the second real place. The proof derives Frobenius congruences from a p -adic argument on the Bruhat–Tits tree and globalizes them with Chebotarev’s theorem. We deduce that the values attached to the maximal order are units at every finite prime, and that the square roots of all values lie in K^{ab} . Using Shintani’s product formula, we show that the values together with i generate K^{ab} . We also prove the rank-one abelian Stark conjecture at a real place of K for $|S| \geq 3$, including Tate’s square-root condition. In one configuration the square-root condition also uses the Fourier-transform identity of Faddeev, Kashaev and Volkov, and there we prove a stronger root law in most cases. Finally, with a quadratic relation of Radchenko and Wheeler, we obtain Zauner-symmetric SICs (symmetric informationally complete quantum measurements) in every dimension $M \geq 5$.

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1. INTRODUCTION

Hilbert’s twelfth problem asks for explicit generators of the abelian extensions of a number field. For $\mathbb{Q}$ they are the values of $e^{2\pi i x}$ at rational x , and for imaginary quadratic fields they are

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special values of modular and elliptic functions at CM points. For real quadratic fields, Stark's conjectures predict that suitable units, given by derivatives of L -functions at $s = 0$, generate abelian extensions, and several authors have proposed analytic functions whose values at real multiplication points should produce these units [Ko, AFK]. Radchenko and Wheeler [RW] studied the special values of Faddeev's quantum dilogarithm at real multiplication points, proved that they are algebraic, and established a large system of identities among them. Assuming one of their identities, the Fourier proportionality relation, we prove a reciprocity law for these values and deduce that the values attached to the maximal order are units, that the values generate the maximal abelian extension, and the rank-one abelian Stark conjecture at a real place. With further identities of Radchenko and Wheeler we obtain Tate's square-root condition in all cases and Zauner-symmetric SICs in every dimension $M \geq 5$.

Notation. Let $K \subset \mathbb{R}$ be a real quadratic field. We call this inclusion the *analytic place* r , write u for the other real place and τ for the nontrivial automorphism of K , and fix $\overline{\mathbb{Q}} \subset \mathbb{C}$, so that $K^{\text{ab}} \subset \mathbb{C}$. Put $K_r^{\text{ab}} := K^{\text{ab}} \cap \mathbb{R}$, the maximal abelian extension of K in which r splits completely. For an integral ideal $\mathfrak{m} \neq O_K$ (a *proper* ideal) let $C_{\mathfrak{m}} := \text{Cl}_{\mathfrak{m}\infty_r\infty_u}(K)$ and $\mathcal{G}_{\mathfrak{m}} := \text{Cl}_{\mathfrak{m}\infty_u}(K)$, the Galois group of the ray class field $H_{\mathfrak{m}\infty_u}$, and put $n_{\mathfrak{m}} := 2/|\ker(C_{\mathfrak{m}} \rightarrow \mathcal{G}_{\mathfrak{m}})| \in \{1, 2\}$; thus $n_{\mathfrak{m}} = 1$ whenever K has no unit of norm -1 . The Artin map Art_K on the ideles $\mathbb{A}_K^\times$ is normalized arithmetically. The idele equal to -1 at a real place v and to 1 elsewhere maps to the complex conjugation ι_v above v , and c_r, c_u denote the corresponding classes.

The values are attached to *data* $d = (I, \varepsilon, z)$: a lattice $I \subset K$ (a $\mathbb{Z}$ -submodule of rank 2), with multiplier ring $\mathcal{O}_I := \text{End } I$ of any conductor; a totally positive unit $\varepsilon \in \mathcal{O}_I$ with $\varepsilon_r > 1$; and $z \in K$ with $(\varepsilon - 1)z \in I$ and $z \notin I$, taken modulo I . Section 2 attaches to d a positive number $W(d)$; in the notation of [RW] it is $|F_\gamma^+(\mathbf{u})|^2$, for the matrix γ of ε in a suitable basis of I and the label $\mathbf{u}$ of z . An idele $s \in \mathbb{A}_K^\times$ with finite part s_f acts on data by $s^{-1}d := (s_f^{-1}I, \varepsilon, s_f^{-1}z)$, where $s_f^{-1}I := K \cap s_f^{-1}\hat{I}$ with $\hat{I} := I \otimes \hat{\mathbb{Z}}$, and $s_f^{-1}z$ is read in $K/s_f^{-1}I \cong \mathbb{A}_{K,f}/s_f^{-1}\hat{I}$. Let $\mathcal{V}_K$ be the set of all values $W(d)$, and χ_K the quadratic character of $K/\mathbb{Q}$.

The values, the identities they satisfy and their algebraicity are due to Radchenko and Wheeler [RW]. Our contribution is the reciprocity law and its consequences.

1.1. Results. Four analytic identities of [RW] enter as hypotheses, and each theorem below names those it needs: the Fourier proportionality [RW, (6)], which is identity (vi) of Section 2; the five-term relation [RW, (7)], together with its form [RW, (36)], which combines (7) with (5); the quadratic relation [RW, Thm. 7, (39)]; and the Fourier-transform identity [RW, (15)], which we call (I1). Radchenko and Wheeler state (I1) after Faddeev and Kashaev [FK94], in the Fourier form of Faddeev, Kashaev and Volkov [FKV], at a real point $\tau > 0$, together with its continuation. Kopp's limit formula [Ko, Thm. 1.1], a preprint, is used only in the alternative route of Appendix A. All other inputs are published or classical: Shintani's product formula in Yamamoto's form [Sh, Ya08, Ya10], Shintani's evaluation of cone zeta functions at $s = 0$ [Sh76], Rademacher's formula for the eta multiplier [RG], Hecke theory, class field theory and Chebotarev's theorem.

Theorem A (Reciprocity law). *Assume identity (6) of [RW]. Every value $W(d)$ is algebraic and lies in K^{ab} , and for every datum d , whatever the multiplier ring of I , and every idele $s \in \mathbb{A}_K^\times$,*

$$(\star) \quad \text{Art}_K(s) W(d) = W(s^{-1}d)^{\text{sgn } s_u}.$$

Equivalently, if $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/K)$ and $\text{Art}_K(s) = \sigma|_{K^{\text{ab}}}$, then $\sigma W(d) = W(s^{-1}d)^{\text{sgn } s_u}$. Moreover $\overline{\sigma W(d)} = \sigma W(d)^{\chi_K(\sigma)}$ for every $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, so the conjugates of $W(d)$ above r are real and those above u lie on the unit circle; the conjugates above r are moreover positive.

The restriction of $(\star)$ to rational ideles reads $\text{Ver}(g)W(I, \varepsilon, z) = W(I, \varepsilon, \chi_{\text{cyc}}(g)z)$ for $g \in \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$, where $\text{Ver}: \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \rightarrow \text{Gal}(K^{\text{ab}}/K)$ is the transfer, which corresponds to the inclusion $\mathbb{A}_{\mathbb{Q}}^{\times} \subset \mathbb{A}_K^{\times}$, and χ_{cyc} is the cyclotomic character with values in $\widehat{\mathbb{Z}}^{\times}$. This restriction, the unit-circle statement and the positivity above r are proved without Theorem DC (the $\det = -1$ covariance of Section 3.1) and without the selection step of Section 3.4.

Put $f_0 := [O_K : \mathbb{Z}[\varepsilon]]$ and $\mathfrak{M}_{\varepsilon} := f_0(\varepsilon - 1)O_K$. For particular ideles, $(\star)$ specializes as follows.

idele s	what $(\star)$ says
finite idele of an integral ideal $\mathfrak{a}$ prime to $\mathfrak{M}_{\varepsilon}$	Artin law $\text{Art}(\mathfrak{a})W(I, \varepsilon, z) = W(\mathfrak{a}_{\mathcal{O}}^{-1}I, \varepsilon, z)$, $\mathfrak{a}_{\mathcal{O}} := \mathfrak{a} \cap \mathcal{O}_I$, for every order
-1 at r	the values are real at r
-1 at u	$\iota_u W = W^{-1}$
$\alpha \in K^{\times}$	homothety law $W(\alpha^{-1}I, \varepsilon, \alpha^{-1}z) = W(I, \varepsilon, z)^{\text{sgn } \alpha_u}$, so the right side of $(\star)$ is $K^{\times}$ -invariant
$a \in \widehat{\mathbb{Z}}^{\times}$	scaling of labels, $W(I, \varepsilon, a^{-1}z)$

For a proper $\mathfrak{m}$ let $\varepsilon_{\mathfrak{m}} > 1$ generate the totally positive units $\equiv 1 \pmod{\mathfrak{m}}$, and put $\mathbf{W}_{\mathfrak{m}} := W(\mathfrak{m}, \varepsilon_{\mathfrak{m}}, 1)$. The maximal-order values with annihilator $\mathfrak{m}$ form a function $W_{\mathfrak{m}}$ on $\mathcal{G}_{\mathfrak{m}}$ with $W_{\mathfrak{m}}(1) = \mathbf{W}_{\mathfrak{m}}$ (Section 3.6). We shall need the following nonvanishing property: for every proper $\mathfrak{f}$ and every primitive character χ of $\mathcal{G}_{\mathfrak{f}}$ with $\chi(c_u) = -1$,

$$(N) \quad \sum_{B \in \mathcal{G}_{\mathfrak{f}}} \bar{\chi}(B) \log W_{\mathfrak{f}}(B) \neq 0.$$

Theorem B (Hilbert's twelfth problem for K). *Assume identity (6) of [RW]. Then $K(\mathcal{V}_K) = K_r^{\text{ab}}$ and $K^{\text{ab}} = K(i, \mathcal{V}_K)$; already the values $\mathbf{W}_{\mathfrak{m}}$, $\mathfrak{m}$ proper, generate K_r^{ab} .*

The proof uses (N), which we prove in Proposition 3.24 from Shintani's product formula in Yamamoto's form [Sh, Prop. 3], [Ya08, Thm. 5.1.1], together with Yamamoto's relation for the class c_u [Ya08, Thm. 5.2.3] and Hecke theory. Appendix A derives (N) also from Kopp's limit formula.

Theorem C (Zauner SICs). *Assume identity (6) of [RW] and the quadratic relation [RW, Thm. 7, (39)]. For every $M \geq 5$ there is a Weyl-Heisenberg SIC in dimension M whose fiducial vector is fixed, up to phase, by an order-three Clifford unitary whose symplectic part is $\text{SL}_2(\mathbb{Z})$ -conjugate to Zauner's matrix $F_{\mathbb{Z}}^{\pm 1}$.*

Theorem D (Units at every finite prime). *Assume identity (6) of [RW]. For every invertible O_K -ideal I , every totally positive $\varepsilon \in O_K^{\times}$ with $\varepsilon_r > 1$ and every z as above, $W(I, \varepsilon, z)$ and the corresponding value $E_{\gamma}(z/v_2)$ of [RW] (Section 2) are units at every finite prime of every number field containing them.*

Theorem E (Square roots). *Assume identity (6) of [RW]. For every datum d , whatever the multiplier ring of I , $\sqrt{W(d)} \in K^{\text{ab}}$.*

For the Stark statements put $Z_{\mathfrak{m}}(s, B) := \zeta_{\mathfrak{m}}(s, B) - \zeta_{\mathfrak{m}}(s, c_u B)$ for $B \in \mathcal{G}_{\mathfrak{m}}$, where $\zeta_{\mathfrak{m}}(s, B)$ is the partial zeta function of B . The *analytic dictionary* is the statement that for every proper

$\mathfrak{m}$ and every $B \in \mathcal{G}_{\mathfrak{m}}$

$$(V) \quad \log W_{\mathfrak{m}}(B) = -n_{\mathfrak{m}} Z'_{\mathfrak{m}}(0, B).$$

We prove (V) from Shintani's product formula in Yamamoto's form (Theorem 4.2). Appendix A derives it also from Kopp's limit formula [Ko, Thm. 1.1]; the two routes share only Shintani's work and Hecke theory.

Let H/K be a finite abelian extension in which r splits completely, and S a set of places of K containing r , u and the primes ramified in H/K , with $|S| \geq 3$. Put $\mathfrak{m} := \mathfrak{f}(H/K) \prod_{\mathfrak{q} \in S \text{ finite}} \mathfrak{q}$. Let $\zeta_S(s, \sigma)$ be the partial zeta function of $\sigma \in \text{Gal}(H/K)$ with the Euler factors at the finite places of S removed, and w_r a fixed place of H above r . The rank-one abelian Stark conjecture $\text{St}(H/K, S)$, in Tate's formulation [Ta, Ch. IV] (see also [Ru]), predicts an element $\varepsilon_H \in H^\times$ with

- (St1) $|\varepsilon_H|_w = 1$ for every place w of H not above r ;
- (St2) $\log |\varepsilon_H^\sigma|_{w_r} = -2\zeta'_S(0, \sigma)$ for all $\sigma \in \text{Gal}(H/K)$;
- (St3) $H(\sqrt{\varepsilon_H})/K$ is abelian.

Call a modulus $\mathfrak{m}'$ *S-adapted* if its support is the set S_f of finite places of S and $\mathfrak{f}(H/K) \mid \mathfrak{m}'$. If every S -adapted $\mathfrak{m}'$ has $n_{\mathfrak{m}'} = 2$ and S contains no prime above 2, we say that (H, S) is in the *open configuration*. Otherwise some S -adapted $\mathfrak{m}'$ has $n_{\mathfrak{m}'} = 1$; this is always the case if K has no unit of norm -1 or if S contains a prime above 2.

Theorem F (Rank-one abelian Stark conjecture at a real place). *Assume identity (6) of [RW], and, if (H, S) is in the open configuration, also (I1). Then $\text{St}(H/K, S)$ holds for every pair (H, S) as above: there is ε_H satisfying (St1), (St2) and (St3).*

The proof uses (V).

In the open configuration let η be the generator of the units $\equiv 1 \pmod{\mathfrak{m}}$ with signs $(-, +)$ at (r, u) , so that $\text{Nm}\eta = -1$ and $\varepsilon_{\mathfrak{m}} = \eta^2$. For $A \in \mathcal{G}_{\mathfrak{m}}$ put $U_A := \exp(-Z'_{\mathfrak{m}}(0, A))$; these are the Stark units, and $U_A = +\sqrt{W_{\mathfrak{m}}(A)}$. In this configuration we take $H := H_{\mathfrak{m}\infty_u}$, and put $t := |\text{tr } \eta| \geq 3$ and $f_\eta := [O_K : \mathbb{Z}[\eta]]$.

Theorem G (Root law). *Assume identity (6) of [RW] and (I1). In the open configuration, $-\eta U_A \in H^{\times 2}$ for every class A in each of the following cases:*

- (a) t odd (then 2 is inert in K), for every f_η and every class number of H , assuming in addition [RW, (7)] at three families of pairs of labels (Section 4.5);
- (b) $f_\eta = 1$, that is $O_K = \mathbb{Z}[\eta]$, for every t , assuming in addition [RW, (36)] at one family of pairs of labels, the slice $\{2c\} \times B$ of Section 4.5.

In both cases the proof of Theorem G uses (V), Shintani's evaluation of two-dimensional cone zeta functions at $s = 0$ [Sh76] (through Theorem 4.3) and Rademacher's formula. The root law for t even and $f_\eta > 1$ remains open. We have checked it in about 130 cases of the open configuration over some 25 fields, exactly once the Stark units are recognized as roots of explicit polynomials. The cases include 2 inert, split and ramified, and t even; there was no failure.

Along the way we prove an archimedean sign law for the values (Theorem 3.1) and a formula for their phases (Theorem 4.3). The deduction of Theorem B from Theorem A and (N) follows the classical pattern of [Ta, Ch. IV] and [St] (compare [Ro]); we claim no novelty for it. Through (V), Theorem A is the reciprocity part of Stark's conjecture for the numbers $\exp(-n_{\mathfrak{m}} Z'_{\mathfrak{m}}(0, B))$.

1.2. Relation to other work. Hilbert’s twelfth problem for totally real fields has a p -adic solution: Dasgupta and Kakde show that Brumer–Stark units, given by an exact p -adic formula [DK1], together with $n - 1$ easily described elements generate the maximal abelian extension of a totally real field of degree n [DK2]. For real quadratic fields, Darmon and Vonk proposed p -adic singular moduli [DV], and Darmon, Pozzi and Vonk proved that the values of the Dedekind–Rademacher cocycle at real multiplication points are p -units in the narrow Hilbert class field [DPV].

The present work is archimedean: the generators are values of one complex-analytic function at real multiplication points, conjectured to be Stark units at a real place [Ko, RW]. Radchenko and Wheeler prove that these values are algebraic [RW, Thm. 1] through an overdetermined system of polynomial equations. The reciprocity law uses one equation of that system, [RW, (6)], and recovers algebraicity (Proposition 3.10). All analytic input to the reciprocity law is in that identity, proved in [RW]; the arithmetic part is a p -adic argument (Section 3.2) that turns the identity into Frobenius congruences. The archimedean rank-one Stark conjecture for real quadratic fields is stated as open in [Ko, §1.5] and [RW, §1]; Kopp obtains the unit property under a further hypothesis: [Ko, Thm. 1.3(2)] assumes [Ko, Conj. 6.9].

Kopp asks in [Ko, §1.5] for an analogue of Shimura’s reciprocity law for the real multiplication values of his Shintani–Faddeev cocycle, and notes that for their squares, in the fundamental-discriminant case, such a law would follow from the Stark conjectures. For maximal-order data $W(d)^{-1}$ is a power of $|\mathrm{Sh}_A^f(\beta)|^2 = \exp(n_{\mathfrak{m}} Z'_{\mathfrak{m}}(0, \mathfrak{A}))$, the modulus in [Ko, Thm. 1.1] (Appendix A), so Theorem A gives such a law for these numbers without assuming the Stark conjectures. We do not relate the values at non-maximal orders to Kopp’s. The shape of $(\star)$ resembles Shimura’s reciprocity law for CM torsion values [Shi, Ch. 5], with the archimedean component acting through $\mathrm{sgn} s_u$; we use this only as an analogy.

Zauner conjectured the existence of SICs with his symmetry in every dimension [Za]. Appleby, Flammia and Kopp proposed a construction through Stark units [AFK], and Radchenko and Wheeler proved, for their values, the quadratic relation on which it rests [RW, §4.4, Thm. 7]. Theorem C supplies the remaining step: a Galois conjugate of the values of modulus one. Radchenko and Wheeler obtain these conjugates independently in [RW2].

While this paper was in preparation, Radchenko and Wheeler released a preprint [RW2] in which they prove, with a different local argument based on the five-term relation [RW, (7)], that all the values are units and satisfy an Artin reciprocity law, and deduce Stark’s reciprocity, the generation of the ray class fields and the unitary conjugates needed for Zauner’s conjecture. On 29 September 2026 we sent them an account of an integrality argument at inert primes, the resulting Frobenius congruences and their application to SICs in even dimensions, and on 30 September an outline of exact root extraction and of a Chebotarev argument placing a Galois conjugate of the values on the unit circle.

1.3. Outline of the proof. Frobenius at a split prime $p = \mathfrak{p}\bar{\mathfrak{p}}$ is visible in the residues of the values. Radchenko and Wheeler’s line product formula, combined with power inheritance, writes $W(\mathfrak{p}^{-1}I, z)$, the value of a lattice one step away, as an exact product of $W(I, z)$ and values of I for the power ε^{k_0} along a line $T_{\mathfrak{p}}$ of labels, where k_0 is the order of ε modulo $\mathfrak{p}$; the k_0 -th root is extracted exactly (Section 3.3). If the residues modulo a prime $\mathfrak{P} \mid \mathfrak{p}$ are constant along that line, the product collapses to

$$W(I, z)^p \equiv W(\mathfrak{p}^{-1}I, z) \pmod{\mathfrak{P}},$$

which is what a reciprocity law predicts for $\mathrm{Frob}_{\mathfrak{p}}$ acting by $I \mapsto \mathfrak{p}^{-1}I$. The proof establishes this congruence, a statement about the p -adic rigidity of the values, and then globalizes it. Figure 1 shows the dependencies.

The local theorem (Section 3.2). Identity (6) of [RW], with the line product formula and reflection, implies that at all but finitely many primes $p \geq 5$ unramified in K every value is a p -adic unit, and that for split p the residues are constant along $T_{\mathfrak{p}}$ or along $T_{\bar{\mathfrak{p}}}$. The matrices linked by the line product formula fill a region of bounded height in the Bruhat–Tits tree (the tube lemma). An exact multiplicative identity, the *defect identity*, makes a defect valuation an average over the next layer outward. Identity (6), through an integrality bound of degree $\lfloor p/3 \rfloor$ on the outermost layer (a *code bound*), turns a pole there into a larger pole one step inward. Together these exclude a pole of maximal order.

Exact root extraction (Section 3.3). The family is built from a power γ^{k_0} , and a congruence between k_0 -th powers would lose a root of unity. The root is therefore taken among positive reals, before reduction, through an exact orbit-product identity. This gives the *Frobenius congruence* with two candidates, $\mathfrak{p}$ and $\bar{\mathfrak{p}}$; nothing local chooses between them.

Chebotarev and selection (Section 3.4). Chebotarev’s theorem turns the congruences into exact statements about Galois elements, still with two candidates. The choice is forced by two global symmetries: reality at the analytic place r , and the homothety law, whose negative-norm case is a new identity (Theorem DC). This proves Theorem A.

The other theorems of Part A follow quickly. Theorem D combines Theorem A at the bad primes with the local argument. Theorem C needs one Galois conjugate of the values on the unit circle, which the Galois form of Theorem A supplies, and Radchenko and Wheeler’s quadratic relation. Theorem B needs in addition the nonvanishing property (N). Theorem E comes from the same local machinery at a second family of primes.

Part B (Section 4). The dictionary (V) makes the values Stark units and gives (St1), (St2) and, when $n_{\mathfrak{m}} = 1$, (St3). When $n_{\mathfrak{m}} = 2$ the Stark units are square roots of values, and Theorem 4.5 gives their own reciprocity law. In the open configuration (St3) follows (Theorem 4.6) from an identity on the lattice $\mathbb{Z}[\eta]$ (Theorem W), which we derive by Radchenko and Wheeler’s contour method from the Fourier-transform identity of Faddeev, Kashaev and Volkov. Theorem G sharpens (St3) to the root law, through a dyadic computation for t odd and through Theorem W on O_K when $O_K = \mathbb{Z}[\eta]$.

No Stark conjecture, no Shimura reciprocity and no moduli interpretation of real multiplication points enter the proof of Theorem A, and the values are never assumed to be algebraic or integral. Appendix C lists what is new and what is imported.

Section 2 defines the values and lists the identities we use. Section 3 proves Theorems A–E, and Section 4 proves Theorems F and G. Section 5 describes how the argument has been checked, and Appendix C lists the imported results and what each theorem depends on. This work was carried out with substantial assistance from AI systems, and all review of it so far has been by AI agents (Section 5).

2. THE VALUES AND THE IDENTITIES THEY SATISFY

We define the positive numbers $W(d)$ and list the identities they satisfy. Identities (i)–(vii) are results of Radchenko and Wheeler [RW]; (ix) follows from [RW, (3)]; (viii) and (x) are ours and are proved in Section 3.1.

Faddeev’s modular quantum dilogarithm. Write $e(x) := e^{2\pi i x}$ and $j(g, \tau) := c_g \tau + d_g$ for $g = \begin{pmatrix} a_g & b_g \\ c_g & d_g \end{pmatrix}$. In this section, and in the analytic formulas of Sections 4.1 and 4.4, τ denotes a complex variable, not the automorphism of K . For $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, $m, n \in \mathbb{Z}$ and $(z, \tau) \in \mathbb{C} \times (\mathbb{C} \setminus \mathbb{R})$, Radchenko and Wheeler [RW, (1)], following Garoufalidis, Kashaev and Zagier [GKZ], put

$$\Phi_{\gamma, m, n}(z; \tau) := \frac{(q^m e(z); q)_{\infty}}{(\tilde{q}^n e(z/j(\gamma, \tau)); \tilde{q})_{\infty}}, \quad q = q_{\tau} := e(\tau), \quad \tilde{q} = \tilde{q}_{\tau} := e(\gamma\tau).$$

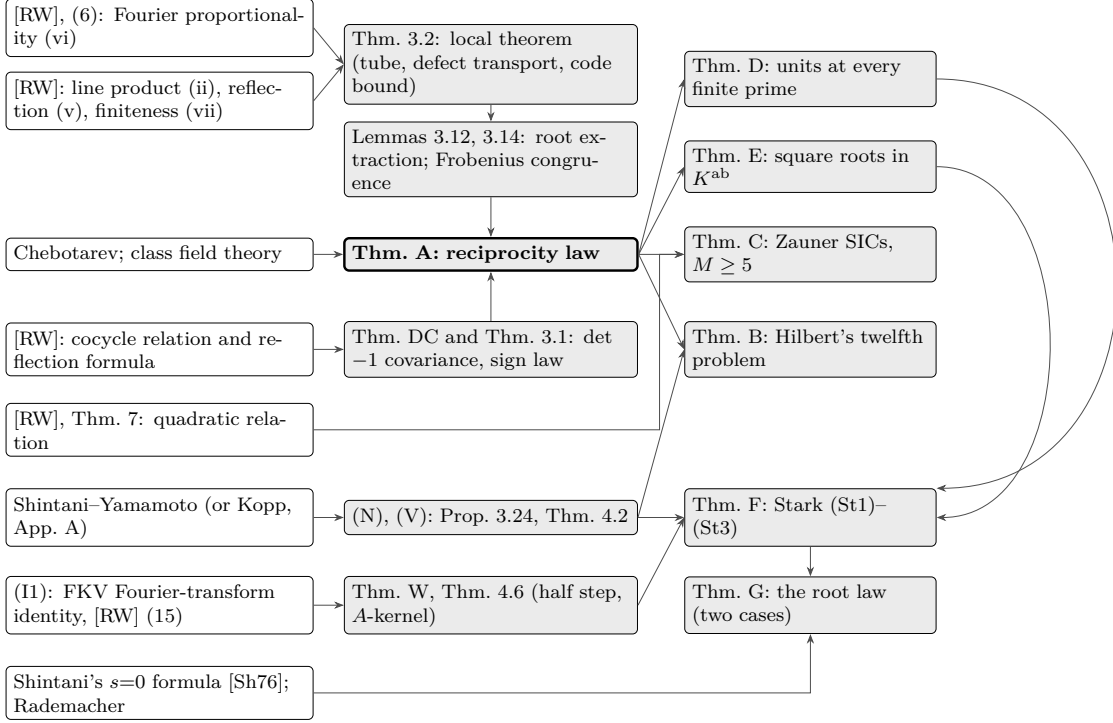


FIGURE 1. Dependencies among the main results. Boxes on the left are imported results, boxes in the middle are the new arguments, and boxes on the right are the main theorems. Theorem G also uses further identities of [RW] (Section 4.5).

For $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ one has $\Phi_{S,m,n}(z; \tau) = \Phi_S(z + m\tau - n; \tau)$ with $\Phi_S := \Phi_{S,0,0}$, and for $\tau > 0$ (after continuation) this is Faddeev's quantum dilogarithm:

$$\Phi_S(z; \tau) = \varphi_{\mathbf{b}}(i\mathbf{b}^{-1}z - i\mathbf{b}/2 + i/(2\mathbf{b})), \quad \mathbf{b} = \sqrt{\tau},$$

where $\log \varphi_{\mathbf{b}}(x) = \int_{\mathbb{R}+i0} e^{-2ixw} dw / (4w \sinh(\mathbf{b}w) \sinh(w/\mathbf{b}))$. By [RW, Prop. 2(i)], $\Phi_{\gamma,m,n}$ is a finite product of factors $\Phi_S(z + (m + \ell)\tau + k\ell; j(\gamma, \tau))$, so it continues meromorphically to $j(\gamma, \tau) \notin \mathbb{R}_{\leq 0}$, in particular to a real fixed point of γ . The *index rules*

$$\Phi_{\gamma,m,n} = (1 - q^m e(z)) \Phi_{\gamma,m+1,n}, \quad \Phi_{\gamma,m,n+1} = (1 - \tilde{q}^n e(z/j(\gamma, \tau))) \Phi_{\gamma,m,n}$$

follow from the definition and $(x; q)_{\infty} = (1 - x)(qx; q)_{\infty}$. Two functional equations of [RW] are used directly: the *cocycle relation* [RW, (17)],

$$\Phi_{gh,m,n}(z; \tau) = \Phi_{g,k,n}(z/j(h, \tau); h\tau) \Phi_{h,m,k}(z; \tau) \quad (k \in \mathbb{Z}),$$

and the *reflection formula* [RW, §2.2]

$$\Phi_{\delta g \delta, 1-m, 1-n}(z; -\tau) = \Phi_{g,m,n}(z; \tau)^{-1}, \quad \delta := \text{diag}(1, -1).$$

Both hold on $\mathbb{C} \times (\mathbb{C} \setminus \mathbb{R})$, and they extend to the real point because the domain of meromorphy is connected.

Admissible matrices and labels. Call $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$ *admissible* if it is hyperbolic with $c > 0$ and $\text{tr } \gamma \geq 3$. Let τ^* be its attracting fixed point. Then $\varepsilon_{\gamma} := c\tau^* + d > 1$ is a

totally positive unit of norm 1 in $K = \mathbb{Q}(\tau^*)$, and $\gamma(\tau^*, 1)^\top = \varepsilon_\gamma(\tau^*, 1)^\top$. Put $I_\gamma := \mathbb{Z}\tau^* + \mathbb{Z}$, $t_\varepsilon := \varepsilon_\gamma^{-1} - 1$ and

$$G_\gamma := t_\varepsilon^{-1} I_\gamma / I_\gamma \cong \mathbb{Z}^2 / \Lambda_\gamma, \quad \Lambda_\gamma := \mathbb{Z}^2(\gamma - I),$$

where $x \in t_\varepsilon^{-1} I_\gamma$ has label $\mathbf{u} \in \mathbb{Z}^2$ (a row vector) defined by $(\varepsilon_\gamma^{-1} - 1)x = u_1 \tau^* + u_2$. The group G_γ has order $N_\gamma = \text{tr } \gamma - 2$, and ε_γ fixes it pointwise.

The values of Radchenko and Wheeler. For $\mathbf{u} \notin \Lambda_\gamma$ put

$$E_\gamma(\mathbf{u}) := \mu_\gamma \Phi_{\gamma, u_1, 0} \left(\frac{u_1 \tau^* + u_2}{\varepsilon_\gamma^{-1} - 1}; \tau^* \right), \quad E_\gamma(0) := \varepsilon_\gamma^{1/2},$$

where $\mu_\gamma \in \mu_{24}$ is the multiplier of Dedekind's eta function, $\eta(\gamma w) = \mu_\gamma \sqrt{cw + d} \eta(w)$. This is $F_\gamma^+(\mathbf{u})$ of [RW, (2), (4)], evaluated at the attracting point as in [RW, (37)]. The first argument is the torsion point of label $\mathbf{u}$, so E_γ is a function on G_γ . We call a pair (m, n) of integers with $m - n = u_1$ *compatible* with the label $\mathbf{u}$. At a fixed point, the index quasi-periodicity [RW, (16)] gives $\Phi_{\gamma, m, n} = \Phi_{\gamma, u_1, 0}$ at the torsion point of label $\mathbf{u}$ for every compatible pair; the intermediate factors $1 - e(\cdot)$ are nonzero because the point does not lie in the lattice.

The Gaussian and its bicharacter are [RW, (3)]

$$q(\mathbf{u}) := (-1)^{u_1 + u_2 + u_1 u_2} e_{2N}(\omega(\mathbf{u}\gamma, \mathbf{u})), \quad b(\mathbf{u}, \mathbf{v}) := e_N(\omega(\mathbf{u}\gamma - \mathbf{u}, \mathbf{v})),$$

with $\omega(\mathbf{v}, \mathbf{w}) = v_1 w_2 - v_2 w_1$, $e_N(x) = e(x/N)$ and $N = N_\gamma$. Both are Λ_γ -periodic. The Gaussian q , not to be confused with the nome q_τ , is even, $b(x, x) = q(x)^2$, $q(nx) = q(x)^{n^2}$, and q takes values in μ_{2e} , where e is the exponent of G_γ . Write $(\mathcal{F}f)(y) := N^{-1/2} \sum_x b(x, y) f(x)$ and, for $x \neq 0$, $w_\gamma(x) := q(x) E_\gamma(x)^2$.

The values $W(d)$. Let $d = (I, \varepsilon, z)$ be a datum (Section 1). A basis (v_1, v_2) of I is a *fixed frame* if v_2 is totally positive and $\theta := v_1/v_2$ satisfies $\theta_r > \theta_u$. Fixed frames exist: take v_2 to be the positive generator of $I \cap \mathbb{Q}$ and adjust the sign of v_1 . The matrix γ of ε in a fixed frame, $\varepsilon(\theta, 1)^\top = \gamma(\theta, 1)^\top$, is admissible, since $c(\theta_r - \theta_u) = \varepsilon_r - \varepsilon_u > 0$; moreover $\tau^* = \theta_r$, $\varepsilon_\gamma = \varepsilon$ and $I_\gamma = v_2^{-1} I$. Put

$$W(d) = W(I, \varepsilon, z) := w_\gamma(z/v_2),$$

reading z/v_2 as a nonzero label of G_γ . Two fixed frames differ by some $g \in \text{SL}_2(\mathbb{Z})$ with $j(g, \theta) = v'_2/v_2 \gg 0$, so (vii) below makes W independent of the frame, and (iv) makes it a positive real. The value W is unchanged under $(I, z) \mapsto (\beta I, \beta z)$ for $\beta \gg 0$; also $w_\gamma(x) = W(I_\gamma, \varepsilon_\gamma, x)$, and $W(I, \varepsilon^k, z) = W(I, \varepsilon, z)^k$ by (iii). When ε is fixed we write $W(I, z)$.

The identities. The following hold for every admissible γ , with no restriction on N ; we write $E = E_\gamma$, $G = G_\gamma$ and $N = N_\gamma$.

- (i) *Regularity.* E is regular at the real multiplication point and nowhere zero on G ; $E(0) = \varepsilon^{1/2}$.
- (ii) *Line product formula.* For a subgroup $T \subset G$ of prime order p there are an admissible sibling γ_T of the same trace, with lattice $I_\gamma + \tilde{T}$ (where $\tilde{T}$ is the preimage of T), and a projection $\pi_T : G \rightarrow G_{\gamma_T}$ with kernel T , such that

$$\prod_{h \in T} E(y + h) = \kappa_T E_{\gamma_T}(\pi_T y)$$

for every coset $y + T$, with $\kappa_T = \mu_\gamma^p / \mu_{\gamma_T} \in \mu_{24}$ independent of the coset, the origin coset included.

- (iii) *Power inheritance.* $E_{\gamma^k} = E_\gamma^k$ and $q_{\gamma^k} = q_\gamma^k$ on $G_\gamma \subset G_{\gamma^k}$.
- (iv) *Real type.* $\overline{E(x)} = q(x) E(x)$; hence $w_\gamma(x) = |E(x)|^2 > 0$.

- (v) *Reflection.* $E(x)E(-x) = q(x)^{-1}$ for $x \neq 0$; hence $w_\gamma(-x) = w_\gamma(x)^{-1}$.
- (vi) *Fourier proportionality.* $\mathcal{F}E(y) = \lambda q(y)E(-y)$ for all y , the origin included. Only the existence of some scalar λ is used. Completing the square gives $(q^{-1}\mathcal{F}^{-1})^3 = \gamma_G \cdot I$ with $\gamma_G = N^{-1/2} \sum_y q(y)^{-1} \in \mu_8$ (Milgram's formula, [MH, Appendix 4]), so (vi) alone forces $\lambda^3 = \gamma_G$ and $\lambda \in \mu_{24}$. Radchenko and Wheeler show that $\lambda = \mu_\gamma$; this is used only for the root law, Section 4.5.
- (vii) *Covariance, finiteness, ε -invariance.* If $g \in \mathrm{SL}_2(\mathbb{Z})$, $j(g, \tau^*) > 0$ and $\gamma' = g\gamma g^{-1}$ is admissible, then $E_{\gamma'}(\mathbf{u}g^{-1}) = (\mu_{\gamma'}/\mu_\gamma)E_\gamma(\mathbf{u})$. The values of all admissible matrices of a given trace form a finite set. Finally, $E_{\gamma^k}(\mathbf{u}\gamma) = E_{\gamma^k}(\mathbf{u})$ on G_{γ^k} .
- (viii) *Homothety law.* $W(\beta I, \varepsilon, \beta z) = W(I, \varepsilon, z)^{\mathrm{sgn} \beta u}$ for $\beta \in K^\times$.
- (ix) *Gaussian under odd changes.* For a sibling along T of odd prime order p , with row-label map M_T (so $\gamma M_T = M_T \gamma_T$ and $\det M_T = p$), one has $q_{\gamma_T}(xM_T) = q_\gamma(x)^p$; for $\pi \in \mathbb{Z}[\varepsilon]$ of odd norm, $q(\pi x) = q(x)^{N\pi}$. Both statements follow from [RW, (3)] and $q(nx) = q(x)^{n^2}$. At $p = 2$, or for even norm, a sign -1 can appear; those cases are not used.
- (x) *Covariance under determinant -1* (Theorem DC in Section 3.1): the analogue of (vii) for $\det g = -1$ and $j(g, \tau^*) < 0$, with complex conjugation.

Sources. For (i): by [RW, §2] (from [RW, Prop. 2]), the zeros and poles of $\Phi_{\gamma, m, n}(\cdot; \tau^*)$ lie in $\mathbb{Z}\tau^* + \mathbb{Z} = I_\gamma$, which the torsion points avoid, and $E_\gamma(0) = \varepsilon^{1/2}$ is their convention [RW, (4), (37)]; that E_γ is a function on G_γ is [RW, Lemma 2]. Identity (ii) is [RW, Prop. 9, (29)]; (iii) and the ε -invariance in (vii) are [RW, Prop. 8]; (iv) is [RW, Prop. 7 and Cor. 1]; (v) is [RW, (5)] (from [RW, Prop. 1]); (vi) is [RW, Thm. 2, (6)] together with [RW, (5)], transported to the attracting point by Lemma FX below; the covariance in (vii) follows from [RW, (17)]; and (ix) follows from [RW, (3)]. As a check of Radchenko and Wheeler's results we have re-derived (i)–(v), (vii) and (ix) from their definitions [RW, (1)–(4)]. For Theorems A and D, the only result of [RW] used without an independent proof is [RW, (6)].

Attracting and repelling fixed points. Radchenko and Wheeler write out the contour proofs of their identities (6) and (7) for $0 < \varepsilon < 1$, that is, at a positive repelling fixed point, and state that the case $\varepsilon > 1$ is similar. Lemma FX, which follows from their cocycle relation and reflection formula, reduces our case, the attracting point, to the case they write out.

Lemma FX (Fixed-point exchange). *Let $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, $J_\tau := j(\gamma, \tau)$ and $\tilde{\gamma} := \delta\gamma^{-1}\delta = \begin{pmatrix} d & b \\ c & a \end{pmatrix}$. Then*

$$\Phi_{\gamma, m, k}(z; \tau) = \Phi_{\tilde{\gamma}, 1-k, 1-m}(z/J_\tau; -\gamma\tau)$$

as meromorphic functions. In value form: for admissible γ with repelling fixed point τ_{rep} , let E_γ^{rep} be defined as E_γ with τ_{rep} in place of τ^ . Then for $\mathbf{u} \notin \Lambda_\gamma$*

$$E_\gamma^{\mathrm{rep}}(\mathbf{u})/\mu_\gamma = E_{\tilde{\gamma}}(\mathbf{u}\delta)/\mu_{\tilde{\gamma}},$$

with $E_{\tilde{\gamma}}$ evaluated at its attracting point $-\tau_{\mathrm{rep}}$.

Proof. The cocycle relation with $(g, h) = (\gamma^{-1}, \gamma)$ and $\Phi_{I, m, m} = 1$ gives $\Phi_{\gamma, m, k}(z; \tau) = \Phi_{\gamma^{-1}, k, m}(z/J_\tau; \gamma\tau)^{-1}$, and by the reflection formula the right side equals $\Phi_{\tilde{\gamma}, 1-k, 1-m}(z/J_\tau; -\gamma\tau)$. Both sides are meromorphic on $\mathbb{C} \times \{j(\gamma, \tau) \notin \mathbb{R}_{\leq 0}\}$. This domain is connected, and $\tau \mapsto -\gamma\tau$ maps it into the corresponding domain of $\tilde{\gamma}$, as $j(\tilde{\gamma}, -\gamma\tau) = 1/J_\tau$. For the value form take $(m, k) = (u_1, 0)$ and $\tau = \tau_{\mathrm{rep}}$. Then z/J_τ is the $\tilde{\gamma}$ -torsion point of label $\mathbf{u}\delta$, and the pair $(1, 1 - u_1)$ is compatible with it. $\square$

Combine Lemma FX with a translation $\begin{pmatrix} 1 & -k \\ 0 & 1 \end{pmatrix}$ that makes both fixed points negative; by (vii), the translation moves E , q and b by one label permutation. The label map $\mathbf{u} \mapsto \mathbf{u}\delta$ preserves q and b , and $\mu_{\tilde{\gamma}} = \mu_{\gamma}$. In this way Lemma FX carries identities (6) and (7) of [RW] from the case written out by Radchenko and Wheeler to the attracting point, with their origin convention $E(0) = \varepsilon^{1/2}$. Their Theorem 3 is therefore used only in its printed case $\tau > 0$.

3. PART A: THE RECIPROCITY LAW, HILBERT'S TWELFTH PROBLEM, SICs AND UNITS

3.1. The archimedean sign law and the homothety law. For the right side of $(\star)$ to depend only on the class of s modulo $K^\times$, the values must satisfy the homothety law (viii) for every $\beta \in K^\times$. For totally positive β this is automatic, and for totally negative β it is reflection. For β of negative norm it needs Theorem DC, which relates values in frames of opposite orientation. We prove Theorem DC and then combine it with Lemma FX and (vii) into one law for all changes of frame.

Theorem DC (Det -1 covariance). *Let γ be admissible, let $g \in \mathrm{GL}_2(\mathbb{Z})$ with $\det g = -1$ be such that $\gamma' := g\gamma g^{-1}$ is admissible, and suppose that $j_g := j(g, \tau^*) < 0$. Then*

$$E_{\gamma'}(\mathbf{u}g^{-1}) = \mu_{\gamma}\mu_{\gamma'} \overline{E_{\gamma}(\mathbf{u})} \quad (\mathbf{u} \neq 0), \quad E_{\gamma'}(0) = E_{\gamma}(0) = \varepsilon^{1/2}.$$

The statement uses four elementary facts, which hold as for $\mathrm{SL}_2(\mathbb{Z})$.

First, $g\tau^*$ is a fixed point of γ' , and the cocycle relation

$$j(\gamma', g\tau^*) j(g, \tau^*) = j(\gamma'g, \tau^*) = j(g\gamma, \tau^*) = j(g, \tau^*) \varepsilon$$

gives $j(\gamma', g\tau^*) = \varepsilon > 1$. So $g\tau^*$ is the attracting point of γ' , and $\varepsilon_{\gamma'} = \varepsilon$.

Second, $g(\tau^*, 1)^T = j_g(g\tau^*, 1)^T$. Since $g \in \mathrm{GL}_2(\mathbb{Z})$, the entries of $g(\tau^*, 1)^T$ span I_{γ} . Hence $I_{\gamma'} = j_g^{-1} I_{\gamma}$.

Third, if $(\varepsilon^{-1} - 1)z = \mathbf{u}(\tau^*, 1)^T$, then

$$(\varepsilon^{-1} - 1)z/j_g = \mathbf{u}g^{-1} \cdot j_g^{-1}g(\tau^*, 1)^T = \mathbf{u}g^{-1}(g\tau^*, 1)^T.$$

So torsion points map by $z \mapsto z/j_g$ and labels by $\mathbf{u} \mapsto \mathbf{u}g^{-1}$.

Fourth, with a prime denoting the conjugate, $g\tau^* - (g\tau^*)' = \det g (\tau^* - \tau^{*'})/(j_g j_g')$. For admissible matrices the attracting point exceeds its conjugate, as $c(\tau^* - \tau^{*'}) = \varepsilon - \varepsilon^{-1} > 0$. So $\mathrm{sgn} \mathrm{Nm} j_g = \det g = -1$.

Proof. Put $h := \delta g \in \mathrm{SL}_2(\mathbb{Z})$. Then $h\tau^* = -g\tau^*$, $j(h, \tau^*) = -j_g > 0$, and $h\gamma h^{-1} = \delta\gamma'\delta =: \gamma'_\delta$. Let z_0 be the torsion point of label $\mathbf{u}$ for γ , let $p' := z_0/j_g$, which is the γ' -torsion point of label $\mathbf{u}' := \mathbf{u}g^{-1}$, and let $k' := u'_1$.

We first expand $\Phi_{h\gamma, u_1, 0} = \Phi_{\gamma'_\delta h, u_1, 0}$ at (z_0, τ^*) by the cocycle relation in the two possible ways:

$$\Phi_{\gamma'_\delta, k', 0}(-p'; -g\tau^*) \Phi_{h, u_1, k'}(z_0; \tau^*) = \Phi_{h, 0, 0}(z_0/\varepsilon; \tau^*) \Phi_{\gamma, u_1, 0}(z_0; \tau^*).$$

The two Φ_h factors are equal. Indeed, write $h = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$. From the definition of Φ in Section 2, with $\tau/j(h, \tau) = D h\tau - B$ and $1/j(h, \tau) = A - C h\tau$, so that $e(\tau/j(h, \tau)) = \tilde{q}^D$ and $e(1/j(h, \tau)) = \tilde{q}^{-C}$, while $e(z + \tau) = q e(z)$, one obtains the translation rules

$$\Phi_{h, m, n}(z + \tau; \tau) = \Phi_{h, m+1, n+D}(z; \tau), \quad \Phi_{h, m, n}(z + 1; \tau) = \Phi_{h, m, n-C}(z; \tau).$$

Since $(\varepsilon^{-1} - 1)z_0 = u_1\tau^* + u_2$, we have $z_0/\varepsilon = z_0 + u_1\tau^* + u_2$, so

$$\Phi_{h, 0, 0}(z_0/\varepsilon; \tau^*) = \Phi_{h, u_1, u_1 D - u_2 C}(z_0; \tau^*), \quad u_1 D - u_2 C = (\mathbf{u}h^{-1})_1 = (\mathbf{u}g^{-1}\delta)_1 = k'.$$

Hence

$$\Phi_{\gamma'_\delta, k', 0}(-p'; -g\tau^*) = E_{\gamma}(\mathbf{u})/\mu_{\gamma}.$$

Next, the reflection formula for γ' with $(m, n) = (1 - k', 1)$ at $z = -p'$ gives $E_{\gamma'}(\mathbf{u})/\mu_{\gamma'} = 1/\Phi_{\gamma', 1-k', 1}(-p'; g\tau^*)$. The point $-p'$ is the γ' -torsion point of label $-\mathbf{u}'$, and the pair $(1 - k', 1)$ has $m - n = (-\mathbf{u}')_1$, so it is compatible with this label. Therefore $\Phi_{\gamma', 1-k', 1}(-p'; g\tau^*) = E_{\gamma'}(-\mathbf{u}')/\mu_{\gamma'}$.

Finally, by (v) and (iv) for γ' , $1/E_{\gamma'}(-\mathbf{u}') = q_{\gamma'}(\mathbf{u}')E_{\gamma'}(\mathbf{u}') = \overline{E_{\gamma'}(\mathbf{u}')}$. So $E_{\gamma}(\mathbf{u}) = \mu_{\gamma}\mu_{\gamma'}\overline{E_{\gamma'}(\mathbf{u}')}$; conjugating and using $|\mu_{\gamma}| = |\mu_{\gamma'}| = 1$ gives the claim. The statement at the origin is the definition, and by (i) every point used lies off the divisor. $\square$

The case $j_g > 0$ is not used. The proof exchanges the sign of j against one application of the reflection formula. The cocycle relation moves the evaluation point to $-g\tau^*$, where γ'_δ has positive automorphy factor; reflection brings it back to $g\tau^*$ at the cost of an inversion, which real type converts into complex conjugation.

Frames and place values. Let I , ε and z be as in a datum. We call any $\mathbb{Z}$ -basis $F = (v_1, v_2)$ of I a *frame*, and put $\theta := v_1/v_2$ and $x := z/v_2$. For $g \in \mathrm{GL}_2(\mathbb{Z})$, the frame $gF := g(v_1, v_2)^T$ has $\theta_{gF} = g\theta$ and second vector $v_2 j(g, \theta)$, and $j(gh, \theta) = j(g, h\theta)j(h, \theta)$. Exactly one unit $\eta_F \in \{\varepsilon, \varepsilon^{-1}\}$ has a matrix γ_F in F with lower-left entry $c > 0$. This matrix is admissible with fixed points θ_r, θ_u , and θ_ν is attracting exactly when $(\eta_F)_\nu > 1$. The label $\mathbf{u}$ of x is defined by $(\eta_F^{-1} - 1)x = u_1\theta + u_2$. For a real place ν , the *place value* $\Phi_\nu(F)$ is $\Phi_{\gamma_F, u_1, 0}$ evaluated at the torsion point of the label $\mathbf{u}$ and at the fixed point θ_ν ; it is the μ -free value of [RW] at θ_ν .

To record how place values change with the frame, put $\mathbf{c}(w) := \bar{w}$ and $\mathbf{i}(w) := w^{-1}$ on $\mathbb{C}^\times$, and define, for $\xi \in K^\times$ and $\bar{\nu}$ the other real place,

$$\sigma_\nu(\xi) := \mathbf{c}^{[\xi_\nu < 0]} \mathbf{i}^{[\xi_{\bar{\nu}} < 0]}.$$

This is a homomorphism $K^\times \rightarrow \langle \mathbf{c}, \mathbf{i} \rangle \cong (\mathbb{Z}/2)^2$ whose kernel consists of the totally positive elements; moreover $\sigma_\nu(-1) = \mathbf{c}\mathbf{i}$ and $\sigma_u(\xi) = \sigma_r(\tau\xi)$.

Theorem 3.1 (Archimedean sign law). *For every frame F , every $g \in \mathrm{GL}_2(\mathbb{Z})$ and each real place ν ,*

$$\Phi_\nu(gF) = \sigma_\nu(j(g, \theta)) \Phi_\nu(F).$$

Equivalently, the place value $\mathfrak{E}_\nu(I, z) := \sigma_\nu(v_2) \Phi_\nu(F)$ does not depend on the frame.

Thus the sign of j at the evaluation place acts by complex conjugation, its sign at the other place acts by inversion, and no root of unity appears.

Proof. Since σ_ν is a homomorphism into a group of commuting involutions and j is a cocycle, the law composes and inverts along chains of frames. It therefore suffices to prove it for four generators, each of which uses one input once.

For $g = -I$ we use reflection and real type. The label becomes $-\mathbf{u}$, and (iv) and (v) give $E(-\mathbf{u}) = 1/\overline{E(\mathbf{u})}$, so the twist is $\mathbf{c}\mathbf{i} = \sigma_\nu(-1)$. At a repelling point the same holds through Lemma FX, since $(-\mathbf{u})\delta = -(\mathbf{u}\delta)$.

For $g = -\delta$, where $j = 1$, we use Lemma FX. Now $\theta \mapsto -\theta$, $\gamma_F \mapsto \tilde{\gamma}_F$, and the label becomes $\mathbf{u}\gamma_F\delta \equiv \mathbf{u}\delta$. Lemma FX, applied to γ_F and to $\tilde{\gamma}_F$, leaves Φ_ν unchanged.

For $g \in \mathrm{SL}_2(\mathbb{Z})$ with $j(g, \theta)$ totally positive and ν attracting we use (vii). The orientation $\theta_r > \theta_u$ is kept, $\gamma_{gF} = g\gamma_F g^{-1}$, the label is $\mathbf{u}g^{-1}$, and (vii) gives $\Phi_\nu(gF) = \Phi_\nu(F)$.

For $\det g = -1$ with $j_\nu < 0 < j_{\bar{\nu}}$ and ν attracting we use Theorem DC. The orientation is kept, since $\det g \cdot \mathrm{sgn Nm} j = +1$, and the label is $\mathbf{u}g^{-1}$. Theorem DC divided by $\mu_{\gamma'}$ gives $\Phi_\nu(gF) = |\mu_\gamma|^2 \overline{\Phi_\nu(F)} = \mathbf{c} \Phi_\nu(F)$.

If θ_ν is repelling, replace F by $(-\delta)F$ and g by $\delta g \delta$; this has the same determinant, and $j(\delta g \delta, -\theta) = j(g, \theta)$. The eight cases ($\det g, \mathrm{sgn} j_\nu, \mathrm{sgn} j_{\bar{\nu}}$) reduce to the last two generators by composing with $-I$ and writing $g = g_1(-\delta)$, where $j(g_1, -\theta) = j(g, \theta)$. $\square$

Theorem 3.1 adds no analytic input to its four generators. It never refers to r , u or to the attracting point, so Radchenko and Wheeler's function at the repelling point obeys the same law.

Evaluating $\mathfrak{E}_\nu$ in convenient frames gives three consequences.

- (a) *Normalization.* In a fixed frame $\sigma_r(v_2) = 1$ and θ_r is attracting, so $\mathfrak{E}_r(I, z) = E_\gamma(\mathbf{u})/\mu_\gamma$ and $|\mathfrak{E}_r(I, z)|^2 = W(I, z)$. Put $W_\nu(I, z) := |\mathfrak{E}_\nu(I, z)|^2$.
- (b) *Place swap.* $\mathfrak{E}_u(I, z) = \mathfrak{E}_r(\tau I, \tau z)$, hence $W_u(I, z) = W(\tau I, \tau z)$. Indeed, τF is a frame of τI with the same matrix and label, and $(\tau\theta)_r = \theta_u$.
- (c) *Homothety.* $\mathfrak{E}_\nu(\beta I, \beta z) = \sigma_\nu(\beta) \mathfrak{E}_\nu(I, z)$ for $\beta \in K^\times$, since βF has the same θ , matrix and label. Taking $|\cdot|^2$ at $\nu = r$ gives the homothety law (viii), $W(\beta I, \beta z) = W(I, z)^{\text{sgn } \beta_u}$; at $\nu = u$ it gives the mirror law $W_u(\beta I, \beta z) = W_u(I, z)^{\text{sgn } \beta_r}$. At the level of E : for $\beta_r < 0 < \beta_u$ and fixed frames of I and βI , $E(\beta I, \beta z)/\mu = \overline{E(I, z)/\mu}$. For such β a fixed frame of βI differs from βF by some g with $\det g = -1$ and $j_g < 0$; this is the case of Theorem DC that enters.

The homothety law supplies (Sel3) in the selection step of Section 3.4, the $K^\times$ -invariance of the right side of $(\star)$, the descent of the maximal-order values from C_m to $\mathcal{G}_m$ (Section 3.6) and the homothety step in the proof of Lemma 3.11(b). Part B uses its form at the level of E at $\beta = \eta$.

Theorem DC is the only input that relates the two orientation classes of frames. The sign of $\text{Nm } v_2$ splits the frames of I into two classes, and a change of frame multiplies it by $\text{Nm } j(g, \theta)$; among the generators only Theorem DC changes the class. So Theorem DC is what makes $K^\times/K_{\text{Nm}>0}^\times \cong \mathbb{Z}/2$ act trivially on the right side of $(\star)$, and it is not a consequence of the other identities. On the $\text{GL}_2(\mathbb{Z})$ -orbit of a fixed point whose order has no unit of norm -1 , replacing $E_\gamma(\mathbf{u})$ by $E_\gamma(-\mathbf{u})$ on the frames of one class preserves (iv)–(vii) and Lemma FX, but violates Theorem DC wherever $W \neq 1$. On orbits whose order has a unit of norm -1 , we have not shown that Theorem DC is independent of the other identities.

3.2. Integrality at a prime: the local theorem. To compare Frobenius at a prime p with the values, we must reduce them modulo a prime above p , and this requires that no value has p in its denominator. We prove this here, together with the finer residue statement needed for the congruences of Section 3.3. In the proof of Theorem A this is the one place where the identities of Section 2 are turned into arithmetic information. It assumes nothing about the values beyond identities (i)–(vii): neither that they are algebraic nor that they are integral at p . Split and inert primes are treated together; only split primes are used for Theorem A.

Setting. Let γ be admissible, $\varepsilon = \varepsilon_\gamma$, and let f be the conductor of $\text{End}(I_\gamma)$, so that $\text{End}(I_\gamma) = \mathbb{Z} + fO_K$. Let $p \geq 5$ be a prime unramified in $K = \mathbb{Q}(\tau^*)$ with $p \nmid N_\gamma f$. Let k_0 be the order of ε modulo p ; then $k_0 \geq 2$, as $p \nmid N_\gamma$. Put $\gamma_0 := \gamma^{k_0}$, $\varepsilon_0 := \varepsilon^{k_0}$ and $R := v_p(\varepsilon_0 - 1) \geq 1$, the largest R with $\varepsilon_0 - 1 \in p^R O_K$. Any R is allowed ($R > 1$ occurs at primes of Wieferich type), and no constant below depends on R .

The *members* are the admissible matrices reachable from γ_0 by repeated siblings along subgroups of order p (identity (ii)). For a member X we write I_X , G_X and E_X for its lattice, labels and values as in Section 2. All members have the trace of γ_0 , hence the same unit ε_0 , and by (vii) they have only finitely many values between them. Fix a field K_{fam} of characteristic 0 containing K , the values of all members, μ_{24} , $\mu_{p^{2R}}$ and the square roots occurring in (vi), and fix any valuation $v : K_{\text{fam}}^\times \rightarrow \mathbb{R}$ with $v(p) = 1$; neither completeness nor discreteness is needed. Let $\mathcal{O}_v$ be its valuation ring. The *residue* of an element of $\mathcal{O}_v$ is its image modulo the

maximal ideal, and $\equiv$ denotes congruence modulo that ideal. For a member X let

$$a(X) := \max(0, \max_y -v(E_X(y)))$$

be its largest pole. By reflection (v), $v(E_X(y)) = -v(E_X(-y))$ for $y \neq 0$, and $E_X(0)^2 = \varepsilon_0$ is a unit. Hence we have the *reflection bound* $|v(E_X(y))| \leq a(X)$ for every y .

Theorem 3.2 (Local theorem). *In this setting the following hold.*

- (U0) *Every value of every member is a v -unit.*
- (U1) *For every member X and every label y , the defect $\Theta_X(y)$ (defined before Lemma 3.7) satisfies $\Theta_X(y) \equiv 1$.*
- (U2) *If $p = \mathfrak{p}\bar{\mathfrak{p}}$ splits, then for every $x \in G_\gamma$ the residue of E_{γ_0} on $x + G_{\gamma_0}[p]$ is invariant under translation by the line $T_{\mathfrak{p}} := \mathfrak{p}_O^{-1}I_\gamma/I_\gamma$, where $\mathfrak{p}_O := \mathfrak{p} \cap \text{End}(I_\gamma)$, or by $T_{\bar{\mathfrak{p}}}$; which of the two may depend on x . If p is inert, the residue is constant on $x + G_{\gamma_0}[p]$.*

The global argument of Section 3.4 decides between the two lines in (U2).

The proof has four ingredients. The first is geometric: members are vertices of a tree whose edges are sibling steps, and the family lives in a region of bounded height, the *tube* (Figure 2). The second is an exact multiplicative identity, the *defect identity*, which carries a large valuation outward through the tube, with no loss at vertices off the crater. The third is an integrality bound on the outermost layer, from the Fourier identity (vi), which turns a pole there into a pole at least three times as large one step inward. The fourth is a comparison of constants showing that these are incompatible with a pole of maximal order. Reducing the same identities modulo the maximal ideal then gives (U1) and (U2).

Lattices at p and the tree. By (ii), the sibling of X along a line $T \subset G_X$ has lattice $I_X + \tilde{T}$, which contains I_X with index p . To see where repeated siblings can lead, we use the classical graph of all lattices at p . Only divisibility by p matters, so we replace a lattice $I \subset K$ by $\Lambda := I \otimes \mathbb{Z}_p \subset K \otimes \mathbb{Q}_p$; equivalently, two lattices are identified when each has index prime to p in their sum. The *Bruhat–Tits tree* [Se] has one vertex for each such Λ up to multiplication by $\mathbb{Q}_p^\times$, and an edge between two vertices whose representatives contain one another with index p . The neighbors of Λ correspond to the lattices $\Lambda \subsetneq \Lambda' \subsetneq p^{-1}\Lambda$, that is, to the $p+1$ lines of $p^{-1}\Lambda/\Lambda \cong \mathbb{F}_p^2$. So every vertex has $p+1$ neighbors, and the graph has no cycles. A member X sits at the vertex $\Lambda_X := I_X \otimes \mathbb{Z}_p$, and every sibling step is an edge.

The tree has a height function. Put $\mathfrak{o} := O_K \otimes \mathbb{Z}_p$ and $\mathfrak{o}_j := \mathbb{Z}_p + p^j\mathfrak{o}$ for $j \geq 0$. Classically, the subrings of $K \otimes \mathbb{Q}_p$ that are lattices are exactly the $\mathfrak{o}_j$, and every lattice is $x \cdot \text{End } \Lambda$ for an invertible $x \in K \otimes \mathbb{Q}_p$, where $\text{End } \Lambda := \{x : x\Lambda \subseteq \Lambda\}$ is its multiplier ring.

For the first statement, write $\mathfrak{o} = \mathbb{Z}_p + \mathbb{Z}_p\vartheta$. A subring R that is a lattice consists of elements integral over $\mathbb{Z}_p$, so $R \subseteq \mathfrak{o}$; the b with $a + b\vartheta \in R$ form an ideal $p^j\mathbb{Z}_p$, and $R = \mathbb{Z}_p + p^j\mathbb{Z}_p\vartheta = \mathfrak{o}_j$. For the second, a lattice Λ contains an invertible element $x \notin p\Lambda$. If p is inert, $K \otimes \mathbb{Q}_p$ is a field and any $x \in \Lambda \setminus p\Lambda$ will do. If p splits, this holds because the non-invertible elements of $\mathfrak{o} \otimes \mathbb{Q}_p \cong \mathbb{Q}_p^2$ form two lines, which cannot cover Λ . Then $x^{-1}\Lambda = \mathbb{Z}_p + \mathbb{Z}_p\theta$ with $\theta \notin \mathbb{Q}_p$. Let $a\theta^2 + b\theta + c = 0$ with coprime $a, b, c \in \mathbb{Z}_p$, $a \neq 0$. If a is a unit, $x^{-1}\Lambda = \mathbb{Z}_p[\theta]$ is a ring. If $p \mid a$ and c is a unit, then $(x\theta)^{-1}\Lambda = \mathbb{Z}_p[\theta^{-1}]$ is a ring. If p divides a and c , then b is a unit, and replacing θ by $\theta + 1$, whose equation has constant term $a - b + c$, leads to the previous case. So Λ is a multiple of a ring, which is its own multiplier ring.

Hence each vertex has a *level* j with $\text{End } \Lambda = \mathfrak{o}_j$. The *crater* $\mathcal{C}$ is the set of vertices of level 0, the lattices with the largest multiplier ring. If p splits, then $\mathfrak{o} \cong \mathbb{Z}_p \times \mathbb{Z}_p$ and $\mathcal{C}$ is an infinite path, the vertices $(p^a, p^b)\mathfrak{o}$; if p is inert, $\mathcal{C}$ is a single vertex. The level is the distance to the crater.

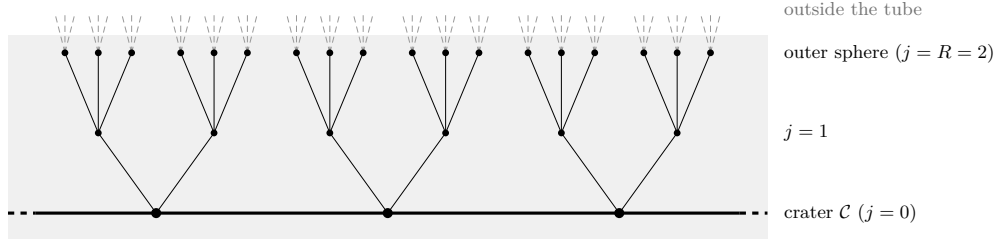


FIGURE 2. The tube for a split prime, drawn with $p = 3$ and $R = 2$ for legibility. Every vertex has $p + 1$ neighbors. A crater vertex has two neighbors along the crater and $p - 1$ outward ones; a vertex at distance 1 has one inward and p outward neighbors, all of them siblings; on the outer sphere only the inward neighbor carries a sibling, and the dashed edges lead out of the tube, where no member lies.

Consider a neighbor $\Lambda + \mathbb{Z}_p p^{-1}y$ of $\Lambda = \mathfrak{o}_j$, with $y \in \mathfrak{o}_j \setminus p\mathfrak{o}_j$. When y is a unit of $\mathfrak{o}_j$, this neighbor is $yp^{-1}\mathfrak{o}_{j+1}$, one step farther out. For $j \geq 1$ the non-units form a single line, which leads to $\mathfrak{o}_{j-1}$. So off the crater a vertex has one neighbor nearer to $\mathcal{C}$ and p farther. A crater vertex has two crater neighbors and $p - 1$ outward ones if p splits, and $p + 1$ outward ones if p is inert. This is the local form of the “volcano” picture of isogeny graphs [Koh, FM], from which we take the name crater. Since $p \nmid f$, the lattice of γ_0 , which is that of γ , lies on the crater.

For a member X let $j(X)$ be the distance of Λ_X from $\mathcal{C}$, let P_X be the p -primary part of G_X , and put $X[p] := G_X[p]$. For a line T of $X[p]$, X_T denotes the sibling of X along T .

Lemma 3.3 (Tube lemma). *Let X be a member.*

- (a) $0 \leq j(X) \leq R$, and $P_X \cong \mathbb{Z}/p^{R+j} \times \mathbb{Z}/p^{R-j}$ with $j = j(X)$.
- (b) If $j(X) < R$, then $X[p] = p^{-1}\Lambda_X/\Lambda_X \cong \mathbb{F}_p^2$: the $p + 1$ lines of $X[p]$ are the $p + 1$ neighbors of Λ_X , and X has a sibling at each. Call a line inward if its sibling is not farther from $\mathcal{C}$, and outward otherwise; write $\text{Out}(X)$ for the set of outward lines and $\nu(X)$ for the number of inward ones. Off the crater there is exactly one inward line D_X , and there are p outward ones. On a split crater the two inward lines are the crater lines $T_{\mathfrak{p}} = \mathfrak{p}_{\mathcal{O}}^{-1}I_X/I_X$ and $T_{\bar{\mathfrak{p}}}$, along which the crater runs; on an inert crater there is none. So $|\text{Out}(X)| = p, p - 1$ or $p + 1$.
- (c) If $j(X) = R$, that is, if X lies on the outer sphere, then P_X is cyclic of order p^{2R} , so $X[p]$ is a single line D_X , and it is inward.
- (d) For an outward line T , the image of $X[p]$ in G_{X_T} is the inward line D_{X_T} .

Proof. All members share ε_0 , the root greater than 1 of $x^2 - (\text{tr } \gamma_0)x + 1$, and ε_0 maps each member’s lattice onto itself. So the question is which vertices ε_0 stabilizes. Write $w := \varepsilon_0^{-1} - 1 = p^R u$ with $u \in \mathfrak{o} \setminus p\mathfrak{o}$.

We first show that $\mathbb{Z}_p[\varepsilon_0] = \mathfrak{o}_R$. Since ε_0 has norm 1, $(1 + p^R u)(1 + p^R \bar{u}) = 1$, so $u + \bar{u} = -p^R u \bar{u} \equiv 0 \pmod{p}$. Moreover u is a unit of $\mathfrak{o}$: if p is inert, because $\mathfrak{o}$ is local with maximal ideal $p\mathfrak{o}$ and $u \notin p\mathfrak{o}$; if p splits, because conjugation swaps the two coordinates of $\mathfrak{o} \cong \mathbb{Z}_p \times \mathbb{Z}_p$ and $u + \bar{u} \equiv 0$ makes both coordinates of u nonzero modulo p . If u were congruent to some $a \in \mathbb{Z}_p$ modulo $p\mathfrak{o}$, then so would be $\bar{u}$, and $2a \equiv 0$. As p is odd, this would give $u \in p\mathfrak{o}$, which is excluded. The same holds for u^{-1} . So $1, u$ is a basis of $\mathfrak{o}$, and $\mathbb{Z}_p[\varepsilon_0] = \mathbb{Z}_p + \mathbb{Z}_p p^R u = \mathfrak{o}_R$. Hence

$$\varepsilon_0 \Lambda = \Lambda \iff \varepsilon_0 \in \text{End } \Lambda = \mathfrak{o}_j \iff \mathfrak{o}_R \subseteq \mathfrak{o}_j \iff j \leq R.$$

Every member therefore lies in the tube $\{j \leq R\}$ around the crater. The labels of X are $w^{-1}I_X/I_X$, whose p -part is $w^{-1}\Lambda_X/\Lambda_X \cong \mathfrak{o}_j/w\mathfrak{o}_j$. This group has order $|\mathrm{Nm} w|_p^{-1} = p^{2R}$, as u is a unit, and exponent p^{R+j} , as $u^{-1} \notin \mathbb{Z}_p + p\mathfrak{o}$. This proves (a).

For $j < R$ we have $p^{R-1}u \in \mathfrak{o}_j$, so $p^{-1}\Lambda \subseteq w^{-1}\Lambda$ and all of $p^{-1}\Lambda/\Lambda$ consists of labels. This proves (b). At $j = R$ the group is cyclic and its single line is fixed by ε_0 . So its sibling is ε_0 -stable, hence not outside the tube, and it is the inward neighbor. This proves (c). For (d), if T is outward, the sibling lattice $\Lambda' := \Lambda + \tilde{T}$ lies in $p^{-1}\Lambda$, and the image of $X[p]$ is $p^{-1}\Lambda/\Lambda'$. Adding it to Λ' gives $p^{-1}\Lambda$, the vertex of X , which is one step nearer to the crater than Λ' . $\square$

Norm one is essential here: in $\mathbb{Q}(\sqrt{5})$, the element $12 + 121\varphi = 1 + 11(1 + 11\varphi)$ is $\equiv 1 \pmod{11}$ but generates only $\mathfrak{o}_2$.

As an example, let $\gamma = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, so that $K = \mathbb{Q}(\sqrt{5})$, $\tau^* = \varphi = (1 + \sqrt{5})/2$, $\varepsilon = \varphi^2$, $I_\gamma = O_K$, $N_\gamma = 1$ and $f = 1$. At $p = 11 = (3 + 2\varphi)(5 - 2\varphi)$ one has $k_0 = 5$, $\gamma_0 = \begin{pmatrix} 89 & 55 \\ 55 & 34 \end{pmatrix}$, $\varepsilon_0 = \varphi^{10} = 55\varphi + 34$ and $\varepsilon_0 - 1 = 11(5\varphi + 3)$ with $5\varphi + 3$ a unit, so $R = 1$. The labels of γ_0 are $\frac{1}{11}O_K/O_K \cong \mathbb{F}_{11}^2$, on the crater. Its 12 lines give 2 siblings along the crater and 10 on the outer sphere, for instance $\begin{pmatrix} 89 & 605 \\ 5 & 34 \end{pmatrix}$ with lattice $\mathbb{Z}\varphi + \frac{1}{11}\mathbb{Z}$. The labels of that sibling form a cyclic group of order 121, generated by $(5\varphi - 8)/121$, and its single line $\{\frac{a}{11}\varphi\}$ leads back to the crater vertex of γ_0 .

Two code bounds from the Fourier identity. Identity (vi) says that the normalized Fourier transform $\mathcal{F}$ on G_X reproduces E_X up to the root of unity $\lambda q(y)$ and the reflection $y \mapsto -y$; applying $\mathcal{F}$ once more, with $\mathcal{F}^2 f(y) = f(-y)$ and q even, gives $\mathcal{F}(qE_X) = \lambda^{-1}E_X$. Because $\mathcal{F}$ divides by $|G_X|^{1/2}$, these identities impose divisibility on weighted sums of the values. Write $G_X = P_X \oplus Q_X$ with Q_X of order prime to p . The pairing and q split along this decomposition. Indeed, for $x \in P_X$ and $y \in Q_X$, $b(x, y)$ has order dividing $|P_X|$ and $|Q_X|$, so $b(x, y) = 1$. The identity $q(x + y) = q(x)q(y)b(x, y)$, which follows from [RW, (3)], then gives $q(x + y) = q(x)q(y)$. As b is nondegenerate on G_X [RW, §4.3], so is its restriction to P_X . Let $\mathcal{F}_P$ and $\mathcal{F}_Q$ be the transforms on P_X and Q_X , normalized by $|P_X|^{-1/2}$ and $|Q_X|^{-1/2}$, let q_P and q_Q be the restrictions of q , and put, with $P := P_X$,

$$\mathfrak{L}(P) := \{f : P \rightarrow \mathcal{O}_v : \mathcal{F}_P f, \mathcal{F}_P(q_P f) \text{ integral}\},$$

with integrality in $\mathcal{O}_v$. We call such integral lattices of functions *codes*, and bounds on the reductions of their elements *code bounds*.

Lemma 3.4 (Column lemma). *Let X be a member and $\beta \in K_{\mathrm{fam}}^\times$. If βE_X is integral, then for every $c \in Q_X$ the column $f_c(x) := \beta E_X(x + c)$, $x \in P_X$, lies in $\mathfrak{L}(P_X)$.*

Proof. By the two identities above, βE_X , $\mathcal{F}(\beta E_X)$ and $\mathcal{F}(q\beta E_X)$ are all integral. Since the pairing and q split, $\mathcal{F} = \mathcal{F}_P \otimes \mathcal{F}_Q$, and for a function F on G_X we have $(\mathcal{F}_P \otimes 1)F = (1 \otimes \mathcal{F}_Q^{-1})\mathcal{F}F$. The kernel of $\mathcal{F}_Q^{-1}$ is $|Q_X|^{-1/2}$ times roots of unity, a v -unit as $p \nmid |Q_X|$. We apply this to $F = \beta E_X$ and $F = q\beta E_X$, whose transforms are integral. Since $q\beta E_X(x + c) = q_Q(c)q_P(x)\beta E_X(x + c)$ with $q_Q(c)$ a root of unity, every column f_c lies in $\mathfrak{L}(P_X)$. $\square$

Lemma 3.4 is the only use of [RW, (6)] in Sections 3.2, 3.3 and 3.4. On the outer sphere P_X is cyclic of order p^{2R} . For a generator g put $\zeta := q(g)$. Then $q(xg) = q(g)^{x^2} = \zeta^{x^2}$ and $b(xg, yg) = q((x + y)g)/(q(xg)q(yg)) = \zeta^{2xy}$. As b is nondegenerate on P_X and p is odd, ζ^2 has order p^{2R} ; and $\zeta^{4R} = q(p^{2R}g) = q(0) = 1$, so ζ has odd order and is a primitive p^{2R} -th root of unity.

Proposition 3.5 (Cyclic code theorem). *Let $P = \mathbb{Z}/p^n$ with $n = 2R$, let ζ be a primitive p^n -th root of unity, $q(x) = \zeta^{x^2}$, $(\mathcal{F}f)(y) = p^{-n/2} \sum_x \zeta^{2xy} f(x)$ and $m := (p^n + 2)/3$. For every*

valuation ring $\mathcal{O}$ of a field of characteristic 0 with $\zeta \in \mathcal{O}$ and residue characteristic $p \geq 5$, the reduction of $\mathfrak{L}_{\mathcal{O}}(P)$ lies in the span of the binomial functions $\binom{x}{i}$ with $i \leq m$.

We use only the consequence for the top base- p digit of i , stated after the proof.

Proof. Write $\mathcal{R}f(x) := f(-x)$, $\mathcal{T}f(x) := f(x-1)$, $\Delta := \mathcal{T} - 1$ and $\mathcal{U} := q^{-1}\mathcal{F}^{-1}$, where q also denotes multiplication by q . For a field k of characteristic p and $j \leq p^n$ let $\mathcal{D}_j \subseteq k^P$ be the span of the functions $x \mapsto \binom{x}{i}$, $0 \leq i < j$, read on the representatives $0 \leq x < p^n$. Steps 1–3 take place over a field of characteristic 0, Steps 4–6 over one discrete valuation ring, and Step 7 passes to an arbitrary valuation ring.

Step 1: Gauss sums and $\mathcal{U}^3 = I$. For $m \geq 0$ and $p \nmid a$ put $G_m(a) := \sum_{x \bmod p^m} \zeta_m^{ax^2}$, with $\zeta_m := \zeta^{p^{n-m}}$ a primitive p^m -th root of unity. For $m \geq 2$ write $x = x_0 + p^{m-1}x_1$ with $0 \leq x_0 < p^{m-1}$ and $x_1 \bmod p$. As $2m-2 \geq m$, we have $ax^2 \equiv ax_0^2 + 2ax_0x_1p^{m-1} \pmod{p^m}$, and the sum over x_1 is p if $p \mid x_0$ and 0 otherwise. Writing $x_0 = py$ gives $G_m(a) = pG_{m-2}(a)$, and $G_0(a) = 1$. As n is even,

$$G(a) := \sum_{x \in P} \zeta^{ax^2} = p^R \quad (p \nmid a).$$

The kernel of $\mathcal{F}$ is $p^{-R}\zeta^{2xy}$. Since $\sum_y \zeta^{2y(x+z)} = p^n$ if $x = -z$ and 0 otherwise, $\mathcal{F}^2 = \mathcal{R}$, so $\mathcal{F}^{-1} = \mathcal{R}\mathcal{F}$ has kernel $p^{-R}\zeta^{-2xy}$. The kernel of $\mathcal{F}^{-1}q^{-1}\mathcal{F}^{-1}$ is therefore

$$p^{-2R} \sum_z \zeta^{-2yz - z^2 - 2zx} = p^{-2R} \zeta^{(x+y)^2} G(-1) = p^{-R} \zeta^{(x+y)^2},$$

which is the kernel of $q\mathcal{F}q$. Hence $\mathcal{F}^{-1}q^{-1}\mathcal{F}^{-1} = q\mathcal{F}q$, and $\mathcal{U}^3 = q^{-1}(\mathcal{F}^{-1}q^{-1}\mathcal{F}^{-1})q^{-1}\mathcal{F}^{-1} = I$. The reflection $\mathcal{R}$ commutes with q , which is even, and with $\mathcal{F}$, whose kernel is invariant under $(x, y) \mapsto (-x, -y)$. Hence it commutes with $\mathcal{U}$.

Step 2: The lattice. For a valuation ring $\mathcal{O}$ as in the statement write $\mathfrak{L}_{\mathcal{O}} := \mathfrak{L}_{\mathcal{O}}(P)$. As $\mathcal{U}^{-1} = \mathcal{U}^2 = \mathcal{F}q$ and $\mathcal{U} = q^{-1}\mathcal{R}\mathcal{F}$, and as $q^{\pm 1}$ and $\mathcal{R}$ preserve $\mathcal{O}^P$, a function f lies in $\mathfrak{L}_{\mathcal{O}}$ if and only if f , $\mathcal{U}f$ and $\mathcal{U}^2f$ lie in $\mathcal{O}^P$. Since $\mathcal{U}^3 = I$, this condition holds for $\mathcal{U}f$ whenever it holds for f . So $\mathfrak{L}_{\mathcal{O}}$ is stable under $\mathcal{U}$, and also under $\mathcal{R}$, which commutes with $\mathcal{U}$ and preserves $\mathcal{O}^P$.

Step 3: Multiplicities. Over a field of characteristic 0 containing ζ and a primitive cube root of unity, the kernels of Step 1 give the kernels

$$\begin{aligned} \mathcal{U} : p^{-R}\zeta^{-y^2-2xy}, & \quad \mathcal{U}^2 = \mathcal{F}q : p^{-R}\zeta^{2xy+x^2}, \\ \mathcal{U}\mathcal{R} = q^{-1}\mathcal{F} : p^{-R}\zeta^{-y^2+2xy}, & \quad \mathcal{U}^2\mathcal{R} = \mathcal{F}^{-1}q : p^{-R}\zeta^{x^2-2xy}. \end{aligned}$$

Their traces are $p^{-R}G(c)$ with $c = -3, 3, 1, -1$, all equal to 1 by Step 1, since $p \neq 3$. The trace of $\mathcal{R}$ is the number of x with $x = -x$, which is 1. For a character $\vartheta = (\xi, \sigma)$ of $\mathbb{Z}/3 \times \mathbb{Z}/2$, with $\xi^3 = 1$ and $\sigma = \pm 1$, let V_{ϑ} be the space of f with $\mathcal{U}f = \xi f$ and $\mathcal{R}f = \sigma f$. The representation $(i, j) \mapsto \mathcal{U}^i \mathcal{R}^j$ of $\mathbb{Z}/3 \times \mathbb{Z}/2$ shows that

$$n_{\vartheta} := \dim V_{\vartheta} = \frac{1}{6} \sum_{i,j} \overline{\vartheta(i,j)} \operatorname{tr}(\mathcal{U}^i \mathcal{R}^j) = \frac{p^n - 1}{6} + \delta_{\vartheta,1},$$

because the five nonidentity traces equal 1, the identity has trace p^n , and $p^n \equiv 1 \pmod{6}$.

Step 4: Decomposition over a discrete valuation ring. Let A_1 be the localization of $\mathbb{Z}[\zeta, \zeta_3]$ at a maximal ideal containing p , with fraction field K_1 and residue field k_1 . It is a discrete valuation ring in which 6 is a unit, as $p \geq 5$. The idempotents

$$e_{\vartheta} := \frac{1}{6} \sum_{i,j} \overline{\vartheta(i,j)} \mathcal{U}^i \mathcal{R}^j$$

have coefficients in A_1 , so they preserve $\mathfrak{L}_{A_1}$ by Step 2. They sum to I , and e_ϑ projects K_1^P onto V_ϑ . Put $\mathfrak{L}_\vartheta := V_\vartheta \cap A_1^P$. If $f \in \mathfrak{L}_\vartheta$, then $\mathcal{U}^{\pm 1}f = \zeta^{\pm 1}f$ is integral, so $\mathfrak{L}_\vartheta \subseteq \mathfrak{L}_{A_1}$. Hence $\mathfrak{L}_{A_1} = \bigoplus_\vartheta \mathfrak{L}_\vartheta$. Each $\mathfrak{L}_\vartheta$ is a free A_1 -module of rank at most n_ϑ , so its reduction $U_\vartheta \subseteq k_1^P$ has dimension at most n_ϑ , and the reduction of $\mathfrak{L}_{A_1}$ is $\sum_\vartheta U_\vartheta$.

Step 5: A reduced Laplacian. Let $\mathcal{M}_b f(x) := \zeta^{2bx} f(x)$. From the kernel of $\mathcal{F}$ one reads off $\mathcal{F}\mathcal{T} = \mathcal{M}_1\mathcal{F}$ and $\mathcal{F}\mathcal{M}_b = \mathcal{T}^{-b}\mathcal{F}$; hence $\mathcal{F}^{-1}\mathcal{T} = \mathcal{M}_{-1}\mathcal{F}^{-1}$ and $\mathcal{T}\mathcal{F}^{-1} = \mathcal{F}^{-1}\mathcal{M}_1$. Moreover $q(x)/q(x-1) = \zeta^{2x-1}$ gives $q\mathcal{T} = \zeta^{-1}\mathcal{M}_1\mathcal{T}q$. Therefore

$$\mathcal{U}\mathcal{T}\mathcal{U}^{-1} = q^{-1}\mathcal{F}^{-1}\mathcal{T}\mathcal{F}q = \mathcal{M}_{-1},$$

$$\mathcal{U}^{-1}\mathcal{T}\mathcal{U} = \mathcal{F}q\mathcal{T}q^{-1}\mathcal{F}^{-1} = \zeta^{-1}\mathcal{F}\mathcal{M}_1\mathcal{F}^{-1}\mathcal{M}_1 = \zeta^{-1}\mathcal{T}^{-1}\mathcal{M}_1,$$

and, taking inverses, $\mathcal{U}\mathcal{T}^{-1}\mathcal{U}^{-1} = \mathcal{M}_1$ and $\mathcal{U}^{-1}\mathcal{T}^{-1}\mathcal{U} = \zeta\mathcal{M}_{-1}\mathcal{T}$. So

$$Z := \sum_{i=0}^2 \mathcal{U}^i(\mathcal{T} + \mathcal{T}^{-1})\mathcal{U}^{-i} = \mathcal{T} + \mathcal{T}^{-1} + \mathcal{M}_{-1} + \mathcal{M}_1 + \zeta^{-1}\mathcal{T}^{-1}\mathcal{M}_1 + \zeta\mathcal{M}_{-1}\mathcal{T}.$$

Conjugation by $\mathcal{U}$ permutes the three summands in the definition, as $\mathcal{U}^3 = I$, so Z commutes with $\mathcal{U}$. It commutes with $\mathcal{R}$, as $\mathcal{R}\mathcal{T}\mathcal{R} = \mathcal{T}^{-1}$. By the second expression Z has entries in $\mathbb{Z}[\zeta]$. Hence Z preserves A_1^P and every V_ϑ , hence every $\mathfrak{L}_\vartheta$, and its reduction $\bar{Z}$ preserves every U_ϑ . As $\zeta \equiv 1$ modulo the maximal ideal, $\mathcal{M}_b$ reduces to the identity and

$$\bar{Z} = 2(\mathcal{T} + 1 + \mathcal{T}^{-1}).$$

Since 2 is invertible in k_1 , the operator $Y := \frac{1}{2}\bar{Z} - 3 = \mathcal{T} - 2 + \mathcal{T}^{-1} = \mathcal{T}^{-1}\Delta^2$ preserves every U_ϑ .

Step 6: Degrees. On k_1^P we have $\Delta^{p^n} = \mathcal{T}^{p^n} - 1 = 0$, so Y is nilpotent, and $Y^{n_\vartheta} = 0$ on U_ϑ , whose dimension is at most n_ϑ . As $\mathcal{T}$ is invertible and commutes with Δ , $\Delta^{2n_\vartheta} = \mathcal{T}^{n_\vartheta}Y^{n_\vartheta}$ kills U_ϑ .

To turn this into a degree bound, we claim that $\ker \Delta^j = \mathcal{D}_j$ for $j \leq p^n$. By Lucas's theorem, for $i < p^n$ the residue of $\binom{x}{i}$ depends only on $x \bmod p^n$, so the identity $\binom{x+1}{i} - \binom{x}{i} = \binom{x}{i-1}$ holds on P . The functions $\binom{x}{i}$, $0 \leq i < p^n$, form a basis of k^P , since the matrix $(\binom{x}{i})_{0 \leq x, i < p^n}$ is unitriangular. So the forward difference $\delta := \mathcal{T}^{-1} - 1$ sends $\binom{x}{i}$ to $\binom{x}{i-1}$ for $i \geq 1$ and $\binom{x}{0}$ to 0, whence $\ker \delta^j = \mathcal{D}_j$. Finally $\Delta = -\mathcal{T}\delta$ gives $\ker \Delta^j = \ker \delta^j$. Therefore $U_\vartheta \subseteq \mathcal{D}_{2n_\vartheta}$. The largest value of $2n_\vartheta$ is $(p^n + 5)/3 = m + 1$, so the reduction of $\mathfrak{L}_{A_1}$ lies in $\mathcal{D}_{m+1}$, the span of the $\binom{x}{i}$ with $i \leq m$.

Step 7: Arbitrary valuation rings. It remains to pass from A_1 to the valuation ring $\mathcal{O}$ of the statement. First, $A_0 := \mathcal{O} \cap \mathbb{Q}(\zeta)$ is a valuation ring of $\mathbb{Q}(\zeta)$ whose maximal ideal contains p . It contains $\mathbb{Z}[\zeta]$, the ring of integers, so it is the localization of $\mathbb{Z}[\zeta]$ at the prime $(1 - \zeta)$, the only prime above p . In particular A_0 is a discrete valuation ring with fraction field $K_0 = A_0[1/p] = \mathbb{Q}(\zeta)$, and the maximal ideal of $\mathcal{O}$ meets A_0 in that of A_0 . Choose A_1 in Step 4 above A_0 , that is, localize at a maximal ideal containing $1 - \zeta$. The residue fields then satisfy $k_0 \subseteq k_1$ and $k_0 \subseteq k_{\mathcal{O}}$, compatibly with reduction.

We next bound the reduction of $\mathfrak{L}_{A_0}$. Since $\mathfrak{L}_{A_0} \subseteq \mathfrak{L}_{A_1}$, Step 6 shows that this reduction lies in $k_0^P \cap (k_1 \otimes \mathcal{D}_{m+1}) = k_0 \otimes \mathcal{D}_{m+1}$. The equality holds because $\mathcal{D}_{m+1} = \ker \Delta^{m+1}$ is cut out by equations with coefficients in $\mathbb{F}_p$.

Finally we extend scalars from A_0 to $\mathcal{O}$. The module $\mathfrak{L}_{A_0}$ is the kernel of

$$A_0^P \rightarrow (K_0/A_0)^{2P}, \quad f \mapsto (\mathcal{F}f, \mathcal{F}(qf)) \bmod A_0.$$

As $\mathcal{O}$ is torsion-free over the discrete valuation ring A_0 , it is flat. Tensoring with $\mathcal{O}$ therefore gives the exact sequence

$$0 \rightarrow \mathcal{O} \otimes \mathfrak{L}_{A_0} \rightarrow \mathcal{O}^P \rightarrow (\mathcal{O}[1/p]/\mathcal{O})^{2P},$$

whose second map is again $f \mapsto (\mathcal{F}f, \mathcal{F}(qf)) \bmod \mathcal{O}$. Hence $\mathfrak{L}_{\mathcal{O}}$ is the $\mathcal{O}$ -span of an A_0 -basis (b_i) of the free module $\mathfrak{L}_{A_0}$, and the reduction of $\sum c_i b_i$ is $\sum \bar{c}_i \bar{b}_i \in k_{\mathcal{O}} \otimes \mathcal{D}_{m+1}$. This argument uses neither the rank of the valuation nor completeness, and the residue field of $\mathcal{O}$ may be transcendental over $\mathbb{F}_p$. $\square$

We use Proposition 3.5 in the following form. Along a coset of the subgroup of order p , the position moves only in its top base- p digit, and every $i \leq m$ has top base- p digit at most $\lfloor m/p^{n-1} \rfloor$, which equals $\lfloor p/3 \rfloor$ (write $p = 3J + r$ with $r \in \{1, 2\}$). So by Lucas's theorem the residue is a polynomial in the coordinate $u \in \mathbb{F}_p$ of the coset of degree at most

$$J := \lfloor p/3 \rfloor; \quad \text{note that } p - 2J \geq 3 \text{ for } p \geq 5.$$

Lemma 3.6 (Envelope). *Let p be any prime, 2 and 3 included, let $P = (\mathbb{Z}/p^r)^2$, and let $\mathcal{O}$ be a valuation ring containing a primitive p^r -th root of unity (so of characteristic 0), with residue characteristic p . If $f : P \rightarrow \mathcal{O}$ has all character sums $\sum_x \chi(x)f(x)$ in $p^r\mathcal{O}$, then the residue of f on each coset of $P[p]$ is a polynomial of total degree $\leq p - 1$ in any affine coordinates on the coset.*

The proof of Theorem D uses Lemma 3.6 at every prime, 2 and 3 included. For a nondegenerate pairing the hypothesis says that f and $\mathcal{F}f$ are integral; on the crater, where $P_X \cong (\mathbb{Z}/p^R)^2$, the lemma applies to every column.

Proof. Let $\zeta \in \mathcal{O}$ be a primitive p^r -th root of unity and $\varpi := \zeta - 1$. In $\mathbb{Z}[\zeta]$, for $1 \leq j < p^r$ the element $\zeta^j - 1$ is a unit times $\varpi^{v_p(j)}$, and p is a unit times $\varpi^{\varphi(p^r)}$. For an element c of $\mathbb{Z}[\zeta]$ of the form unit $\cdot \varpi^e$ we write $\text{ord}(c) := e$.

The characters of P are $x \mapsto \zeta^{-(sx_1 + tx_2)}$, $(s, t) \in P$, so the hypothesis says that $\mathcal{F}f(s, t) := p^{-r} \sum_x \zeta^{-(sx_1 + tx_2)} f(x)$ is integral for all (s, t) . Translating f multiplies every character sum by a root of unity, so it suffices to treat the coset $P[p]$.

For $0 \leq a < p^r$ put

$$D_a := \prod_{i < a} (\zeta^a - \zeta^i), \quad g_a(x) := \frac{1}{D_a} \prod_{i < a} (\zeta^x - \zeta^i) \quad (x \in \mathbb{Z}/p^r).$$

On the representatives $0 \leq x < p^r$, $g_a(x) = 0$ for $x < a$ and $g_a(a) = 1$, and for $x \geq a$

$$g_a(x) = \prod_{j=1}^a \frac{\zeta^{x-a+j} - 1}{\zeta^j - 1} = \left[\begin{matrix} x \\ a \end{matrix} \right]_{\zeta} \in \mathbb{Z}[\zeta],$$

the Gaussian binomial coefficient at ζ . So the matrix $(g_a(x))_{x,a}$ is unitriangular with entries in $\mathbb{Z}[\zeta]$, and the functions $g_a \otimes g_b : (x_1, x_2) \mapsto g_a(x_1)g_b(x_2)$ form a basis of $\mathcal{O}^P$. Thus $f = \sum c_{ab} g_a \otimes g_b$ is integral if and only if every c_{ab} is integral.

To express the integrality of $\mathcal{F}f$ in the same coordinates, write $\prod_{i < a} (Z - \zeta^i) = \sum_{k=0}^a T_{ka} Z^k$, so that $T_{ka} \in \mathbb{Z}[\zeta]$, $T_{aa} = 1$ and $T_{ka} = 0$ for $k > a$. Then $g_a(x) = D_a^{-1} \sum_k T_{ka} \zeta^{kx}$, and for $0 \leq s < p^r$ orthogonality gives $\sum_x \zeta^{-sx} g_a(x) = p^r T_{sa} / D_a$. Hence

$$\mathcal{F}f(s, t) = \sum_{a,b} T_{sa} T_{tb} \frac{p^r c_{ab}}{D_a D_b}.$$

The matrix (T_{sa}) is unitriangular over $\mathbb{Z}[\zeta]$, so $\mathcal{F}f$ is integral if and only if every $p^r c_{ab}/(D_a D_b)$ is integral. Now $\zeta^a - \zeta^i = \zeta^i(\zeta^{a-i} - 1)$, so $\text{ord}(D_a) = \text{wt}(a) := \sum_{j=1}^a p^{v_p(j)}$, while $\text{ord}(p^r) = r\varphi(p^r)$. Therefore f and $\mathcal{F}f$ are integral if and only if

$$c_{ab} \in \varpi^{e_{ab}} \mathcal{O}, \quad e_{ab} := \max(0, \text{wt}(a) + \text{wt}(b) - r\varphi(p^r)).$$

We now restrict to $P[p] = \{(p^{r-1}s, p^{r-1}t) : 0 \leq s, t < p\}$. Modulo ϖ we have $\zeta \equiv 1$, so $g_a(x) \equiv \binom{x}{a}$; for $x < a$ both sides vanish. By Lucas's theorem $\binom{p^{r-1}s}{a} \equiv 0 \pmod{p}$ unless $a = p^{r-1}A$ with $0 \leq A < p$, in which case it is $\equiv \binom{s}{A}$. For such a , counting the $j \leq p^{r-1}A$ divisible by p^k for $1 \leq k \leq r-1$ gives

$$\text{wt}(p^{r-1}A) = Ap^{r-1} + \sum_{k=1}^{r-1} (p^k - p^{k-1}) Ap^{r-1-k} = Ap^{r-2}(r(p-1) + 1).$$

If $A + B \geq p$, then

$$\text{wt}(p^{r-1}A) + \text{wt}(p^{r-1}B) - r\varphi(p^r) \geq p^{r-1}(r(p-1) + 1) - rp^{r-1}(p-1) = p^{r-1} > 0,$$

so the corresponding coefficient has residue 0. Hence the residue of f on $P[p]$ is

$$\bar{f}(p^{r-1}s, p^{r-1}t) = \sum_{A+B \leq p-1} \bar{c}_{p^{r-1}A, p^{r-1}B} \binom{s}{A} \binom{t}{B}.$$

Here $\binom{s}{A}$ is a polynomial of degree A in s over $\mathbb{F}_p$, as $A < p$. So the residue is a polynomial of total degree at most $p-1$. Such a polynomial has partial degrees less than p , so it is the unique reduced representative of its function on $\mathbb{F}_p^2$, and an $\mathbb{F}_p$ -linear change of coordinates does not raise its total degree. No step divides by 2 or 3. $\square$

The defect identity. A pole cannot be followed directly from a member to an outward sibling. By (ii) the sibling's value has the sum of the valuations along a line, and values of positive valuation on that line can cancel the pole. The proof therefore follows a different quantity. For a line T put $L_T(y) := \prod_{h \in T} E_X(y+h)$, put $\Pi_X(y) := \prod_{h \in X[p]} E_X(y+h)$, and recall that $\nu(X)$ is the number of inward lines. The *defect* is

$$\Theta_X(y) := E_X(y)^{-p^2} \Pi_X(y)^{1-\nu(X)} \prod_{T \text{ inward}} L_T(y)^p,$$

that is, $(L_{D_X}(y)/E_X(y)^p)^p$ off the crater, $(L_{T_p} L_{T_{\bar{p}}})^p / (E_X^{p^2} \Pi_X)$ on a split crater, and $\Pi_X / E_X^{p^2}$ on an inert crater. Numerator and denominator contain the same number of values, so $\Theta_X(y) = 1$ when E_X is constant on $y + X[p]$. Off the crater, $\Theta_X(y)$ measures how far the values along the inward line through y are from being equal.

Lemma 3.7 (Defect identity). *If $j(X) < R$, then*

$$\Theta_X(y)^p = \prod_{T \text{ outward}} \Theta_{X_T}(\pi_T y),$$

an identity in $K_{\text{fam}}^\times$.

Proof. We use two exact identities. The first is the *vertex identity*: the $p+1$ lines through y cover y exactly $p+1$ times and every other point of $y+X[p]$ once, so $\prod_T L_T(y) = E_X(y)^p \Pi_X(y)$. The second is the *edge identity*. For an outward line T , the p translates of T inside $y+X[p]$ map onto the inward line of X_T through $\pi_T y$ (Lemma 3.3(d)), and (ii), with the same κ_T on every translate, gives

$$\Pi_X(y) = \kappa_T^p L_{D_{X_T}}(\pi_T y), \quad L_T(y) = \kappa_T E_{X_T}(\pi_T y).$$

By the vertex identity and the count of outward lines, $\Theta_X(y) = \prod_{T \text{ outward}} \Pi_X(y)/L_T(y)^p$. By the edge identity the p -th power of each factor is $\Theta_{X_T}(\pi_T y)$, the roots of unity cancelling. $\square$

The exponent $1 - \nu(X)$ is what makes Π_X appear once for every outward line. Nothing is reduced and nothing is assumed about the values; the identity needs only that κ_T does not depend on the translate, which is part of (ii).

The defect is needed because the sharp bound $J = \lfloor p/3 \rfloor$ requires P_X to be cyclic, which happens only on the outer sphere. Elsewhere only Lemma 3.6 is available, and its bound $p - 1$ along a line gives $p - 2(p - 1) < 0$, which yields nothing. This is why a pole must first be turned into a defect and carried outward.

Proof of the local theorem. Put $\theta_X(y) := v(\Theta_X(y))$, a real number since no value is 0 by (i).

Lemma 3.8 (Transport). *If $j(X) < R$, then $p\theta_X(y) = \sum_{T \text{ outward}} \theta_{X_T}(\pi_T y)$. Consequently there are a member Y on the outer sphere and a label y' of Y with $\theta_Y(y') \geq \theta_X(y)$ if $j(X) \geq 1$, and $\theta_Y(y') \geq \frac{p}{|\text{Out}(X)|} \theta_X(y)$ if $j(X) = 0$.*

Proof. The formula follows by taking valuations in Lemma 3.7. Off the crater there are p outward lines, so $\theta_X(y)$ is the mean of the outward values, and some outward sibling has θ at least as large. On the crater the first step multiplies the bound by $p/(p - 1)$ (split) or $p/(p + 1)$ (inert). Every outward neighbor is again a member, so iterating reaches the outer sphere. $\square$

Lemma 3.9 (Outer-boundary inequality). *If $j(X) = R$ and $a := a(X) > 0$, then $(p - 2J)a \leq a(X_D)$, where X_D is the sibling of X along D_X .*

Proof. Choose y_0 with $v(E_X(y_0)) = -a$ and put $\beta := E_X(y_0)^{-1}$, so that βE_X is integral. Along $y_0 + D$ the residue of βE_X is a polynomial of degree $\leq J$ (Proposition 3.5 and Lucas's theorem) taking the value 1 at y_0 . So it vanishes at most J times, and $v(E_X) = -a$ at no fewer than $p - J$ points of $y_0 + D$. At the remaining points $v(E_X) \leq a$ by the reflection bound. By (ii) the product along $y_0 + D$ is a root of unity times a value of the inward sibling X_D , so $-a(X_D) \leq -(p - J)a + Ja$. $\square$

Proof of (U0). Let $A := \max_X a(X)$, a maximum over finitely many values, and suppose that $A > 0$, attained at (X_0, y_0) . By the reflection bound, $|v(E_X(y))| \leq A$ everywhere. Below the outer sphere also $|v(\Pi_X(y))| \leq A$: by the edge identity and (ii) at X_T , $\Pi_X(y)$ is a root of unity times a single value of the member whose lattice is $p^{-1}I_X$, a homothetic copy of I_X .

The member X_0 is not on the outer sphere, since there Lemma 3.9 would give a pole of size at least $(p - 2J)A \geq 3A$. Its defect at y_0 is large. Off the crater, $\theta_{X_0}(y_0) = p v(E_{X_D}) - p^2 v(E_{X_0}(y_0)) \geq p(p - 1)A$; on an inert crater, $\theta_{X_0}(y_0) \geq (p^2 - 1)A$; on a split crater, $\theta_{X_0}(y_0) \geq (p^2 - 2p - 1)A$. By Lemma 3.8 there are Y on the outer sphere and y' with $\theta_Y(y') \geq cA$, where $c = p(p - 1)$ when starting off the crater or on an inert crater, and $c = p(p^2 - 2p - 1)/(p - 1)$ when starting on a split crater.

This produces a pole on the outer sphere. There $\theta_Y(y') = p v(L_{D_Y}(y')) - p^2 v(E_Y(y')) \leq pA - p^2 v(E_Y(y'))$, so $a(Y) \geq \rho A$ with $\rho = c/p^2 - 1/p$, that is, $\rho = (p - 2)/p$ or $\rho = (p - 3)/(p - 1)$. One step inward, Lemma 3.9 at Y gives

$$a(Y_D) \geq (p - 2J)\rho A \geq 3\rho A > A,$$

since $3(p - 2)/p > 1$ and $3(p - 3)/(p - 1) > 1$ for $p \geq 5$. This contradicts the choice of A . So $A = 0$, and with the reflection bound $v(E_X(y)) = 0$ for every value. $\square$

For $p = 7$, starting off the crater, the numbers are $\theta \geq 42A$, then $a(Y) \geq \frac{5}{7}A$, then $a(Y_D) \geq \frac{15}{7}A$. Apart from Lemma 3.9, the argument uses only that v turns products into sums

and that a maximum is at least a mean. The hypothesis $p \geq 5$ is used twice: 6 must be a unit in Proposition 3.5, and $p - 2J \geq 3$ (with $3\rho > 1$) closes the contradiction; at $p = 3$, $p - 2J = 1$.

Proof of (U1) and (U2). By (U0) every defect is a unit. Suppose that some $\Theta_Y \neq 1$, and take such a Y with $j(Y)$ maximal.

On the outer sphere this is impossible. The residues of E_Y along the translates of D_Y through y and through $-y$ are polynomials of degree $\leq J$, and reflection makes their product (one of them read backwards) a nonzero constant. Indeed $E_Y(y+h)E_Y(-y-h) = q(y+h)^{-1}$, and for $h \in P_Y$ the ratio $q(y+h)/q(y)$ is a root of unity of p -power order, hence $\equiv 1$; where the origin occurs, use $E_Y(0)^2 = \varepsilon_0 \equiv 1$. We use the *pairing lemma*: two polynomials over a field whose degrees add up to less than p , and whose product takes the same nonzero value at all p points of $\mathbb{F}_p$, are constant. As $2J < p$, the residue of E_Y is constant along D_Y , and $\Theta_Y \equiv 1$.

Below the outer sphere the outward siblings satisfy $\Theta \equiv 1$ by maximality, so Lemma 3.7 gives $\Theta_Y(z)^p \equiv 1$, hence $\Theta_Y(z) \equiv 1$, since $x \mapsto x^p$ is injective on a field of characteristic p . This proves (U1).

Off the crater, (U1) says that $L_{D_X}(y) \equiv E_X(y)^p$. The left side is the same at every point of $y + D_X$, and $x \mapsto x^p$ is injective, so the residues of the values are constant along each translate of the inward line. On an inert crater the same argument gives constancy on each translate of $X[p]$.

On a split crater write the points of $y + X[p]$ as $y + s + t$ with $s \in T_p$ and $t \in T_{\bar{p}}$. As L_T is T -invariant, (U1) gives $(\text{red } E_X)^{p^2}(y+s+t) = \alpha(s)\beta(t)$ with α and β nowhere zero. Lemma 3.6 bounds the total degree of $\text{red } E_X$ by $p-1$. Raising to the power p does not raise it: as a function on $\mathbb{F}_p^2$, the p -th power of a polynomial is the polynomial with the p -th powers of its coefficients. So $\alpha(s)\beta(t)$, as a reduced polynomial, has total degree $\deg \alpha + \deg \beta \leq p-1$, and likewise for the factors α', β' at $-y$. Reflection makes $\alpha(s)\beta(t)\alpha'(-s)\beta'(-t)$ a nonzero constant, so $\alpha(s)\alpha'(-s)$ and $\beta(t)\beta'(-t)$ are each constant. By the pairing lemma a nonconstant pair would have degree sum at least p , and two such pairs would exceed $2(p-1)$. So α or β is constant, which is (U2). $\square$

Nothing local decides which of the two lines occurs. Every function on P_X that depends only on the coordinate along one crater line satisfies all three integrality conditions, so the dichotomy is the most the column lattice can give. The choice is made globally, in Section 3.4.

Proposition 3.10 (Algebraicity). *Every value of [RW] is algebraic.*

Proof. Fix a split prime $p \geq 5$ with $p \nmid N_\gamma f$. Suppose that a value ξ were transcendental. Let L_0 be a number field containing K and the roots of unity and square roots of the setting, and let F be the field generated over L_0 by ξ and the values of the members; these are finitely many, by (vii), which does not use algebraicity. Extend ξ to a transcendence basis of F over L_0 . The weighted Gauss valuation over a prime of L_0 above p , normalized by $v(p) = 1$ and with weight $v(\xi) < 0$, is a discrete valuation of the rational function field, and it extends to a discrete valuation v of the finite extension F . It satisfies the hypotheses of the setting, since no integrality of the values was assumed. Then (U0) would make ξ^{k_0} a unit, by (iii), a contradiction. $\square$

This gives an independent proof of [RW, Thm. 1].

3.3. Exact root extraction and the Frobenius congruence. Theorem 3.2 concerns the values of $\gamma_0 = \gamma^{k_0}$. The lines $T_p, T_{\bar{p}}$ consist of labels of γ_0 , not of γ (as $p \nmid N_\gamma$), and on the labels of γ the values of γ_0 are k_0 -th powers, by (iii). A congruence between k_0 -th powers determines its base modulo $\mathfrak{P} \mid p$ only up to a k_0 -th root of unity, and this ambiguity does

occur, since $k_0 \mid p - 1$. So we take the root before reducing, among positive reals, where k_0 -th roots are unique. The identities below are exact equalities of positive reals, and only the last step of Lemma 3.12 reduces.

Lemma 3.11 (Lattice distribution and positive orbit products). (a) *Suppose that ε' is a totally positive unit with $\varepsilon'_r > 1$. Let $I \subseteq I'$ be lattices with $\varepsilon'I = I$ and $I'/I \subseteq t_{\varepsilon'}^{-1}I/I$. Then for every $z \in t_{\varepsilon'}^{-1}I$ with $z \notin I'$,*

$$\prod_{y \in I'/I} W(I, \varepsilon', z + y) = W(I', \varepsilon', z).$$

(b) *Let $I \subseteq I'$ be lattices of finite index and u a unit with $uI = I$, $uI' = I'$, $|u_r| > 1$ and $u_u > 0$. Put $m(u) := 1$ if $u_r > 0$ and $m(u) := 2$ otherwise, and let $\mathcal{W}(I, u, x) := W(I, u^{m(u)}, x)^{1/m(u)}$ be the positive root. Let $z \in K$ with $(u - 1)z \in I$ and $z \notin I'$. For each $\langle u \rangle$ -orbit O on I'/I choose $h_O \in O$ and put $d_O := |O|$. Then*

$$\mathcal{W}(I', u, z) = \prod_O \mathcal{W}(I, u^{d_O}, z + h_O),$$

an identity of positive reals, independent of the representatives h_O .

Proof of (a). Since ε' fixes $t_{\varepsilon'}^{-1}I/I$ pointwise, every lattice between I and I' is ε' -stable. A composition series of I'/I , with the product grouped by cosets, therefore reduces (a) to one step $L \subset L'$ of prime index p . Let (v_1, v_2) be a fixed frame of L and γ the matrix of ε' in it, so that $L = v_2 I_\gamma$, and let $T \subset G_\gamma$ be the line L'/L read in I_γ . Taking absolute squares in the line product formula (ii), with $|\kappa_T| = 1$ and $|E|^2 = w$ by (iv), gives $\prod_{h \in T} w_\gamma(x + h) = |E_{\gamma_T}(\pi_T x)|^2$. The content of the lemma is that the right side is $W(L', \varepsilon', z)$, with the same z and exponent $+1$; this needs the sibling's frame. In [RW, Prop. 9], the sibling is $\gamma_T = N_T \gamma N_T^{-1}$ with

$$N_T = \begin{pmatrix} 1 & f/y \\ 0 & h/y \end{pmatrix}, \quad h, y \geq 1, \quad hy = p, \quad f \in \mathbb{Z},$$

and π_T sends the label (ℓ, j) to $(h\ell, yj - f\ell)$. Then $N_T(\tau^*, 1)^\top = (h/y)(\tau', 1)^\top$ with $\tau' := (y\tau^* + f)/h$, so τ' is the attracting fixed point of γ_T , and γ_T is the matrix of ε' on $L'_0 := \mathbb{Z}v'_1 + \mathbb{Z}v'_2$, where $v'_1 := \tau'/y$ and $v'_2 := 1/y$. This frame is fixed: v'_2 is a positive rational, and $\tau'_r - \tau'_u = (y/h)(\tau_r^* - \tau_u^*) > 0$; both facts come from $\det N_T > 0$ and the lower row $(0, h/y)$ of N_T . The label map is the identity on elements: if $(\varepsilon'^{-1} - 1)x = \ell\tau^* + j$, the image label gives $h\ell\tau' + (yj - f\ell) = y(\ell\tau^* + j)$, which is x read in the frame with second vector $1/y$. Finally $\tau^* = hv'_1 - fv'_2$ and $1 = yv'_2$, so $I_\gamma \subset L'_0$ with index p , and $\ker \pi_T = T$; hence $L'_0 = I_\gamma + \tilde{T}$. So $\prod_{h \in T} W(I_\gamma, \varepsilon', x + h) = W(I_\gamma + \tilde{T}, \varepsilon', x)$, and multiplying by $v_2 \gg 0$ gives the step. $\square$

The orientation of the sibling frame matters. If its second vector were not totally positive, or if $\theta_r < \theta_u$, the frame rules of Section 3.1 would replace the right side by $W(L', \varepsilon', \pm z)^{\pm 1}$; the upper triangular N_T with $\det N_T > 0$ excludes this. Part (a) uses only (ii), (iv) and (vii).

Proof of (b). Let n be divisible by $m(u)$ and by the order of u on I'/I . Then u^n is totally positive with $u_r^n > 1$, fixes I'/I pointwise and satisfies $(u^n - 1)z \in I$. So (a) at $\varepsilon' = u^n$ gives $W(I', u^n, z) = \prod_{h \in I'/I} W(I, u^n, z + h)$; all these points lie off I , as $z \notin I'$. By (iii) the left side is $\mathcal{W}(I', u, z)^n$. Homothety by u ((viii), with $uI = I$ and $u_u > 0$) makes $W(I, u^n, z + \cdot)$ constant on $\langle u \rangle$ -orbits, because $u(z + h) \equiv z + uh \pmod{I}$. On an orbit O , u^{d_O} fixes $z + h_O$ modulo I , so $z + h_O$ is a label of the totally positive unit $u^{d_O m(u^{d_O})}$, and $d_O m(u^{d_O})$ divides n : if $m(u^{d_O}) = 2$, then $u_r < 0$ and d_O is odd, while $2 = m(u)$ and d_O both divide n . By (iii), $W(I, u^n, z + h_O) = \mathcal{W}(I, u^{d_O}, z + h_O)^{n/d_O}$, and the d_O equal factors of O contribute $\mathcal{W}(I, u^{d_O}, z + h_O)^n$. Take positive n -th roots. $\square$

When u is a power of the unit ε whose matrix is used, as in Lemma 3.12, homothety by u is the ε -invariance in (vii), and Theorem DC is not used. Section 4 applies (b) to a unit of norm -1 .

Setting for root extraction. Let γ be the matrix of ε in a fixed frame (v_1, v_2) of I , let $\mathcal{O} := \text{End}(I)$, of conductor f , and let $p = \mathfrak{p}\bar{\mathfrak{p}}$ be a split prime with $p \geq 5$ and $p \nmid N_\gamma f$. Let $\mathfrak{q} \in \{\mathfrak{p}, \bar{\mathfrak{p}}\}$ and $\mathfrak{q}_\mathcal{O} := \mathfrak{q} \cap \mathcal{O}$, an invertible prime of degree one. Since ε has norm 1, $\varepsilon \bmod \bar{\mathfrak{p}}$ is the conjugate of $(\varepsilon \bmod \mathfrak{p})^{-1}$, so the order of ε in $(\mathcal{O}/\mathfrak{q}_\mathcal{O})^\times \cong \mathbb{F}_p^\times$ is the k_0 of Section 3.2; in particular $k_0 \mid p - 1$. As $\varepsilon^{k_0} \equiv 1 \pmod{\mathfrak{q}_\mathcal{O}}$, the line

$$T := \mathfrak{q}_\mathcal{O}^{-1} I_\gamma / I_\gamma \subset G_{\gamma_0}$$

is the crater line $T_\mathfrak{q}$ of Theorem 3.2. It is a one-dimensional $\mathcal{O}/\mathfrak{q}_\mathcal{O}$ -vector space on which ε acts as a scalar of order k_0 , so ε acts freely on $T \setminus 0$ with $\ell := (p - 1)/k_0$ orbits; let $h_1, \dots, h_\ell$ represent them. Moreover $k_0 > 1$, since $p \nmid N_\gamma = -\text{Nm}(\varepsilon - 1)$, while ε fixes G_γ pointwise; hence $T \cap G_\gamma = 0$.

Lemma 3.12 (Root extraction). *For $x \in G_\gamma \setminus 0$ and $z := v_2 x$,*

$$W(\mathfrak{q}_\mathcal{O}^{-1} I, z) = W(I, z) \cdot \prod_{i=1}^{\ell} w_{\gamma_0}(x + h_i),$$

an identity of positive reals, independent of the representatives h_i ; no root of unity and no Gaussian factor appears. If, at a prime $\mathfrak{P} \mid p$ of a field containing the values involved, these values are $\mathfrak{P}$ -units and the residue of E_{γ_0} is constant on $x + T$, then

$$W(\mathfrak{q}_\mathcal{O}^{-1} I, z) \equiv W(I, z)^p \pmod{\mathfrak{P}}.$$

Proof. By a totally positive homothety we may take $I = I_\gamma$ and $v_2 = 1$. The identity is Lemma 3.11(b) with $u = \varepsilon$ and $I' = \mathfrak{q}_\mathcal{O}^{-1} I_\gamma$, so that $I'/I = T$. Since $T \cap G_\gamma = 0$, we have $z \notin I'$. The $\langle \varepsilon \rangle$ -orbits on T are $\{0\}$, contributing $W(I, \varepsilon, z) = w_\gamma(x) =: w_x$, and the ℓ orbits of the h_i , of length k_0 , contributing $W(I, \varepsilon^{k_0}, z + h_i) = w_{\gamma_0}(x + h_i)$, as γ_0 is the matrix of ε^{k_0} in the same frame.

For the congruence, $w_{\gamma_0}(x + h) = q_{\gamma_0}(x + h) E_{\gamma_0}(x + h)^2$ and $q_{\gamma_0}(x + h) = q_{\gamma_0}(x) q_{\gamma_0}(h) b(x, h)$, where the last two factors are roots of unity of p -power order (as $ph = 0$), hence $\equiv 1$. With the constancy of the residue on $x + T$ and (iii), $w_{\gamma_0}(x + h_i) \equiv w_{\gamma_0}(x) = w_x^{k_0}$. So the right side is $\equiv w_x^{1+k_0\ell} = w_x^p$. $\square$

The exponent count $1 + k_0\ell = p$ is exact: it counts the source point x and ℓ orbits of length k_0 . Before any root is taken, (ii) on the single coset $x + T$ reads $(E_\gamma(x) \Pi_x)^{k_0} = \kappa E_{\gamma_\mathfrak{q}}(x')^{k_0}$, where $\Pi_x := \prod_i E_{\gamma_0}(x + h_i)$, κ is a root of unity, and $\gamma_\mathfrak{q}$ is the matrix of ε on $\mathfrak{q}_\mathcal{O}^{-1} I$; by (iii), $\gamma_\mathfrak{q}^{k_0}$ is the sibling of γ_0 along T . Lemma 3.12 takes the positive k_0 -th root of its absolute value, so the value of κ never enters.

Definition 3.13 (Good primes). Fix a lattice I and a unit ε as in a datum, with matrix γ in a fixed frame, and a finite Galois extension L/K containing the values of all admissible matrices of trace $\text{tr } \varepsilon$; these form a finite set by (vii), and they are algebraic by Proposition 3.10. A rational prime p is *good* for (ε, L) if $p \geq 5$, p splits in K , $p \nmid N_\gamma f_0$ (so that $p \nmid f$, as $f \mid f_0$), and p is unramified in L .

This notion depends on I only through ε , and all but finitely many split primes are good. The root extraction, the Frobenius congruence and every Chebotarev step below need only such a cofinite set.

Lemma 3.14 (Frobenius congruence). *Let $p = \mathfrak{p}\bar{\mathfrak{p}}$ be good for (ε, L) and let $\mathfrak{P} \mid \mathfrak{p}$ be a prime of L . Then for every $z \in t_\varepsilon^{-1}I \setminus I$ the values involved are $\mathfrak{P}$ -units, and*

$$W(I, z)^p \equiv W(\mathfrak{p}_\mathcal{O}^{-1}I, z) \quad \text{or} \quad W(I, z)^p \equiv W(\bar{\mathfrak{p}}_\mathcal{O}^{-1}I, z) \pmod{\mathfrak{P}}.$$

Proof. Extend $v_\mathfrak{p}$, normalized by $v(p) = 1$, to a valuation v of a field $K_{\text{fam}} \supseteq L$ as in the setting of Theorem 3.2, whose hypotheses hold ($p \geq 5$ split, $p \nmid N_\gamma f$). By (U0) every value of every member is a v -unit; in particular $E_\gamma(x)$, whose k_0 -th power is $E_{\gamma_0}(x)$, is one. By (U2) at $x = z/v_2$ there is $\mathfrak{q} \in \{\mathfrak{p}, \bar{\mathfrak{p}}\}$ such that the residue of E_{γ_0} is invariant under $T_\mathfrak{q}$, hence constant on $x + T_\mathfrak{q}$. Lemma 3.12 gives $W(\mathfrak{q}_\mathcal{O}^{-1}I, z) \equiv W(I, z)^p$ modulo the maximal ideal of v . Both sides lie in L , so the congruence holds modulo $\mathfrak{P}$. $\square$

As $\text{Frob}_\mathfrak{P}$ acts on $\mathcal{O}_L/\mathfrak{P}$ as the p -th power map, Lemma 3.14 says that $\text{Frob}_\mathfrak{P}W(I, z)$ is congruent to one of two candidates. The statement is symmetric under $\mathfrak{p} \leftrightarrow \bar{\mathfrak{p}}$, and the candidate can belong to $\bar{\mathfrak{p}}$ although $\mathfrak{P}$ lies above $\mathfrak{p}$.

3.4. The reciprocity law. We first set up the action of ideles on data. The lattice $s_f^{-1}I$ of Section 1.1 has the same multiplier ring as I . Indeed, the multiplier ring of a lattice $\bar{L}$ is $K \cap \prod_\ell \text{End}(L_\ell)$, where $L_\ell := L \otimes \mathbb{Z}_\ell$. The completion of $s_f^{-1}I$ at ℓ is $s_\ell^{-1}I_\ell$, and multiplication by the unit s_ℓ^{-1} of the commutative ring $K \otimes \mathbb{Q}_\ell$ does not change the multiplier ring. Moreover $(\varepsilon - 1)s_f^{-1}z = s_f^{-1}(\varepsilon - 1)z$ lies in $s_f^{-1}\hat{I}$, and $s_f^{-1}z \notin s_f^{-1}\hat{I}$. So $s^{-1}d$ is again a datum, and $(st)^{-1}d = t^{-1}(s^{-1}d)$.

Fix ε and put $\mathfrak{M} := \mathfrak{M}_\varepsilon = f_0(\varepsilon - 1)\mathcal{O}_K$. Let $U_\mathfrak{M}$ be the group of ideles s with $s_\mathfrak{q} \in 1 + \mathfrak{M}\mathcal{O}_\mathfrak{q}$ for $\mathfrak{q} \mid \mathfrak{M}$ and $s_\mathfrak{q} \in \mathcal{O}_\mathfrak{q}^\times$ otherwise, with arbitrary infinite components; let $U_\mathfrak{M}^+$ be its subgroup with $s_r, s_u > 0$, and $U_{\mathfrak{M},f}$ its finite part. Then

$$C_\mathfrak{M} = \mathbb{A}_K^\times / K^\times U_\mathfrak{M}^+ \cong \mathbb{A}_{K,f}^\times / K_{\gg 0}^\times U_{\mathfrak{M},f},$$

and Art identifies this group with $\text{Gal}(H_{\mathfrak{M}\infty_r\infty_u}/K)$. Let c_r, c_u be the classes of $(1; -1, 1)$ and $(1; 1, -1)$; then $\text{Art}(c_v)$ is the complex conjugation ι_v restricted to $H_{\mathfrak{M}\infty_r\infty_u}$. Let Ω_ε be the set of data with unit ε , modulo $z \mapsto z + I$ and $(I, z) \mapsto (\beta I, \beta z)$ for $\beta \gg 0$. It is finite: finitely many orders contain $\mathbb{Z}[\varepsilon]$, each has finitely many narrow classes of invertible $\mathcal{O}$ -ideals, and each lattice has finitely many labels. The value W is a function on Ω_ε .

Lemma 3.15 (Ray classes acting on data). *The following hold for data of every order.*

- (a) $\mathfrak{M}$ is τ -stable, and $\mathfrak{M}z \subseteq I$ for every datum (I, ε, z) .
- (b) $d \mapsto s_f^{-1}d$ induces an action of $C_\mathfrak{M}$ on Ω_ε . If s is the finite idele of an integral ideal $\mathfrak{a}$ prime to $\mathfrak{M}$, then $s^{-1}(I, \varepsilon, z) = (\mathfrak{a}_\mathcal{O}^{-1}I, \varepsilon, z)$, where $\mathfrak{a}_\mathcal{O} := \mathfrak{a} \cap \mathcal{O}_I$.
- (c) For $a \in \mathbb{Z}$ prime to $\mathfrak{M}$ and $\pi \gg 0$ with $\pi \equiv a \pmod{\mathfrak{M}}$, $[(\pi)] \cdot (I, z) = (I, az)$. For $\beta \equiv 1 \pmod{\mathfrak{M}}$ with $\beta_r < 0 < \beta_u$, $c_r \cdot (I, z) = (\beta^{-1}I, z)$. Moreover $c_r c_u \cdot (I, z) = (I, -z)$.
- (d) The involution $c \mapsto \bar{c}$ of $C_\mathfrak{M}$ induced by τ satisfies $\bar{c}_r = c_u$ and $[\bar{\mathfrak{p}}] = [\mathfrak{p}]$.

Proof. (a) We have $\tau(\varepsilon - 1) = -\varepsilon^{-1}(\varepsilon - 1)$, and $f_0\mathcal{O}_K \subseteq \mathbb{Z}[\varepsilon] \subseteq \mathcal{O}_I$ gives $\mathfrak{M}z \subseteq I$.

(b) $K_{\gg 0}^\times$ acts trivially, since the action commutes with homotheties. An element $u \in U_{\mathfrak{M},f}$ has $u_\ell^{\pm 1} \in \mathcal{O}_{I,\ell}$ at every ℓ (at $\ell \mid f_0$ because $1 + \mathfrak{M}_\ell \subseteq \mathbb{Z}[\varepsilon]_\ell$), so $u^{-1}\hat{I} = \hat{I}$ and $(u^{-1} - 1)z \in \mathfrak{M}z \subseteq I$. For $\mathfrak{a}$ prime to $\mathfrak{M}$ the idele is 1 wherever $z_\ell \neq 0$ or ℓ divides the conductor of I , which gives $(\mathfrak{a}_\mathcal{O}^{-1}I, z)$ [Cox, Prop. 7.20].

(c) $[(\pi)] \cdot (I, z) = (\pi^{-1}I, z) \sim (I, \pi z) = (I, az)$, as $(\pi - a)z \in \mathfrak{M}z \subseteq I$. The principal idele β is the product of the idele of (β) , an element of $U_{\mathfrak{M}}^+$, and $(1; \beta_r, \beta_u)$; hence $[(\beta)] = c_r$. For $\beta \equiv 1$ totally negative, $\pi := -\beta$ is the case $a = -1$.

(d) By (a), τ preserves $U_\mathfrak{M}$ and $K^\times$, so it induces an involution of $C_\mathfrak{M}$. It exchanges the two real places, so it maps $(1; -1, 1)$ to $(1; 1, -1)$, and it maps the idele of $\mathfrak{p}$ to that of $\tau\mathfrak{p} = \bar{\mathfrak{p}}$. $\square$

Lemma 3.16 (The ordering bit). *Let $\mathbb{A}_K^\times$ act on $(data) \times \{\pm 1\}$ by $s \star (d, \epsilon) := (s^{-1}d, \epsilon \operatorname{sgn} s_u)$, and put $V(d, \epsilon) := W(d)^\epsilon$, so that the right side of $(\star)$ is $V(s \star (d, 1))$. Then $V(s \star x) = V(x)$ for $s \in U_{\mathfrak{M}}^+$, for $s = (1; s_r, s_u)$ with $s_u > 0$, and for $s = \alpha \in K^\times$ with $\operatorname{Nm} \alpha > 0$. Moreover $V(\alpha \star x) = V(x)$ for all $\alpha \in K^\times$ if and only if the homothety law (viii) holds. Hence, given (viii), $V(s \star x)$ depends only on the class of s in $C_{\mathfrak{M}}$.*

Proof. $U_{\mathfrak{M}}^+$ acts trivially by Lemma 3.15(b), and archimedean ideles do not move d . With $\beta = \alpha^{-1}$, $V(\alpha \star (d, \epsilon)) = W(\beta I, \beta z)^{\epsilon \operatorname{sgn} \alpha_u}$, which equals $W(d)^\epsilon$ exactly when (viii) holds for β . For $\operatorname{Nm} \beta > 0$ this is frame independence, with reflection if $\beta \ll 0$. $\square$

The Chebotarev argument below determines the Galois action on each value only up to a choice between two candidates. The following lemma removes the ambiguity.

Lemma 3.17 (Selector). *Let a finite abelian group $\tilde{C}$, with an involutive automorphism $c \mapsto \bar{c}$ and elements c_r, c_u with $\bar{c}_r = c_u$, act on a finite set Ω . Let $F \subset \mathbb{R}$ be a field, $W : \Omega \rightarrow F_{>0}$, and $\varphi : \tilde{C} \rightarrow \operatorname{Aut}(F)$ a homomorphism with*

- (Sel1) $\varphi(c)W(\omega) \in \{W(c\omega), W(\bar{c}\omega)\}$ for all c, ω ;
- (Sel2) $\varphi(c_r) = 1$;
- (Sel3) $W(c_r\omega) = W(\omega)$ and $W(c_u\omega) = W(\omega)^{-1}$ for all ω .

Then $\varphi(c)W(\omega) = W(c\omega)$ for all c, ω .

Proof. Put $x := \varphi(c)W(\omega)$. By (Sel2), $x = \varphi(cc_r)W(\omega)$, and (Sel1) at cc_r , with $\overline{cc_r} = \bar{c}c_u$ and (Sel3), gives $x \in \{W(c\omega), W(\bar{c}\omega)^{-1}\}$. By (Sel1) at c , $x \in \{W(c\omega), W(\bar{c}\omega)\}$. If $x \neq W(c\omega)$, then $x = W(\bar{c}\omega) = W(\bar{c}\omega)^{-1}$ is a positive real equal to its inverse, so $x = 1$ and $W(\omega) = \varphi(c)^{-1}(1) = 1$. The same dichotomy for c^{-1} at $c\omega$ gives $\varphi(c)^{-1}W(c\omega) \in \{W(\omega), 1\} = \{1\}$, so $W(c\omega) = 1 = x$, a contradiction. $\square$

Both (Sel2) and (Sel3) are needed. The analogous values at the repelling fixed point satisfy every hypothesis except (Sel3). Let m be the least common multiple of 24 and the orders of the values of q ; when $K \cap \mathbb{Q}(\mu_m) = \mathbb{Q}$, the inverted conjugates $1/\sigma W$, for $\sigma|_K = \tau$ with $\sigma(\zeta) = \zeta^{-1}$ on μ_m , satisfy every hypothesis except (Sel2).

Theorem RL (Artin law on Ω_ε). *Let $F_\varepsilon := K(W(\omega) : \omega \in \Omega_\varepsilon)$.*

- (RL1) $F_\varepsilon \subseteq H_{\mathfrak{M}\infty_u}$; in particular F_ε/K is abelian and ι_r is trivial on F_ε .
- (RL2) $\operatorname{Art}(c)W(\omega) = W(c\omega)$ for every τ -fixed class c . In particular $\operatorname{Art}([\pi])W(I, z) = W(I, az)$ for $\pi \gg 0$, $\pi \equiv a \pmod{\mathfrak{M}}$ with $a \in \mathbb{Z}$, and $\operatorname{Art}(c_u)W(\omega) = W(\omega)^{-1}$.
- (RL3) $\operatorname{Art}(c)W(\omega) = W(c\omega)$ for every $c \in C_{\mathfrak{M}}$ and every ω .

Proof. For $\omega = (I, z)$ the matrix γ of ε has $\operatorname{Nm}(\varepsilon - 1) = -N_\gamma$, so $\operatorname{Nm} \mathfrak{M} = f_0^2 N_\gamma$ and every good prime is prime to $\mathfrak{M}$. Let L/K be a finite Galois extension containing $H_{\mathfrak{M}\infty_r\infty_u}$ and every value of trace $\operatorname{tr} \varepsilon$. The same good primes then work for every $\omega \in \Omega_\varepsilon$.

Step 1: Chebotarev on a product of differences. Let $g \in \operatorname{Gal}(L/K)$ and $c(g) \in C_{\mathfrak{M}}$ with $\operatorname{Art}(c(g)) = g|_{H_{\mathfrak{M}\infty_r\infty_u}}$. By Chebotarev's theorem [Ch] there are infinitely many good degree-one primes $\mathfrak{p}$ with a prime $\mathfrak{P} \mid \mathfrak{p}$ of L such that $\operatorname{Frob}_{\mathfrak{P}} = g$; for these, $[\mathfrak{p}] = c(g)$ and $[\bar{\mathfrak{p}}] = \bar{c}(g)$. At each such $\mathfrak{P}$ we have $gW(\omega) \equiv W(\omega)^p$, and Lemma 3.14, with Lemma 3.15(b), gives $W(\omega)^p \equiv W(c(g)\omega)$ or $W(\omega)^p \equiv W(\bar{c}(g)\omega)$. So the fixed algebraic number $(gW(\omega) - W(c(g)\omega))(gW(\omega) - W(\bar{c}(g)\omega))$ lies in infinitely many primes, and is therefore 0:

$$(\text{Cand}) \quad gW(\omega) \in \{W(c(g)\omega), W(\bar{c}(g)\omega)\} \quad (\omega \in \Omega_\varepsilon).$$

No separation of distinct values is needed.

Step 2: Containment. If g is trivial on $H_{\mathfrak{M}\infty_r\infty_u}$, then $c(g) = 1$ and g fixes every $W(\omega)$. So $F_\varepsilon \subseteq H_{\mathfrak{M}\infty_r\infty_u}$.

Step 3: Reality at r . Each $W(\omega)$ is a positive real and $K^{\text{ab}} \subset \mathbb{C}$ lies above r , so $\text{Art}(c_r)$, which is complex conjugation, fixes F_ε . Hence $F_\varepsilon \subseteq H_{\mathfrak{M}\infty_u}$, which is (RL1).

Step 4: τ -fixed classes. If $\bar{c} = c$, the two candidates in (Cand) coincide, so $\text{Art}(c)W(\omega) = W(c\omega)$. Rational classes are τ -fixed and act by $(I, z) \mapsto (I, az)$. The class $c_r c_u$ is τ -fixed and acts by $z \mapsto -z$; by reflection (v) it therefore acts by inversion. With Step 3, $\text{Art}(c_u) = \text{Art}(c_r c_u) \text{Art}(c_r)^{-1}$ inverts every $W(\omega)$. This is (RL2).

Step 5: Selection. Put $\varphi := \text{Art}(\cdot)|_{F_\varepsilon}$. As $\text{Gal}(L/K) \rightarrow C_{\mathfrak{M}}$ is onto, (Cand) is (Sel1) for $\Omega = \Omega_\varepsilon$, $\tilde{C} = C_{\mathfrak{M}}$ and the involution of Lemma 3.15(d); Step 3 is (Sel2). For (Sel3), choose $\beta \equiv 1 \pmod{\mathfrak{M}}$ with $\beta_r < 0 < \beta_u$; it exists, since $1 + \mathfrak{M}$ is a translate of a full lattice in $\mathbb{R}^2$. Then $c_r \cdot (I, z) = (\beta^{-1}I, z)$, and the homothety law (viii) for β , which is the single use of Theorem DC, gives $W(\beta^{-1}I, z) = W(I, \beta z) = W(I, z)$, as $(\beta - 1)z \in \mathfrak{M}z \subseteq I$. Since $c_u = c_r c_u \cdot c_r$, also $W(c_u \omega) = W(c_r \omega)^{-1} = W(\omega)^{-1}$. Lemma 3.17 now gives (RL3). $\square$

No hypothesis $c_r \neq 1$ is needed: if $c_r = 1$ then $c_u = 1$, and (RL2) forces $W \equiv 1$. The τ -fixed classes form in general a proper subgroup of $C_{\mathfrak{M}}$, so Step 5 is needed for the full law.

Proof of Theorem A. Algebraicity is Proposition 3.10. Let d be a datum with unit ε , ω its class in Ω_ε , and $s \in \mathbb{A}_K^\times$. By (RL1), $W(d) \in H_{\mathfrak{M}\infty_u} \subseteq K^{\text{ab}}$.

Step 1. Choose $\alpha \in K^\times$ with $\text{sgn } \alpha_r = \text{sgn } s_r$ and $\text{sgn } \alpha_u = \text{sgn } s_u$ (one of $\pm 1, \pm \sqrt{d_K}$, with $\sqrt{d_K} > 0$ at r), and put $s' := \alpha^{-1}s$. Then $s'_r, s'_u > 0$ and $\text{Art}_K(s) = \text{Art}_K(s')$.

Step 2. Since $(1; s'_r, s'_u) \in U_{\mathfrak{M}}^+$, the class c of s' in $C_{\mathfrak{M}}$ is the class of its finite part s'_f , and $c\omega$ is the class of $(s'_f)^{-1}d$.

Step 3. By (RL3), $\text{Art}_K(s')W(d) = W(c\omega) = W((s'_f)^{-1}d)$.

Step 4. By Lemma 3.16, that is, by (viii) for α ,

$$W(s^{-1}d)^{\text{sgn } s_u} = V(\alpha \star (s' \star (d, 1))) = V(s' \star (d, 1)) = W((s'_f)^{-1}d).$$

Steps 3 and 4 give $(\star)$. $\square$

Rational and Galois forms. Since $W(d) \in K^{\text{ab}}$, the Galois form of Theorem A is $(\star)$ itself.

Consider first rational ideles $s \in \mathbb{A}_{\mathbb{Q}}^\times \subset \mathbb{A}_K^\times$. Then $s_r = s_u$, so Step 1 of the proof of Theorem A takes $\alpha = \pm 1$, Step 4 needs only the positive-norm case of Lemma 3.16, and c is τ -fixed, so Step 3 needs only (RL2). Thus the restriction of $(\star)$ to rational ideles uses neither Theorem DC nor Lemma 3.17. The transfer Ver corresponds to the inclusion $\mathbb{A}_{\mathbb{Q}}^\times \subset \mathbb{A}_K^\times$. For $a \in \hat{\mathbb{Z}}^\times$ we have $a^{-1}z = a'z$ with $a' \equiv a^{-1}$ an integer, and $\chi_{\text{cyc}}(\text{Art}_{\mathbb{Q}}(a)) = a^{-1}$; the real idele -1 gives $W(I, z)^{-1} = W(I, -z)$. Hence

$$\text{Ver}(g) W(I, \varepsilon, z) = W(I, \varepsilon, \chi_{\text{cyc}}(g) z) \quad (g \in \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})).$$

Now let $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ be arbitrary. If $\sigma|_K = \text{id}$, then $\sigma F_\varepsilon = F_\varepsilon \subset \mathbb{R}$ by (RL1), and (Cand) makes $\sigma W(\omega)$ a positive real. If $\sigma|_K = \tau$, then $\sigma^{-1}\iota\sigma$, with ι complex conjugation, is the complex conjugation of the embedding σ , which lies above u . On F_ε it is $\text{Art}(c_u)$, so $\iota\sigma W = \sigma(W^{-1})$ by (RL2), that is, $|\sigma W| = 1$. Hence

$$\overline{\sigma W(d)} = \sigma W(d)^{\chi_K(\sigma)}.$$

Only Steps 1–4 of the proof of Theorem RL enter here, so Theorem DC and Lemma 3.17 are not used. Section 3.7 uses this form.

Remark 3.18 (What reciprocity cannot give). Let $\mathfrak{m}$ be proper with $c_u \neq 1$ in $\mathcal{G}_{\mathfrak{m}}$, let $H := H_{\mathfrak{m}\infty_u}$, and let $y \in H^\times$ be totally positive. Then

$$Y(c) := \text{Art}(c)(y/\text{Art}(c_u)y) \quad (c \in C_{\mathfrak{m}})$$

satisfies every clause that $(\star)$ imposes on a family indexed by $C_{\mathfrak{m}}$. Yet $y = 1$ gives $Y \equiv 1$; and $y = \pi^2$ with $(\pi) = \mathfrak{Q}^{h(H)}$, $\mathfrak{Q}$ above a prime split completely in H , gives $v_{\mathfrak{Q}}(Y(1)) = 2h(H) \neq 0$. So a family compatible with $(\star)$ need not generate anything beyond K , nor consist of units. Generation needs an input about the numbers themselves, namely (N); the unit property needs local ideles that change the lattice, as in the proof of Theorem D.

3.5. Units at every finite prime. For maximal-order data Theorem D covers every finite prime, including 2, 3, ramified primes and primes dividing $\varepsilon - 1$. The proof has two parts. The *anchors* make a value a unit at $\mathfrak{q}$ whenever $\mathfrak{q}$ divides the annihilator of its label, using $(\star)$ for ideles at $\mathfrak{q}$. *Propagation* carries this to all other labels, using (vi) at the prime in question.

Throughout this subsection lattices are invertible O_K -ideals. Fix a totally positive $\varepsilon \in O_K^\times$ with $\varepsilon_r > 1$, and let $\mathcal{M}_\varepsilon$ be the set of admissible γ with $\text{tr } \gamma = \text{tr } \varepsilon$ and $\text{End } I_\gamma = O_K$, that is, the matrices of ε in fixed frames of invertible ideals. Their values form a finite set of algebraic numbers, by (vii) and Proposition 3.10. Let F be a number field containing these values, the values of the finitely many intermediate orders used below, μ_{24} , μ_N (where $N = N_\varepsilon$) and the square roots $\sqrt{|P|}$, $\sqrt{|Q|}$ below. Fix a rational prime p , a prime $\mathfrak{Q} \mid p$ of F with valuation v , and put $\mathfrak{q} := \mathfrak{Q} \cap O_K$. Write $G_\gamma = P \oplus Q$ for the decomposition into p -primary and prime-to- p parts. A label $x \in G_\gamma = t_\varepsilon^{-1}I_\gamma/I_\gamma$ has a component $x_{\mathfrak{q}'} \in K_{\mathfrak{q}'}/I_{\mathfrak{q}'}$ at each prime $\mathfrak{q}'$, nonzero if and only if $\mathfrak{q}' \mid \text{Ann}(x)$. In product identities only, we put $W(I, 0) := \varepsilon$; this is consistent with (ii) on the origin coset, which gives $\prod_{0 \neq h \in T} W(I, h) = 1$.

Proposition 3.19 (Idelic anchors). *Let $\gamma \in \mathcal{M}_\varepsilon$ and $x \in G_\gamma \setminus 0$ with $\mathfrak{q} \mid \text{Ann}(x)$. Then $W(I_\gamma, \varepsilon, x)$ and $E_\gamma(x)$ are $\mathfrak{Q}$ -units.*

Proof. Write $I = I_\gamma$ and $W(I, x) = W(I, \varepsilon, x)$.

Step 1: Local units. A finite idele supported at $\mathfrak{q}$ has its Artin image in the decomposition group of $\mathfrak{Q}$, which preserves v , and has exponent $+1$ in $(\star)$. For $a \in O_{\mathfrak{q}}^\times$ the idele a preserves I and replaces $x_{\mathfrak{q}}$ by $a^{-1}x_{\mathfrak{q}}$. So $v(W(I, x))$ is constant on $O_{\mathfrak{q}}^\times$ -orbits. Since $I_{\mathfrak{q}}$ is free over $O_{\mathfrak{q}}$, these orbits are the sets of labels with given other components and given $\mathfrak{q}$ -annihilator exponent.

Step 2: Uniformizer. Let s be the idele of a uniformizer at $\mathfrak{q}$, so that $s^{-1}I = J := \mathfrak{q}^{-1}I$. If $x_{\mathfrak{q}} = 0$, then $s^{-1}x$ and x define the same class in K/J : at $\mathfrak{q}$ both lie in $J_{\mathfrak{q}}$, and $s = 1$ elsewhere. Hence, for such a nonzero label,

$$(L) \quad v(W(J, x)) = v(\text{Art}(s)W(I, x)) = v(W(I, x)).$$

Step 3: Distribution. If $\mathfrak{q} \mid \text{Ann}(x)$, then $T := J/I \subset G_\gamma$ has order $N_{\mathfrak{q}}$, and Lemma 3.11(a) for $I \subset J$, extended to the origin coset by the convention above, gives

$$(D) \quad \prod_{h \in T} W(I, x + h) = W(J, x).$$

For $N_{\mathfrak{q}} = p^2$ this is two line products, through an intermediate lattice whose order may be non-maximal.

Step 4: Induction on $r := v_{\mathfrak{q}}(\text{Ann } x) \geq 1$. If $r = 1$, the coset $x + T$ contains exactly one label x_0 with $(x_0)_{\mathfrak{q}} = 0$, and the other $N_{\mathfrak{q}} - 1$ labels form a single $O_{\mathfrak{q}}^\times$ -orbit, of common valuation $a = v(W(I, x))$. Taking valuations in (D) and using (L) for $x_0 \equiv x \pmod{J}$,

$$v(W(I, x_0)) + (N_{\mathfrak{q}} - 1)a = v(W(J, x_0)) = v(W(I, x_0)),$$

so $a = 0$; if $x_0 = 0$, both outer terms are $v(\varepsilon) = 0$. If $r \geq 2$, all $N\mathfrak{q}$ labels $x+h$ have $\mathfrak{q}$ -exponent r and the same other components, so they form one orbit, of common valuation a . In J the label x has $\mathfrak{q}$ -exponent $r-1 \geq 1$, so the induction hypothesis and (D) give $N\mathfrak{q} \cdot a = v(W(J, x)) = 0$.

Finally $E_\gamma(x)^2 = W/q(x)$, and $q(x)$ is a root of unity. $\square$

The argument is the same for dyadic, ramified and inert $\mathfrak{q}$. It needs local ideles that change the lattice, through (L): the families Y of Remark 3.18 obey the reciprocity law at a single modulus and are not units.

Proof of Theorem D. Fix p , Ω and $\mathfrak{q}$ as above. By reflection (v) and $E(0)^2 = \varepsilon$, it suffices to show that every value is Ω -integral.

Step 1: Columns at every prime. Lemma 3.4 uses only (vi) with $\lambda \in \mu_{24}$, the orthogonality $b(P, Q) = 1$, and $p \nmid |Q|$; so it holds at every prime. If $\beta \in F^\times$ makes βE_γ integral on G_γ , then for each $c \in Q$ the column $f_c(y) := \beta E_\gamma(y + c)$, $y \in P$, has f_c and $\mathcal{F}_P f_c$ integral. If $P \neq 0$ then $v(\sqrt{|P|}) > 0$.

Step 2: $\mathfrak{q}$ is the only prime above p (inert or ramified), and $P \neq 0$. Suppose that some $E_\gamma(x)$ has a pole, and choose x with $v(E_\gamma(x)) = -A$ minimal, $A > 0$. If $x_P \neq 0$, then $\mathfrak{q} \mid \text{Ann}(x)$ and $E_\gamma(x)$ is a unit by Proposition 3.19; so $x \in Q$. Take $\beta := E_\gamma(x)^{-1}$ and $c := x$. Then $f_c(0) = 1$, while for $y \in P \setminus 0$ the label $y + x$ has nonzero P -component, so $E_\gamma(y + x)$ is an anchor unit and $v(f_c(y)) = A > 0$. Hence

$$\sqrt{|P|} \mathcal{F}_P f_c(w) = \sum_{y \in P} b(y, w) f_c(y) \equiv 1$$

is a unit for every w , and $v(\mathcal{F}_P f_c(w)) = -v(\sqrt{|P|}) < 0$, contradicting Step 1.

Step 3: $p = \mathfrak{q}\bar{\mathfrak{q}}$ splits, and $P \neq 0$. Since $\tau(\varepsilon - 1) = -\varepsilon^{-1}(\varepsilon - 1)$, the integer $R := v_{\mathfrak{q}}(\varepsilon - 1)$ equals $v_{\bar{\mathfrak{q}}}(\varepsilon - 1)$. So $P = P_{\mathfrak{q}} \oplus P_{\bar{\mathfrak{q}}} \cong (\mathbb{Z}/p^R)^2$, with the two crater lines $T_{\mathfrak{q}} := P_{\mathfrak{q}}[p]$ and $T_{\bar{\mathfrak{q}}} := P_{\bar{\mathfrak{q}}}[p]$, and $b|_P$ is nondegenerate. Let

$$A := \max(0, \max\{-v(E_\gamma(x)) : \gamma \in \mathcal{M}_\varepsilon, x \in G_\gamma\}),$$

a maximum over finitely many values, and suppose that $A > 0$ is attained at (γ, x) . By Proposition 3.19, $x_{\mathfrak{q}} = 0$. Put $\beta := E_\gamma(x)^{-1}$. By Step 1 and nondegeneracy, the column through x satisfies the hypothesis of Lemma 3.6, so the residue of βE_γ on $x + P[p]$ is a polynomial of total degree at most $p - 1$. Consider the point $x + s + t$ with $s \in T_{\mathfrak{q}}$, $t \in T_{\bar{\mathfrak{q}}}$. If $s \neq 0$, the label has nonzero $\mathfrak{q}$ -component, the value is an anchor unit, and the residue is 0; at $s = t = 0$ the residue is 1. In coordinates (X, Y) on $T_{\mathfrak{q}} \times T_{\bar{\mathfrak{q}}}$, a reduced polynomial vanishing wherever $X \neq 0$ is $(1 - X^{p-1})g(Y)$, and total degree at most $p - 1$ forces g to be constant, equal to 1. So the pole spreads along the whole crater line: $v(E_\gamma(x + t)) = -A$ for every $t \in T := T_{\bar{\mathfrak{q}}}$. (If $x \in T$, this already contradicts $v(E(0)) = 0$.) The line product (ii) gives $\prod_{t \in T} E_\gamma(x + t) = \kappa_T E_{\gamma_T}(\pi_T x)$ with $\kappa_T \in \mu_{24}$. The sibling lattice is $\bar{\mathfrak{q}}^{-1} I_\gamma$, again an invertible O_K -ideal, so $\gamma_T \in \mathcal{M}_\varepsilon$. Then $v(E_{\gamma_T}(\pi_T x)) = -pA < -A$: a pole p times larger, contradicting the choice of A .

Step 4: $P = 0$. Choose $k \geq 1$ with $\varepsilon^k \equiv 1 \pmod{pO_K}$. Then ε^k has nonzero p -primary part, Steps 2 and 3 make the values of $\mathcal{M}_{\varepsilon^k}$ units, and (iii) gives $E_{\gamma^k} = E_\gamma^k$ on G_γ . $\square$

Maximal orders enter twice: transitivity in the anchors needs $I_{\mathfrak{q}}$ free over $O_{\mathfrak{q}}$, and in Step 3 the sibling along a crater line must stay in $\mathcal{M}_\varepsilon$. The argument does not give units for arbitrary non-maximal data. The argument is not circular: $(\star)$ uses the local theory only at good split primes (Section 3.4), and propagation uses only (ii), (iii), (v)–(vii), Lemma 3.4 and Lemma 3.6. At the bad primes the argument rests on identity (6) of [RW], which through Lemma 3.4 carries the unit property from the anchor labels to the others.

3.6. Hilbert's twelfth problem. Theorem B needs two things. First, the maximal-order values must be indexed by ray classes, so that Fourier analysis on a ray class group measures the field they generate. Second, that Fourier transform must be nonzero at every odd primitive character; this is property (N), which no reciprocity statement can give (Remark 3.18).

Indexing. From here on lattices are invertible O_K -ideals. Fix a proper integral ideal $\mathfrak{m}$, let $\varepsilon_{\mathfrak{m}} > 1$ generate the totally positive units $\equiv 1 \pmod{\mathfrak{m}}$, and write $W(I, z) := W(I, \varepsilon_{\mathfrak{m}}, z)$. Let $\mathcal{P}_{\mathfrak{m}}$ be the set of pairs (I, z) , with I an invertible fractional ideal and $z \in K$ such that $\text{Ann}_{O_K}(z \bmod I) = \mathfrak{m}$, modulo $(I, z) \sim (I, z + i)$ for $i \in I$ and $(I, z) \sim (\beta I, \beta z)$ for $\beta \gg 0$. The annihilator condition reads $v_{\mathfrak{q}}(I) - v_{\mathfrak{q}}(z) = v_{\mathfrak{q}}(\mathfrak{m})$ for $\mathfrak{q} \mid \mathfrak{m}$, and $v_{\mathfrak{q}}(I) - v_{\mathfrak{q}}(z) \leq 0$ otherwise.

Lemma 3.20 (Torsor). (a) *Every class $z + I$ contains totally positive elements z_+ . The ideal $(z_+)\mathfrak{m}I^{-1}$ is prime to $\mathfrak{m}$, and its class $\tilde{\mathfrak{A}}(I, z) \in C_{\mathfrak{m}}$ depends only on the class of (I, z) in $\mathcal{P}_{\mathfrak{m}}$.*

(b) $\tilde{\mathfrak{A}} : \mathcal{P}_{\mathfrak{m}} \rightarrow C_{\mathfrak{m}}$ is a bijection.

(c) *For integral $\mathfrak{a}$ prime to $\mathfrak{m}$, $\tilde{\mathfrak{A}}(\mathfrak{a}^{-1}I, z) = [\mathfrak{a}] \tilde{\mathfrak{A}}(I, z)$.*

(d) *The image of $\tilde{\mathfrak{A}}(I, z)$ in $\mathcal{G}_{\mathfrak{m}}$ is $\mathfrak{A}_z := [(z_u)\mathfrak{m}I^{-1}]$ for any representative z_u positive at u .*

Proof. (a) The class $z + I$ is a translate of a full lattice in $\mathbb{R}^2$, so it meets the positive quadrant. Two totally positive representatives have ratio $\equiv 1 \pmod{\mathfrak{m}}$.

(b) For $\mathfrak{a}$ integral and prime to $\mathfrak{m}$, the pair $(\mathfrak{a}^{-1}\mathfrak{m}, 1)$ satisfies the annihilator conditions and has class $[(1)\mathfrak{m}\mathfrak{m}^{-1}\mathfrak{a}] = [\mathfrak{a}]$. So $\tilde{\mathfrak{A}}$ is onto.

For injectivity, a pair (I, z) is equivalent to $(J, 1)$ with $J := z_+^{-1}I$, and $\tilde{\mathfrak{A}}(J, 1) = [\mathfrak{m}J^{-1}]$. If $[\mathfrak{m}J^{-1}] = [\mathfrak{m}J'^{-1}]$, then $J' = \beta J$ with $\beta \gg 0$, $\beta \equiv 1 \pmod{\mathfrak{m}}$, and $(J', 1) \sim (J, \beta^{-1})$. It remains to see that $\beta^{-1} - 1 \in J$, prime by prime. For $\mathfrak{q} \mid \mathfrak{m}$, $v_{\mathfrak{q}}(\beta^{-1} - 1) \geq v_{\mathfrak{q}}(\mathfrak{m}) = v_{\mathfrak{q}}(J)$ by the annihilator condition for $(J, 1)$. For $\mathfrak{q} \nmid \mathfrak{m}$, $v_{\mathfrak{q}}(\beta^{-1} - 1) \geq \min(-v_{\mathfrak{q}}(\beta), 0)$, and both $-v_{\mathfrak{q}}(\beta) \geq v_{\mathfrak{q}}(J)$ and $0 \geq v_{\mathfrak{q}}(J)$ hold, by the annihilator conditions for $(J', 1)$ and for $(J, 1)$.

(c) The class $z + I$ lies in $z + \mathfrak{a}^{-1}I$, so z_+ serves for both pairs, and $(z_+)\mathfrak{m}(\mathfrak{a}^{-1}I)^{-1} = \mathfrak{a}(z_+)\mathfrak{m}I^{-1}$. The annihilator conditions persist, as $\mathfrak{a}$ is integral and prime to $\mathfrak{m}$.

(d) If z_u is positive at u , then $z_u/z_+ \equiv 1 \pmod{\mathfrak{m}}$ is positive at u , so the two ideals differ by the class 1 or c_r , which is trivial in $\mathcal{G}_{\mathfrak{m}}$. $\square$

So the maximal-order values with unit $\varepsilon_{\mathfrak{m}}$ and annihilator $\mathfrak{m}$ form a function $W(\mathfrak{c}) := W(\tilde{\mathfrak{A}}^{-1}(\mathfrak{c}))$ on $C_{\mathfrak{m}}$.

Corollary 3.21 (Sign law and Artin law). (a) *Let $\beta \in O_K$ with $\beta \equiv 1 \pmod{\mathfrak{m}}$. Then $W([\beta]\mathfrak{c}) = W(\mathfrak{c})^{\text{sgn } \beta_u}$. In particular $W(c_r\mathfrak{c}) = W(\mathfrak{c})$ and $W(c_u\mathfrak{c}) = W(\mathfrak{c})^{-1}$, so W descends to*

$$W_{\mathfrak{m}} : \mathcal{G}_{\mathfrak{m}} \rightarrow \mathbb{R}_{>0}, \quad W_{\mathfrak{m}}(c_u B) = W_{\mathfrak{m}}(B)^{-1}.$$

(b) $F_{\mathfrak{m}} := K(W_{\mathfrak{m}}(B) : B \in \mathcal{G}_{\mathfrak{m}}) \subseteq H_{\mathfrak{m}\infty_u}$, and $\text{Art}(b)W_{\mathfrak{m}}(B) = W_{\mathfrak{m}}(bB)$ for $b, B \in \mathcal{G}_{\mathfrak{m}}$.

(c) *Every maximal-order value with annihilator $\mathfrak{m}$ is $W(I, \varepsilon, x) = W_{\mathfrak{m}}(\mathfrak{A}_x)^k$, where $\varepsilon = \varepsilon_{\mathfrak{m}}^k$.*

Proof. (a) For $\mathfrak{c} = \tilde{\mathfrak{A}}(I, z)$ we have $\beta z \equiv z \pmod{I}$, so $[(\beta)]\mathfrak{c} = \tilde{\mathfrak{A}}(\beta^{-1}I, z)$, and (viii) gives $W(\beta^{-1}I, z) = W(I, \beta z)^{\text{sgn } \beta_u} = W(I, z)^{\text{sgn } \beta_u}$. The four sign patterns of β give 1, c_r , c_u , $c_r c_u$; for $c_r c_u$ only reflection enters, for c_r and c_u Theorem DC. Part (b) follows from (RL3) and Lemma 3.20(c). (c) A unit fixes x if and only if it is $\equiv 1 \pmod{\text{Ann } x}$; then use (iii). $\square$

Put $\mathbf{W}_{\mathfrak{m}} := W(\mathfrak{m}, 1) = W_{\mathfrak{m}}(1)$; the second equality holds because $\tilde{\mathfrak{A}}(\mathfrak{m}, 1)$ is the trivial class.

Fourier analysis on $\mathcal{G}_m$. Put $\ell_m := \log W_m$ and $\hat{\ell}_m(\chi) := \sum_B \bar{\chi}(B) \ell_m(B)$ for $\chi \in \hat{\mathcal{G}}_m$.

Lemma 3.22 (Field equals support). *Under the Artin map, the character group of $\text{Gal}(F_m/K)$, viewed inside $\hat{\mathcal{G}}_m$, is the subgroup generated by the support of $\hat{\ell}_m$.*

Proof. By the Artin law (Corollary 3.21(b)), $\text{Art}(b)$ fixes F_m if and only if $\ell_m(bB) = \ell_m(B)$ for all B , that is, if and only if $(\chi(b) - 1)\hat{\ell}_m(\chi) = 0$ for all χ . $\square$

So (N) asks that every odd primitive χ lie in the support of $\hat{\ell}_f$. This condition is insensitive to replacing ℓ by $-\ell$, $k\ell$, $\ell(B^{-1})$ or a translate, so it does not depend on sign or orientation conventions. It does depend on the place: the analogous function of the conjugate data, $C \mapsto Z'_{\tau f}(0, \text{pr } \tau C)$, is c_u -invariant.

Shintani's formula. Fix a proper $\mathfrak{f}$ and $C \in C_f$. Following Yamamoto [Ya08, §2.1], after Zagier's reduction theory [Zag75], choose an integral $\mathfrak{a} \in C$, a lattice $\mathfrak{b} = \mathbb{Z}\omega + \mathbb{Z}$ with $0 < \omega' < 1 < \omega$, where $\omega' := \tau\omega$, and $z \gg 0$ with $\mathfrak{a}\mathfrak{b} = (z)\mathfrak{f}$. Then $(\mathfrak{b}, z) \in \mathcal{P}_f$ and $\tilde{\mathfrak{A}}(\mathfrak{b}, z) = C$. Let $\omega_0 = \omega$, $\omega_n = b_n - 1/\omega_{n+1}$ with $b_n = \lceil \omega_n \rceil \geq 2$, be the purely periodic reduced continued fraction, of period ℓ . Let $(x_n, y_n) \in (0, 1) \times [0, 1)$ be the point z carried along it, defined by $x_0\omega + y_0 \equiv z \pmod{\mathfrak{b}}$, $x_{n+1} \equiv b_n x_n + y_n \pmod{\mathbb{Z}}$ and $y_{n+1} = 1 - x_n$, and put $\xi_n := x_n \omega_n + y_n$. Let $\varepsilon_+ > 1$ generate the totally positive units of O_K and write $\varepsilon_f = \varepsilon_+^r$; the sequence (x_n, y_n) has period $N := r\ell$ [Ya08, §2.1], [RW, §2.3], and indices are read modulo N . Shintani's formula [Sh, Prop. 3], in Yamamoto's form [Ya08, Thm. 5.1.1] (see also [RW, Prop. 4]), reads

$$X_1(C) X_2(C) = \exp h(C), \quad h(C) := -\zeta'(0, C) + \zeta'(0, c_r c_u C),$$

$$X_1(C) = \prod_{n=1}^N \mathcal{S}(\omega_n, \xi_n), \quad X_2(C) = \prod_{n=1}^N \mathcal{S}(\omega'_n, x_n \omega'_n + y_n).$$

Here $\zeta(s, C)$ is the partial zeta function of $C \in C_f$, and $\mathcal{S}(\omega, w)$ is Shintani's double sine function, a quotient of double gamma functions with periods $(1, \omega)$. For $\omega > 0$ and $0 < w < 1 + \omega$ it is the modulus of the basic factor of Radchenko and Wheeler's functions ([Sh, Prop. 5]; the proof of [RW, Prop. 5]).

Lemma 3.23 (First Shintani factor). *The product $\Gamma = M_0 \cdots M_{N-1}$, $M_n = \begin{pmatrix} b_n & -1 \\ 1 & 0 \end{pmatrix}$, is the matrix of ε_f in the fixed frame $(\omega, 1)$ of $\mathfrak{b}$, with attracting fixed point ω ; the point z has a label $\mathbf{u}$ in G_Γ ; and $X_1(C) = |E_\Gamma(\mathbf{u})|$. Hence*

$$X_1(C)^2 = W(\mathfrak{b}, z) = W_f(C).$$

The factor $X_2(C)$ is the corresponding modulus at the repelling fixed point ω' .

Proof. Step 1: The matrix. By Zagier's reduction theory [Zag75] (see the proof of [Ya08, Lemma 2.1.3]), $\omega_1 \cdots \omega_\ell = \varepsilon_+$, so $\omega_1 \cdots \omega_N = \varepsilon_+^r = \varepsilon_f$. Next, $M_n(\omega_{n+1}, 1)^\top = \omega_{n+1}(\omega_n, 1)^\top$, as $b_n \omega_{n+1} - 1 = \omega_{n+1} \omega_n$. So $\Gamma(\omega, 1)^\top = \omega_1 \cdots \omega_N (\omega, 1)^\top = \varepsilon_f(\omega, 1)^\top$, as $\omega_N = \omega_0 = \omega$. Applying τ gives $\Gamma(\omega', 1)^\top = \varepsilon_f^{-1}(\omega', 1)^\top$. Writing $\Gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we get $c\omega + d = \varepsilon_f$ and $c\omega' + d = \varepsilon_f^{-1}$, so $c(\omega - \omega') = \varepsilon_f - \varepsilon_f^{-1} > 0$ and $\text{tr } \Gamma = \varepsilon_f + \varepsilon_f^{-1} \geq 3$. Hence Γ is admissible with attracting fixed point ω , and it is the matrix of ε_f in $(\omega, 1)$, which is a fixed frame since $\omega > \omega'$.

Step 2: The label. Put $\mathbf{s} := (x_0, y_0)$ and $z_0 := x_0\omega + y_0 \equiv z \pmod{\mathfrak{b}}$. As $\text{Ann}(z \pmod{\mathfrak{b}}) = \mathfrak{f}$ and $\varepsilon_f \equiv 1 \pmod{\mathfrak{f}}$, we have $(\varepsilon_f^{-1} - 1)z_0 \in \mathfrak{b}$, and

$$(\varepsilon_f^{-1} - 1)z_0 = \mathbf{s}(\Gamma^{-1} - I)(\omega, 1)^\top.$$

So z_0 is the torsion point of label $\mathbf{u} := \mathbf{s}(\Gamma^{-1} - I) \in \mathbb{Z}^2$, and $u_1 = m := (d - 1)x_0 - cy_0$.

Step 3: The product. To factor $E_\Gamma(\mathbf{u})$ along the continued fraction, we follow the proof of [RW, Prop. 5]. For T near ω put $T_N := T$, $Z_N := x_0 T + y_0$, $T_{j-1} := b_{j-1} - 1/T_j$ and $Z_{j-1} := Z_j/T_j$. The product formula [RW, (20)] states that for all integers $m_1, \dots, m_{N-1}$

$$\Phi_{\Gamma, m_N, m_0}(Z_N; T) = \prod_{j=1}^N \Phi_S(Z_j + m_j T_j - m_{j-1}; T_j).$$

It is proved for T in the upper half plane and continues to a connected neighborhood of ω , since $c\omega + d > 0$ and every $T_j(\omega) = \omega_j$ is positive.

We now choose the integers m_j . At $T = \omega$ write $Z_j = u_j \omega_j + v_j$. Then $(u_N, v_N) = (x_0, y_0)$, $u_{j-1} = -v_j$ and $v_{j-1} = u_j + b_{j-1} v_j$, because $1/\omega_j = b_{j-1} - \omega_{j-1}$. By induction on $N - j$, using $y_j = 1 - x_{j-1}$ and $x_j \equiv b_{j-1} x_{j-1} + y_{j-1} \pmod{\mathbb{Z}}$, we get $u_j \equiv x_j$ and $v_j \equiv y_j \pmod{\mathbb{Z}}$. Choose $m_N := 0$ and $m_j := x_j - u_j \in \mathbb{Z}$ for $0 \leq j < N$. Then for $1 \leq j \leq N$

$$Z_j + m_j \omega_j - m_{j-1} = x_j \omega_j + v_j - (x_{j-1} + v_j) = x_j \omega_j + y_j - 1 = \xi_j - 1.$$

Moreover $Z_0 = z_0/\varepsilon_{\mathfrak{f}}$ has coordinates $(u_0, v_0) = \mathbf{s}\Gamma^{-1}$, so $m_0 = x_0 - u_0 = -m$.

As $x_j \in (0, 1]$ and $y_j \in [0, 1)$, the argument $\xi_j - 1$ lies in $(-1, \omega_j)$, which avoids the poles $\mathbb{Z}_{\geq 0} + \omega_j \mathbb{Z}_{\geq 1}$ and the zeros $\mathbb{Z}_{\leq -1} + \omega_j \mathbb{Z}_{\leq 0}$ of $\Phi_S(\cdot; \omega_j)$ [RW, §2]. Finally the pairs $(0, -m)$ and $(m, 0)$ are both compatible with $\mathbf{u}$, so the index quasi-periodicity of Section 2 replaces $\Phi_{\Gamma, 0, -m}$ by $\Phi_{\Gamma, m, 0}$ at z_0 . We obtain the exact identity

$$(3.1) \quad E_\Gamma(\mathbf{u})/\mu_\Gamma = \Phi_{\Gamma, m, 0}(z_0; \omega) = \prod_{j=1}^N \Phi_S(\xi_j - 1; \omega_j).$$

Step 4: Absolute values. Taking absolute values, $|E_\Gamma(\mathbf{u})| = \prod_j \mathcal{S}(\omega_j, \xi_j) = X_1(C)$. As $(\omega, 1)$ is a fixed frame with second vector 1, $X_1(C)^2 = w_\Gamma(\mathbf{u}) = W(\mathbf{b}, z)$, and this is $W_{\mathfrak{f}}(C)$ because $\tilde{\mathfrak{A}}(\mathbf{b}, z) = C$. The same computation at $T = \omega'$, where $T_j(\omega') = \omega'_j \in (0, 1)$ and $c\omega' + d > 0$, gives $|\Phi_{\Gamma, m, 0}(x_0 \omega' + y_0; \omega')| = \prod_j \mathcal{S}(\omega'_j, x_j \omega'_j + y_j) = X_2(C)$. The proof uses only that N is a period of (ω_n) and of (x_n, y_n) , so it applies verbatim to Γ^k , with kN factors. $\square$

The identification of the first factor is due to Radchenko and Wheeler [RW, Prop. 5]: their cocycle relation and product formulas [RW, (17), (19), (20)] factor E_Γ along the continued fraction into basic factors whose arguments stay off the pole and zero sets, at either fixed point. Our contribution here is the indexing that places this value at the class C , and the reading of $X_2(C)$ as a value (Section 4.1).

Proposition 3.24 (Nonvanishing via Shintani–Yamamoto). *For every proper $\mathfrak{f}$ and every primitive character χ of $\mathcal{G}_{\mathfrak{f}}$ with $\chi(c_u) = -1$,*

$$\hat{\ell}_{\mathfrak{f}}(\chi) = -\frac{n_{\mathfrak{f}}}{\pi} \mathbf{w}(\bar{\chi}) \sqrt{d_K N_{\mathfrak{f}}} L(1, \chi) \neq 0,$$

where $\mathbf{w}(\bar{\chi})$ is the root number of $\bar{\chi}$. In particular (N) holds.

Proof. Yamamoto's [Ya08, Cor. 5.2.5(1)], which follows from his Theorems 5.1.1 and 5.2.3 (the latter his relation for the class c_u) and the functional equation, gives

$$L(1, \chi) = -\frac{\pi \mathbf{w}(\chi)}{\sqrt{d_K N_{\mathfrak{f}}}} \sum_{C \in C_{\mathfrak{f}}} \bar{\chi}(C) \log X_1(C).$$

By Lemma 3.23, $\log X_1 = \frac{1}{2} \log W_{\mathfrak{f}}$, which is constant on c_r -cosets by the sign law (Corollary 3.21(a)); as $|\ker(C_{\mathfrak{f}} \rightarrow \bar{\mathcal{G}}_{\mathfrak{f}})| = 2/n_{\mathfrak{f}}$, we get $\sum_{C \in C_{\mathfrak{f}}} = (2/n_{\mathfrak{f}}) \sum_{B \in \mathcal{G}_{\mathfrak{f}}}$. Since $|\mathbf{w}(\chi)| = 1$ and $\mathbf{w}(\bar{\chi}) = \overline{\mathbf{w}(\chi)}$, we have $\mathbf{w}(\chi)^{-1} = \mathbf{w}(\bar{\chi})$. Hecke's theorem gives $L(1, \chi) \neq 0$. $\square$

This uses Shintani's formula at every class, including the boundary case $x_n = 1$, whose computation Yamamoto omits in [Ya08]; it is covered by [Ya10, Thms. 3.5, 4.10]. Appendix A derives (N), and the dictionary (V), alternatively from Kopp's limit formula [Ko, Thm. 1.1]; that route shares only Shintani's work and Hecke theory with this one.

Proof of Theorem B, given (N). By Theorem A, $K(\mathcal{V}_K) \subseteq K_r^{\text{ab}}$. Conversely, let ψ be a finite-order character of $\text{Gal}(K_r^{\text{ab}}/K)$; it is even at r . Suppose first that ψ is odd at u . Then its finite conductor $\mathfrak{f}$ is proper, since characters of $\text{Cl}_{\infty_u} = \text{Cl}$ are even at u , and ψ is primitive on $\mathcal{G}_{\mathfrak{f}}$. By (N) and Lemma 3.22, ψ is a character of $\text{Gal}(F_{\mathfrak{f}}/K)$, and $F_{\mathfrak{f}} = K(\mathbf{W}_{\mathfrak{f}})$ by the Artin law. If ψ is even at u , write $\psi = (\psi\chi_0)\chi_0^{-1}$ with χ_0 the character of $K(\sqrt{\sqrt{d_K}})/K$, which is odd at u . So the $\mathbf{W}_{\mathfrak{m}}$ generate K_r^{ab} , the fixed field of complex conjugation, and $K^{\text{ab}} = K_r^{\text{ab}}(i)$. $\square$

The deduction is the classical one (Section 1.1). Roblot [Ro] carries it out assuming Stark's conjecture, for the maximal totally real abelian extension. We do not claim $K(\mathbf{W}_{\mathfrak{m}}) = H_{\infty_u}$ for every $\mathfrak{m}$; this fails when $c_u = 1$ in a nontrivial $\mathcal{G}_{\mathfrak{m}}$.

3.7. Zauner SICs. For $M \geq 5$ let $\alpha = \begin{pmatrix} M-1 & -1 \\ 1 & 0 \end{pmatrix}$ and $\gamma = \alpha^3$, an admissible matrix with $N_{\gamma} = M^2(M-3)$, and let $K = \mathbb{Q}(\sqrt{(M-3)(M+1)})$, a real quadratic field. Let $\varepsilon_{\alpha} > 1$ be the eigenvalue of α , $\nu := \varepsilon_{\alpha}^{1/2}$ and $r_M := \sqrt{M+1} = \nu + \nu^{-1}$. On $\mathbb{C}^M$ let $D_{ij} = p_1^i q_1^j$ be the Weyl–Heisenberg displacement operators, $(p_1 f)(y) = f(y+1)$, $(q_1 f)(y) = \zeta_M^{-y} f(y)$, indexed by the M -torsion labels $h_{ij} = (i\varepsilon_{\alpha} + j)/M \in G_{\gamma}$, and put

$$P := M^{-1} \sum_{i,j} E_{\gamma}(h_{ij}) \theta(i,j) D_{ij}, \quad \theta(i,j) = (-1)^{i+j} e_{2M}(i^2 + j^2).$$

A *Weyl–Heisenberg SIC* in dimension M is a unit vector $\psi \in \mathbb{C}^M$, the *fiducial vector*, with $|\langle \psi, D_{ij} \psi \rangle|^2 = 1/(M+1)$ for all $(i,j) \neq (0,0)$; its M^2 translates $D_{ij} \psi$ then span equiangular lines. A *Clifford unitary* is a unitary that normalizes the group generated by the D_{ij} ; it permutes them, up to scalars, through a matrix of $\text{SL}_2(\mathbb{Z}/M)$, its *symplectic part*. *Zauner's matrix* is $F_Z := \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$, of order three; modulo M it agrees with the matrix F_z of [AFK, Def. 3.12]. Every element of order three in $\text{SL}_2(\mathbb{Z})$ is conjugate to F_Z or F_Z^{-1} , so the conclusion below does not depend on whether labels are read as rows or columns.

Two facts about P and K are needed. The first is Radchenko and Wheeler's *quadratic relation* [RW, Thm. 7, (39)]: $(P - \nu)(P + \nu^{-1}) = 0$ for every $M \geq 4$. They prove it from [RW, (36), (38), (5)], where (36) is [RW, (7)] combined with [RW, (5)]. The second is *field disjointness*: $K \cap \mathbb{Q}(\zeta_{2M}) = \mathbb{Q}$ for $M \geq 5$, by the conductor–discriminant formula.

Lemma 3.25 (Unit modulus gives a SIC). *Let $\sigma \in \text{Aut}(\overline{\mathbb{Q}})$ satisfy $|\sigma E_{\gamma}(h)| = 1$ for all $h \neq 0$ in $G_{\gamma}[M]$.*

- (a) *The matrix $\Pi := \sigma((\nu I - P)/r_M)$ is a Hermitian idempotent of trace one, $\Pi = |\psi\rangle\langle\psi|$, with $|\langle \psi, D_{ij} \psi \rangle|^2 = 1/(M+1)$ for $(i,j) \neq (0,0)$.*
- (b) *If σ also fixes $\mathbb{Q}(\zeta_{2M})$, then ψ is an eigenvector of an order-three Clifford unitary whose symplectic part is $\text{SL}_2(\mathbb{Z})$ -conjugate to $F_Z^{\pm 1}$.*

Proof. (a) By the quadratic relation, $\Pi_0 := (\nu I - P)/r_M$ satisfies $r_M^2 \Pi_0^2 = (\nu^2 + 1)I - (\nu + \nu^{-1})P = r_M^2 \Pi_0$, and $\text{tr} \Pi_0 = (M\nu - \nu^3)/r_M = 1$, as $\text{tr} P = E_{\gamma}(0) = \nu^3$. Both properties survive σ . Expanding, $\Pi = M^{-1}I + \sum_{(i,j) \neq 0} c_{ij} D_{i,j}$, where $\sigma(\zeta_M) = \zeta_M^a$ and $|c_{ij}| = |\sigma E_{\gamma}(h_{ij})|/(Mr_M) = 1/(Mr_M)$, since $|\sigma(r_M)| = r_M$ and $\sigma\theta \in \mu_{2M}$. Orthogonality

$\text{tr}(D_b^* D_c) = M \delta_{bc}$ gives

$$\text{tr}(\text{III}\Pi^*) = M \left(\frac{1}{M^2} + \frac{M^2 - 1}{(M + 1)M^2} \right) = 1.$$

For any idempotent Π of trace one, $\|\Pi - \Pi^*\|_{\text{HS}}^2 = 2 \text{tr}(\text{III}\Pi^*) - 2 \text{Re tr}(\Pi^2) = 2 \text{tr}(\text{III}\Pi^*) - 2$, so $\text{tr}(\text{III}\Pi^*) \geq 1$, with equality if and only if $\Pi = \Pi^*$. Here equality holds, so Π is a rank-one orthogonal projection, and $|\langle \psi, D_b \psi \rangle| = |\text{tr}(\Pi D_b)| = M/(Mr_M) = (M + 1)^{-1/2}$.

(b) The unit ε_α acts on the labels with order three and fixes E_γ by (vii). $\square$

Proof of Theorem C. By field disjointness we may choose σ with $\sigma|_K = \tau$ and σ trivial on $\mathbb{Q}(\zeta_{2M})$. The Galois form of Theorem A (Section 3.4), which uses only (RL1) and (RL2), gives $|\sigma W| = 1$ for $W = q(h)E_\gamma(h)^2$; since $q(h)$ is a root of unity, $|\sigma E_\gamma(h)| = 1$. Now apply Lemma 3.25. $\square$

Only the idele -1 at both real places matters here, and $\mathbb{Z}[\varepsilon_\alpha]$ need not be the maximal order.

3.8. Square roots of the values.

Proof of Theorem E. Let $d = (I, \varepsilon, z)$ have matrix γ and label x , choose a root of unity ρ with $\rho^2 = q(x)$, and put $Y := \rho E_\gamma(x)$; by (iv), Y is real with $Y^2 = W := W(d)$. Let e be the exponent of G_γ and $d_1 := \text{lcm}(24, 4e)$, so that $\rho^{d_1} = 1$. Let $n_B \in \mathbb{Z}$ with $d_1 \mid n_B$ and $n_B O_K \subseteq f_0(\varepsilon^{-1} - 1)O_K$, so that multiplication by any $\pi \equiv 1 \pmod{n_B O_K}$ fixes every label. Let H_{big} be a ray class field $H_{n'_B O_K \infty_u}$, $n_B \mid n'_B$, containing W (by (RL1)), $K(\sqrt{\varepsilon})$ and $K(\zeta_{d_1})^+$. Let L be the normal closure of $K(\sqrt{W})$ over K , and $L' := LH_{\text{big}}$; this field is real, since all K -conjugates of W are positive at r . We show that $Y \in H_{\text{big}}$.

Step 1: Choice of primes. Let $\sigma \in \text{Gal}(L'/H_{\text{big}})$. It agrees with the inversion $\zeta_{d_1} \mapsto \zeta_{d_1}^{-1}$ on $L' \cap K(\zeta_{d_1}) \subseteq K(\zeta_{d_1})^+ \subseteq H_{\text{big}}$, so Chebotarev's theorem on the compositum gives infinitely many good degree-one primes $\mathfrak{p}$ of K , split completely in H_{big} , with $p \equiv -1 \pmod{d_1}$ and Frobenius σ on L' . Splitting in H_{big} gives $\mathfrak{p} = (\pi)$ with $\pi \equiv 1 \pmod{n_B O_K}$ and $\pi_u > 0$, so $\text{Nm} \pi \equiv 1 \pmod{d_1}$. Since $p \equiv -1$, this forces $\text{Nm} \pi = -p$, and therefore $\pi_r < 0$. Splitting in $K(\sqrt{\varepsilon})$ makes ε a square modulo $\mathfrak{p}$, so k_0 divides $(p - 1)/2$, which is odd since $p \equiv 3 \pmod{4}$. As $p \equiv 2 \pmod{3}$, $\gcd(k_0, 24) = 1$.

Step 2: The constants are 1. On the origin coset, (ii) and reflection give $\kappa_T \in \mu_{p^2} \cap \mu_{24} = 1$ at the level $\gamma_0 = \gamma^{k_0}$. For an admissible A we have $\mu_{A^k} = \mu_A^k$. Indeed, applying the transformation law $\eta(Aw) = \mu_A \sqrt{j(A, w)} \eta(w)$ k times and comparing with the law for A^k gives $\mu_{A^k} \sqrt{j(A^k, w)} = \mu_A^k \prod_{i < k} \sqrt{j(A, A^i w)}$ on the upper half-plane, with principal square roots of numbers of positive imaginary part. The ratio of the two square-root factors is continuous with square 1, as $j(A^k, w) = \prod_{i < k} j(A, A^i w)$, hence constant; as w tends to the attracting fixed point every j tends to a positive real, so the ratio is 1. For $\mathfrak{q} \in \{\mathfrak{p}, \bar{\mathfrak{p}}\}$ and the line $T := \mathfrak{q}^{-1}I/I$ (for any order $\mathcal{O}$), let $\gamma_{\mathfrak{q}}$ be the matrix of ε on the sibling lattice, so that $\gamma_{\mathfrak{q}}^{k_0}$ is the sibling of γ_0 along T (Lemma 3.12). Then $(\mu_{\gamma}^p / \mu_{\gamma_{\mathfrak{q}}})^{k_0} = 1$, and $\gcd(k_0, 24) = 1$ gives $\mu_{\gamma_{\mathfrak{q}}} = \mu_{\gamma}^p$. Since $24 \mid p + 1$, $\mu_{\gamma} \mu_{\gamma_{\mathfrak{q}}} = 1$.

Step 3: The target values exactly. The sibling lattice is $\pi^{-1}I$ or $\bar{\pi}^{-1}I$, up to positive homothety, and $\pi^{-1}z \equiv z \pmod{\pi^{-1}I}$ because $(\pi - 1)z \in I$; likewise for $\bar{\pi}$. Apply Theorem 3.1 to the homotheties π^{-1} , with signs $(-, +)$ at (r, u) , and $\bar{\pi}^{-1}$, with signs $(+, -)$; these are elements of negative norm, so Theorem DC is the generator that enters. This gives $E_{\gamma_{\mathfrak{p}}}(x') =$

$\mu_\gamma \mu_{\gamma_{\bar{p}}} \overline{E_\gamma(x)}$ and $E_{\gamma_{\bar{p}}}(x'') = \mu_\gamma \mu_{\gamma_{\bar{p}}} E_\gamma(x)^{-1}$. By Step 2 the constants are 1:

$$E_{\gamma_{\bar{p}}}(x') = \overline{E_\gamma(x)}, \quad E_{\gamma_{\bar{p}}}(x'') = E_\gamma(x)^{-1}.$$

Step 4: Lemma 3.12 at the level of E . With Λ as in the discussion following Lemma 3.12, (ii), (iii) and (vii) on $x + T$ give $(E_\gamma(x)\Lambda)^{k_0} = \kappa E_{\gamma_q}(x')^{k_0}$, with κ a root of unity. Put $\zeta' := E_{\gamma_q}(x')/(E_\gamma(x)\Lambda)$. By real type and (ix), conjugation gives $\zeta'^2 \in \mu_{p^\infty}$; as $\zeta'^{k_0} = \kappa^{-1}$ has order prime to p , $\zeta'^2 = 1$. Here $\kappa = 1$ and k_0 is odd, so $\zeta' = 1$: the identity $E_{\gamma_q}(x') = E_\gamma(x)\Lambda$ holds exactly, before absolute values are taken.

Step 5: Congruence. At a prime $\mathfrak{P} \mid \mathfrak{p}$ of a field containing the values, (U0) of Theorem 3.2 makes all values units, and by (U2) the residue of E_{γ_0} is constant on $x + T$ for $T = T_{\mathfrak{p}}$ or $T = T_{\bar{\mathfrak{p}}}$. On that line $\Lambda \equiv E_{\gamma_0}(x)^\ell = E_\gamma(x)^{k_0\ell}$, so $E_{\gamma_q}(x') \equiv E_\gamma(x)^{1+k_0\ell} = E_\gamma(x)^p$. With Step 3, $E_\gamma(x)^p \equiv \overline{E_\gamma(x)} = \rho^2 E_\gamma(x)$ or $E_\gamma(x)^p \equiv E_\gamma(x)^{-1}$. Since $\rho^p = \rho^{-1}$, multiplying by ρ^p gives

$$Y^p \equiv Y \quad \text{or} \quad Y^p \equiv Y^{-1} \pmod{\mathfrak{P}}.$$

Step 6: Descent. Frobenius at the prime of L' below $\mathfrak{P}$ is σ , so $(\sigma Y - Y)(\sigma Y - Y^{-1})$ lies in infinitely many primes and is 0. If $\sigma Y = Y^{-1}$, then $W = \sigma W = W^{-1}$, so $W = 1$ and $Y = \pm 1 = Y^{-1}$. In every case $\sigma Y = Y$, so $Y \in H_{\text{big}} \subseteq K^{\text{ab}}$. $\square$

The square roots need not lie in $H_{\mathfrak{m}\infty_u}$: in $\mathbb{Q}(\sqrt{3})$ with $\mathfrak{m} = (5)$ or $\mathfrak{m} = \mathfrak{p}_{71}$, an exact computation shows that no $W_{\mathfrak{m}}(A)$ is a square there. The proof needs the identities at the level of E , including the phase in Theorem DC; the choice $p \equiv -1 \pmod{d_1}$ makes that phase exactly 1.

4. PART B: STARK UNITS AT A REAL PLACE

When $n_{\mathfrak{m}} = 2$, the Stark unit is the positive square root U of a value, and it needs its own reciprocity law (Theorem 4.5). Condition (St3) asks for $\sqrt{U} \in K^{\text{ab}}$. Theorem E gives this when $n_{\mathfrak{m}} = 1$, and Theorem 4.6, through the identity (W), gives it in the open configuration (Section 4.4).

4.1. The dictionary and the phase. We keep the notation of Shintani's formula in Section 3.6. The second Shintani factor is also a value: it is the place value at u of the same datum, which the place swap of Theorem 3.1 turns into a value of the conjugate datum.

Lemma 4.1 (Second Shintani factor). $X_2(C)^2 = W(\hat{\tau}C)$, where $\hat{\tau} : C_{\mathfrak{f}} \rightarrow C_{\tau\mathfrak{f}}$ is induced by τ . The map $\hat{\tau}$ exchanges c_r and c_u , and $\varepsilon_{\tau\mathfrak{f}} = \varepsilon_{\mathfrak{f}}$.

Proof. Each factor of the cycle product has modulus $\mathcal{S}(\omega'_n, \cdot)$ at ω' , so $X_2(C) = |\Phi_u(F)|$ for the frame $F = (\omega, 1)$. As $v_2 = 1$, the place swap gives

$$X_2(C)^2 = |\mathfrak{E}_u(\mathfrak{b}, z)|^2 = |\mathfrak{E}_r(\tau\mathfrak{b}, \tau z)|^2 = W(\tau\mathfrak{b}, \tau z),$$

and $\tilde{\mathfrak{A}}(\tau\mathfrak{b}, \tau z) = [\tau\mathfrak{a}] = \hat{\tau}C$. Passing from τF to a fixed frame of $\tau\mathfrak{b}$ uses only the fixed-point exchange (Lemma FX), not Theorem DC. Finally, τ swaps the sign patterns $(-, +)$ and $(+, -)$. $\square$

Shintani's formula therefore reads $|\mathfrak{E}_r(\mathfrak{b}, z)|^2 |\mathfrak{E}_u(\mathfrak{b}, z)|^2 = \exp 2h(C)$, a product of the two place values of one datum. The sign law (Corollary 3.21) separates them.

Theorem 4.2 (The dictionary). *For every proper $\mathfrak{f}$ and every $\mathfrak{c} \in C_{\mathfrak{f}}$,*

$$\log W(\mathfrak{c}) = h(\mathfrak{c}) + h(c_r \mathfrak{c}) = -n_{\mathfrak{f}} Z'_{\mathfrak{f}}(0, \text{pr } \mathfrak{c}).$$

This is (V).

The proof uses Shintani's formula in Yamamoto's form (including its boundary case), the fixed-point exchange and the sign law.

Proof. By Shintani's formula, $X_1^2 = W$ (Lemma 3.23) and Lemma 4.1, we have $\log W(C) + \log W(\hat{\tau}C) = 2h(C)$. At Cc_r the sign law gives $W(Cc_r) = W(C)$ for $\mathfrak{f}$ and, as $\hat{\tau}c_r = c_u$, $W(\hat{\tau}C c_u) = W(\hat{\tau}C)^{-1}$ for $\tau\mathfrak{f}$. Hence $\log W(C) - \log W(\hat{\tau}C) = 2h(Cc_r)$. Adding the two relations gives $\log W(C) = h(C) + h(Cc_r)$. If $c_r \neq 1$ (that is, $n_{\mathfrak{f}} = 1$), then $\zeta_{\mathfrak{f}}(s, \text{pr } C) = \zeta(s, C) + \zeta(s, Cc_r)$, and the right side is $-2Z'_{\mathfrak{f}}(0, \text{pr } C)$. If $c_r = 1$ (that is, $n_{\mathfrak{f}} = 2$), it is $2(-\zeta'(0, C) + \zeta'(0, c_u C)) = -2Z'_{\mathfrak{f}}(0, \text{pr } C)$. $\square$

The same exact product determines the *phase* of a value of [RW], and not only its modulus. Put

$$B_{22}(X \mid \tau, 1) := \frac{X^2 - (1 + \tau)X}{\tau} + \frac{\tau^2 + 3\tau + 1}{6\tau},$$

the second Bernoulli polynomial of the cone with periods $\tau, 1$, and $\mathfrak{S}(C) := \sum_{n=1}^N B_{22}(\xi_n \mid \omega_n, 1)$.

Theorem 4.3 (Modulus and phase). *In the setting of Shintani's formula (Section 3.6), $\mathfrak{S}(C) = 2\zeta(0, C) \in \mathbb{Q}$. For $k \geq 1$ let Γ be the matrix of $\varepsilon_{\mathfrak{f}}^k$ in the frame $(\omega, 1)$ and $\mathbf{u}$ the label of z . Then*

$$E_{\Gamma}(\mathbf{u})/\mu_{\Gamma} = \sqrt{W(\mathbf{b}, \varepsilon_{\mathfrak{f}}^k, z)} \cdot \exp(-\pi i k \zeta(0, C)), \quad \text{equivalently} \quad q(\mathbf{u})\mu_{\Gamma}^2 = e(k \zeta(0, C)).$$

Thus, for maximal-order data, the modulus of a value of [RW] is given by the derivative of a partial zeta function at $s = 0$, and its phase by the value there. The proof uses Shintani's formula, Shintani's evaluation of two-dimensional cone zeta functions at $s = 0$ [Sh76] (with [RW, Prop. 3]) and Hecke theory. Kopp [Ko, Props. 7.21–7.23] treats the phase of his cocycle values; a detailed comparison is not attempted here.

Proof. Step 1: The phase of one factor. Let $\tau > 0$, $X \in (0, 1 + \tau)$ and $\mathbf{b} = \sqrt{\tau}$. By the formula of Section 2, $\Phi_S(X - 1; \tau) = \varphi_{\mathbf{b}}(iy)$ with $y := (X - (1 + \tau)/2)/\mathbf{b} \in \mathbb{R}$, and

$$\log \varphi_{\mathbf{b}}(iy) = \int_{\mathbb{R}+i0} f(w) dw, \quad f(w) := \frac{e^{2yw}}{4w \sinh(\mathbf{b}w) \sinh(w/\mathbf{b})}.$$

The integral converges because $2|y| < \mathbf{b} + \mathbf{b}^{-1}$, which is the condition $0 < X < 1 + \tau$. To find the phase of $\varphi_{\mathbf{b}}(iy)$, we compute the imaginary part of this integral. The integrand is real on $\mathbb{R} \setminus 0$. From $\sinh(\mathbf{b}w) \sinh(w/\mathbf{b}) = w^2(1 + (\tau + \tau^{-1})w^2/6 + O(w^4))$, its Laurent expansion at 0 is $a_{-3}w^{-3} + a_{-2}w^{-2} + a_{-1}w^{-1} + O(1)$ with real coefficients and

$$a_{-1} = \frac{1}{4}(2y^2 - (\tau + \tau^{-1})/6).$$

Take the contour along $\mathbb{R}$ outside $(-\delta, \delta)$ and along the upper half circle of radius δ . The two rays contribute real numbers. On the half circle, oriented from $-\delta$ to δ , $\int w^{-1}dw = -\pi i$, while $\int w^{-2}dw = -2/\delta$ and $\int w^{-3}dw = 0$ are real, and the regular part contributes $O(\delta)$. Letting $\delta \rightarrow 0$, $\text{Im} \log \varphi_{\mathbf{b}}(iy) = -\pi a_{-1}$. Expanding $y^2 = (X^2 - (1 + \tau)X)/\tau + (1 + \tau)^2/(4\tau)$ gives

$$2y^2 - \frac{\tau + \tau^{-1}}{6} = 2 \frac{X^2 - (1 + \tau)X}{\tau} + \frac{3(1 + \tau)^2 - (1 + \tau^2)}{6\tau} = 2B_{22}(X \mid \tau, 1),$$

so $a_{-1} = \frac{1}{2}B_{22}(X \mid \tau, 1)$ and

$$\Phi_S(X - 1; \tau) \in \mathbb{R}_{>0} \cdot e(-B_{22}(X \mid \tau, 1)/4).$$

Step 2: The exact product. Let Γ_1 be the matrix of $\varepsilon_{\mathfrak{f}}$ in $(\omega, 1)$, so that $\Gamma = \Gamma_1^k$. The proof of Lemma 3.23, applied to Γ_1^k with the period kN , gives the identity (3.1) before absolute values

are taken, with kN factors $\Phi_S(\xi_n - 1; \omega_n)$, $\xi_n \in (0, 1 + \omega_n)$, the indices read modulo N . By Step 1 and Lemma 3.23,

$$E_\Gamma(\mathbf{u})/\mu_\Gamma = X_1(C)^k e(-k\mathfrak{S}(C)/4), \quad X_1(C)^k = \sqrt{W(\mathfrak{b}, \varepsilon_{\mathfrak{f}}^k, z)}.$$

Step 3: Rationality. By real type (iv), $\overline{E_\Gamma(\mathbf{u})} = q(\mathbf{u})E_\Gamma(\mathbf{u})$, so $q(\mathbf{u})E_\Gamma(\mathbf{u})^2 = |E_\Gamma(\mathbf{u})|^2$ and

$$(E_\Gamma(\mathbf{u})/\mu_\Gamma)^2 = |E_\Gamma(\mathbf{u})|^2/(q(\mathbf{u})\mu_\Gamma^2).$$

Comparing with Step 2 gives $q(\mathbf{u})\mu_\Gamma^2 = e(k\mathfrak{S}(C)/2)$. The left side is a root of unity, as q takes values in roots of unity and $\mu_\Gamma \in \mu_{24}$. Hence $k\mathfrak{S}(C)/2 \in \mathbb{Q}$. Since $\mathfrak{S}(C)$ is real, $\mathfrak{S}(C) \in \mathbb{Q}$. As every ω_n and ξ_n lies in K , $\mathfrak{S}(C)$ is an element of K . Being rational, it equals its conjugate $\mathfrak{S}'(C) := \sum_{n=1}^N B_{22}(x_n\omega'_n + y_n \mid \omega'_n, 1)$.

Step 4: Shintani's evaluation at $s = 0$. Shintani's cone decomposition, in the form [RW, Prop. 3] (Yamamoto's [Ya08, Prop. 2.1.4]), writes

$$\zeta(s, C) = (d_K^{1/2} \text{Nf})^{-s} \sum_{n=1}^N Z_{Q_n}(s, x_n, y_n), \quad Q_n(x, y) := \frac{(x\omega_n + y)(x\omega'_n + y)}{\omega_n - \omega'_n},$$

with $Z_Q(s, x, y) := \sum_{a, b \geq 0} Q(x + a, y + b)^{-s}$. Shintani's evaluation of these cone zeta functions at $s = 0$ [Sh76] is

$$Z_Q(0, x, y) = \frac{1}{4} [B_{22}(x\omega + y \mid \omega, 1) + B_{22}(x\omega' + y \mid \omega', 1)]$$

for $Q(x, y) = (x\omega + y)(x\omega' + y)/(\omega - \omega')$. Summing over n gives $\zeta(0, C) = (\mathfrak{S}(C) + \mathfrak{S}'(C))/4 = \mathfrak{S}(C)/2$ by Step 3. Substituting $\mathfrak{S}(C) = 2\zeta(0, C)$ into Steps 2 and 3 gives $E_\Gamma(\mathbf{u})/\mu_\Gamma = \sqrt{W(\mathfrak{b}, \varepsilon_{\mathfrak{f}}^k, z)} \exp(-\pi i k \zeta(0, C))$ and $q(\mathbf{u})\mu_\Gamma^2 = e(k\zeta(0, C))$. $\square$

4.2. Reduction to ray class fields and square-root reciprocity. We first normalize. Let $\mathfrak{m}$ have support S_f , with $|S| \geq 3$. For $A \in \mathcal{G}_{\mathfrak{m}}$, $\zeta_S(s, A) + \zeta_S(s, c_u A)$ is a partial zeta function of $\text{Cl}_{\mathfrak{m}}$. The characters of $\text{Cl}_{\mathfrak{m}}$ are even at both real places, so their $L_S(s, \chi)$ vanish to order ≥ 2 at $s = 0$. Hence

$$\zeta'_S(0, A) = \frac{1}{2} Z'_{\mathfrak{m}}(0, A),$$

and (St2) for $H_{\mathfrak{m}\infty_u}$ says that $U_A := \exp(-Z'_{\mathfrak{m}}(0, A))$ satisfies $\text{Art}(A)U_{[1]} = U_A$. For every admissible $\mathfrak{m}$, norms from $H_{\mathfrak{m}\infty_u}$ to H carry (St1)–(St3) for $U_{[1]}$ to (H, S) , by additivity of partial zeta functions; on both sides the number $|\mu(\cdot)|$ of roots of unity in the field is 2. If $n_{\mathfrak{m}} = 1$, then $U_A = W_{\mathfrak{m}}(A)$ by (V). If $n_{\mathfrak{m}} = 2$, then $W_{\mathfrak{m}}(A) = U_A^2$, and the Artin law for U_A is a separate statement.

A datum $d = (I, \varepsilon, z)$ is η -type if some $\eta \in \mathcal{O}_I^\times$ has $\text{Nm}\eta = -1$, $\eta_r < -1$, $\eta_u > 0$, $\eta^2 = \varepsilon$ and $\eta z \equiv z \pmod{I}$. For such data put $U(d) := +\sqrt{W(d)}$. Ideles preserve η -type data. Suppose $n_{\mathfrak{m}} = 2$ and $\mathfrak{m}$ is odd. Then the units $\equiv 1 \pmod{\mathfrak{m}}$ form an infinite cyclic group, since $-1 \not\equiv 1$. It contains a unit of signs $(-, +)$, since $n_{\mathfrak{m}} = 2$ means $c_r = 1$ in $C_{\mathfrak{m}}$. Signs are multiplicative, so that unit is an odd power of a generator, and the generator η with $|\eta_r| > 1$ has signs $(-, +)$. Moreover $\varepsilon_{\mathfrak{m}} = \eta^2$, as the totally positive units $\equiv 1$ are the even powers of η . Every datum $(I, \varepsilon_{\mathfrak{m}}, z)$ on an invertible $\mathcal{O}_K$ -ideal with annihilator $\mathfrak{m}$ is η -type, as $\eta \equiv 1 \pmod{\text{Ann } z}$, and $U = U_A$ for its class A , since $W = W_{\mathfrak{m}}(A) = U_A^2$ by (V).

Corollary 4.4 (Sign of E/μ). *If $c_r = 1$ or $c_u = 1$ in $C_{\mathfrak{f}}$, then $\zeta(0, C) = 0$ for every $C \in C_{\mathfrak{f}}$, since each $L(s, \chi)$ acquires a $\Gamma_{\mathbb{R}}$ -factor at an unramified real place. Hence, for data d on invertible $\mathcal{O}_K$ -ideals, $E_\gamma(\mathbf{u})/\mu_\gamma = +U(d) > 0$ if d is η -type, and $E_\gamma(\mathbf{u})/\mu_\gamma = +1$ if some unit $\equiv 1$ modulo the annihilator of z has signs $(+, -)$.*

Proof. If $c_r = 1$, every character χ of $C_{\mathfrak{f}}$ is even at r ; if $c_u = 1$, every χ is even at u . Then $L(s, \chi)$, with the Euler factors at the primes dividing $\mathfrak{f}$ removed, vanishes at $s = 0$: for nontrivial χ by the $\Gamma_{\mathbb{R}}$ -factor of its functional equation at that real place, and for trivial χ because $\zeta_K(0) = 0$. Hence $\zeta(0, C) = |C_{\mathfrak{f}}|^{-1} \sum_{\chi} \bar{\chi}(C) L(0, \chi) = 0$.

Now let $d = (I, \varepsilon, z)$ be a datum on an invertible O_K -ideal, $\mathfrak{f} := \text{Ann}(z)$ and $\varepsilon = \varepsilon_{\mathfrak{f}}^k$. By Lemma 3.20 the class C of d is represented by a pair $(\mathbf{b}, z)$ as in Shintani's formula. By the normalization (a) after Theorem 3.1, $E_{\gamma}(\mathbf{u})/\mu_{\gamma}$ is the same for both, since it is the place value $\mathfrak{E}_r$, which is invariant under translation and totally positive homothety. So Theorem 4.3 gives $E_{\gamma}(\mathbf{u})/\mu_{\gamma} = \sqrt{W(d)}$ whenever $\zeta(0, C) = 0$. If d is η -type, then $\eta \equiv 1 \pmod{\mathfrak{f}}$ has signs $(-, +)$, so $c_r = 1$ and $E_{\gamma}(\mathbf{u})/\mu_{\gamma} = \sqrt{W(d)} = U(d)$. If a unit $\beta \equiv 1 \pmod{\mathfrak{f}}$ has signs $(+, -)$, then $c_u = [(\beta)] = 1$, and Corollary 3.21(a) gives $W(I, \varepsilon_{\mathfrak{f}}, z) = W(I, \varepsilon_{\mathfrak{f}}, z)^{-1}$, so $W(d) = 1$ and $E_{\gamma}(\mathbf{u})/\mu_{\gamma} = 1$. $\square$

Theorem 4.5 (Square-root reciprocity). *Assume identity (6) of [RW]. For every η -type datum d , of every order, and every idele s ,*

$$\text{Art}_K(s)U(d) = U(s^{-1}d)^{\text{sgn } s_u}.$$

Hence $U(d) \in K^{\text{ab}}$, its conjugates above r are positive, and those above u lie on the unit circle. In particular, if $n_{\mathfrak{m}} = 2$, then $U_A \in H_{\mathfrak{m}\infty_u}$, $\text{Art}(B)U_A = U_{AB}$ for $B \in \mathcal{G}_{\mathfrak{m}}$, and every conjugate of U_A above r is a positive U_C . This last assertion also uses (V).

Proof. Fix ε , and let $\Omega_{\varepsilon}^{\eta} \subseteq \Omega_{\varepsilon}$ be the set of classes of η -type data with unit ε , where $\eta^2 = \varepsilon$ is fixed. It is stable under $C_{\mathfrak{M}}$, since $s_f^{-1}I$ has the same multiplier ring as I , and $\eta s_f^{-1}z - s_f^{-1}z = s_f^{-1}(\eta - 1)z \in s_f^{-1}I$. Let $\mathcal{S}_U := \{U(\omega) : \omega \in \Omega_{\varepsilon}^{\eta}\}$, a finite set of positive reals, algebraic since $U^2 = W$. By Step 5 of the proof of Theorem RL, $W(c_r\omega) = W(\omega)$ and $W(c_u\omega) = W(\omega)^{-1}$. So $U(c\omega)$ is defined for $c \in C_{\mathfrak{M}}/\langle c_r \rangle$, and $U(c_u\omega) = U(\omega)^{-1}$. We identify this group with $\text{Gal}(H_{\mathfrak{m}\infty_u}/K)$ by Art. For g in a Galois group over K let $c(g) \in C_{\mathfrak{M}}/\langle c_r \rangle$ correspond to the restriction of g to $H_{\mathfrak{m}\infty_u}$.

The field L_0 . Let L_0 be the normal closure over K of $K(\mathcal{S}_U)H_{\mathfrak{m}\infty_u}$, and fix an embedding of $\overline{\mathbb{Q}}$ into $\mathbb{C}$ above r . For $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/K)$ and $\omega \in \Omega_{\varepsilon}^{\eta}$, (RL1) and (RL3) of Theorem RL give $(\sigma U(\omega))^2 = \sigma W(\omega) = W(c(\sigma)\omega) > 0$, so $\sigma U(\omega)$ is real. The field $H_{\mathfrak{m}\infty_u}$ is real, since complex conjugation acts on it as $\text{Art}(c_r) = 1$. As L_0 is generated over K by the $\sigma U(\omega)$ and by $H_{\mathfrak{m}\infty_u}$, it is real, and being Galois over K it is real at every place above r . The values E , whose field need not have this property, enter only through local fields in Step 1.

Step 1: Frobenius at primes with k_0 even. Let $\omega \in \Omega_{\varepsilon}^{\eta}$ be the class of (I, z) , let γ be the matrix of ε in a fixed frame (v_1, v_2) of I , $\mathcal{O} := \text{End}(I)$, and $x := z/v_2 \in G_{\gamma}$. Let $L \supseteq L_0$ be a finite Galois extension of K containing the values of all admissible matrices of trace $\text{tr } \varepsilon$. Let $p = \mathfrak{p}\bar{\mathfrak{p}}$ be good for (ε, L) (Definition 3.13) and such that the order k_0 of ε modulo $\mathfrak{p}$ is even.

We first compute the orbits of η on the crater lines. For $\mathfrak{q} \in \{\mathfrak{p}, \bar{\mathfrak{p}}\}$ we use the setting for root extraction of Section 3.3. There $T := \mathfrak{q}_{\mathcal{O}}^{-1}I/I$ is a line, $T \cap G_{\gamma} = 0$, and $\eta \in \mathcal{O}$ acts on T by a scalar $\bar{\eta} \in (\mathcal{O}/\mathfrak{q}_{\mathcal{O}})^{\times} = \mathbb{F}_p^{\times}$, so that ε acts by $\bar{\eta}^2$, of order k_0 for both choices of $\mathfrak{q}$. If n is the order of $\bar{\eta}$, then $k_0 = n/\gcd(n, 2)$, so k_0 even forces $n = 2k_0$. Hence $\ell = (p-1)/k_0$ is even, and the $\langle \eta \rangle$ -orbits on T are $\{0\}$ and $\ell/2$ orbits of length $2k_0$, each the union of the ε -orbits of some h_j and of ηh_j . Let $\tilde{h}_j \in \mathfrak{q}_{\mathcal{O}}^{-1}I$ lift h_j .

Next we apply Lemma 3.11(b) with $u = \eta$, the lattices $I \subseteq I' := \mathfrak{q}_{\mathcal{O}}^{-1}I$, and z . Its hypotheses hold: $\eta I = I$, $\eta I' = I'$, $|\eta_r| > 1$, $\eta_u > 0$, $(\eta - 1)z \in I$, and $z \notin I'$ because $T \cap G_{\gamma} = 0$ and $z \notin I$. As $m(\eta) = 2$, we have $\mathcal{W}(I', \eta, z) = U(I', \varepsilon, z)$ and $\mathcal{W}(I, \eta, z) = U(I, \varepsilon, z)$. As $\eta^{2k_0} = \varepsilon^{k_0}$ is

totally positive, $\mathcal{W}(I, \eta^{2k_0}, \cdot) = \mathcal{W}(I, \varepsilon^{k_0}, \cdot)$. So

$$U(\mathfrak{q}_{\mathcal{O}}^{-1}I, \varepsilon, z) = U(I, \varepsilon, z) \prod_{j=1}^{\ell/2} W(I, \varepsilon^{k_0}, z + \tilde{h}_j),$$

an identity of positive reals. Its homothety step is (viii) at $\beta = \eta$, a case of Theorem DC.

Now let $\mathfrak{P} \mid \mathfrak{p}$ be a prime of L_0 and extend $v_{\mathfrak{P}}$, normalized by $v(p) = 1$, to a valuation v of a field $K_{\text{fam}} \supseteq L$ as in the setting of Theorem 3.2. By (U0) all values involved are v -units. By (U2) there is $\mathfrak{q} \in \{\mathfrak{p}, \bar{\mathfrak{p}}\}$ such that the residue of E_{γ_0} is constant on $x + T_{\mathfrak{q}}$. For this $\mathfrak{q}$, the second paragraph of the proof of Lemma 3.12 gives $W(I, \varepsilon^{k_0}, z + \tilde{h}_j) = w_{\gamma_0}(x + h_j) \equiv w_x^{k_0} = U(\omega)^{2k_0}$, so the identity above gives

$$U([\mathfrak{q}]\omega) \equiv U(\omega)^{1+k_0\ell} = U(\omega)^p,$$

where $[\mathfrak{q}]\omega$ is the class of $(\mathfrak{q}_{\mathcal{O}}^{-1}I, z)$ (Lemma 3.15(b)). Both sides lie in L_0 , so the congruence holds modulo $\mathfrak{P}$.

Let $g \in \text{Gal}(L_0/K)$, and suppose that infinitely many such primes $\mathfrak{p}$, of degree one, have a prime $\mathfrak{P} \mid \mathfrak{p}$ of L_0 with $\text{Frob}_{\mathfrak{P}} = g$. At each of them $gU(\omega) \equiv U(\omega)^p \equiv U([\mathfrak{q}]\omega) \pmod{\mathfrak{P}}$, and $[\mathfrak{q}]\omega \in \Omega_{\varepsilon}^{\eta}$. So the algebraic number $\prod_{y \in \mathcal{S}_U} (gU(\omega) - y)$ lies in infinitely many primes, and is 0. Hence $gU(\omega) \in \mathcal{S}_U$ is positive. Its square is $gW(\omega) = W(c(g)\omega)$ by $(\star)$, so

$$gU(\omega) = U(c(g)\omega) \quad (\omega \in \Omega_{\varepsilon}^{\eta}).$$

Step 2: These Frobenius elements generate. None of -1 , η , $-\eta$ is a square in K , since -1 and η are negative at r and $-\eta$ is negative at u . The product of any two of them is the third up to a square. So $K(i)$, $K(\sqrt{\eta})$ and $K(\sqrt{-\eta})$ are distinct quadratic extensions of K , and $L_{\eta} := K(i, \sqrt{\eta})$ has degree 4 over K . The place r is complex in $K(i)$ and in $K(\sqrt{\eta})$, and L_0 is real above r , so $L_0 \cap L_{\eta}$ is K or $K(\sqrt{-\eta})$.

Let $\tau_B \in \text{Gal}(L_{\eta}/K)$ fix i and move $\sqrt{\eta}$. If a degree-one prime $\mathfrak{p}$, unramified in L_{η} , has Frobenius τ_B on L_{η} , then $p \equiv 1 \pmod{4}$ and $\bar{\eta}$ is a non-square in $\mathbb{F}_p^{\times}$. Then the 2-part of the order of $\bar{\eta}$ equals that of $p - 1$, which is at least 4, so k_0 , the order of $\bar{\eta}^2$, is even.

Let Φ be the set of $g \in \text{Gal}(L_0/K)$ that extend to an element of $\text{Gal}(L_0 L_{\eta}/K)$ restricting to τ_B on L_{η} . Every $g \in \Phi$ satisfies the hypothesis of Step 1. Indeed, by Chebotarev's theorem [Ch] for $L_0 L_{\eta}/K$, each extension $\tilde{g}$ of g with $\tilde{g}|_{L_{\eta}} = \tau_B$ is the Frobenius of a prime of $L_0 L_{\eta}$ above infinitely many good degree-one primes $\mathfrak{p}$, and these have k_0 even. If $L_0 \cap L_{\eta} = K$, then $\Phi = \text{Gal}(L_0/K)$. If $L_0 \cap L_{\eta} = K(\sqrt{-\eta})$, then Φ is the set of g moving $\sqrt{-\eta} = i\sqrt{\eta}$, the nontrivial coset of an index-two subgroup, which generates $\text{Gal}(L_0/K)$.

Step 3: Assembly. The set of $g \in \text{Gal}(L_0/K)$ with $gU(\omega) = U(c(g)\omega)$ for all $\omega \in \Omega_{\varepsilon}^{\eta}$ is closed under products, as $\Omega_{\varepsilon}^{\eta}$ is $C_{\mathfrak{M}}$ -stable and c is a homomorphism. So it is a subgroup. It contains Φ by Steps 1 and 2, and is therefore all of $\text{Gal}(L_0/K)$. If g is trivial on $H_{\mathfrak{M}\infty_u}$, then $c(g) = 1$ and g fixes $\mathcal{S}_U$. Hence $\mathcal{S}_U \subset H_{\mathfrak{M}\infty_u} \subseteq K^{\text{ab}}$, and $\text{Art}(c)U(\omega) = U(c\omega)$ for all $c \in C_{\mathfrak{M}}$.

The idelic form follows as in Steps 1–4 of the proof of Theorem A. For an idele s , choose $\alpha \in K^{\times}$ with the signs of s at r and u , and put $s' := \alpha^{-1}s$. Then $\text{Art}_K(s) = \text{Art}_K(s')$, the class of s' in $C_{\mathfrak{M}}$ is that of s'_f , and $\text{Art}_K(s')U(d) = U(s'^{-1}_f d)$. Taking positive square roots in (viii) gives $U(\beta I, \varepsilon, \beta z) = U(I, \varepsilon, z)^{\text{sgn } \beta u}$ for η -type data, and β -homothety preserves η -type. With $\beta = \alpha^{-1}$ this gives $U(s^{-1}d)^{\text{sgn } s_u} = U(s'^{-1}_f d)$.

The conjugates of $U(d)$ above r are the $\sigma U(d)$ with $\sigma|_K = \text{id}$. They lie in $\mathcal{S}_U$ and are positive. If $\sigma|_K = \tau$, then $\sigma^{-1}\iota\sigma$, with ι complex conjugation, is the complex conjugation of the embedding σ , which lies above u . On $H_{\mathfrak{M}\infty_u}$ it is $\text{Art}(c_u)$, which inverts U . So $\iota\sigma U = \sigma(U^{-1})$, that is, $|\sigma U| = 1$.

Finally let $n_{\mathfrak{m}} = 2$, so that $\varepsilon = \varepsilon_{\mathfrak{m}} = \eta^2$ and every datum with annihilator $\mathfrak{m}$ on an invertible O_K -ideal is η -type. By (V), $U_A = U(\mathfrak{A}^{-1}(A'))$ for any $A' \in C_{\mathfrak{m}}$ above A . As in Corollary 3.21(b), the action of $C_{\mathfrak{m}}$ on these data factors through $C_{\mathfrak{m}}$, since ideles in $U_{\mathfrak{m}}^+$ fix them, and by Lemma 3.20(c) it is multiplication in $C_{\mathfrak{m}}$. With c_r acting trivially, the law $\text{Art}(c)U(\omega) = U(c\omega)$ gives $U_A \in H_{\mathfrak{m}\infty_u}$ and $\text{Art}(B)U_A = U_{AB}$ for $B \in \mathcal{G}_{\mathfrak{m}}$. In particular every conjugate of U_A above r is U_C for some C , and is positive. $\square$

4.3. Assembly. By Theorem D, every $W_{\mathfrak{m}}(B)$, and hence every $U_A = W_{\mathfrak{m}}(A)^{1/n_{\mathfrak{m}}}$, is a unit at every finite prime. By the Galois form of Theorem A (together with Theorem 4.5 when $n_{\mathfrak{m}} = 2$), $|\sigma U_A| = 1$ for every embedding σ above u . By Theorem E, $\sqrt{U_A} = W_{\mathfrak{m}}(A)^{1/2} \in K^{\text{ab}}$ when $n_{\mathfrak{m}} = 1$.

Suppose S_f contains a prime $\mathfrak{q} \mid 2$. Raising its exponent makes the order of the fundamental unit modulo $\mathfrak{q}^k$ divisible by arbitrarily high powers of 2, since a power of the fundamental unit lies in the pro-2 group $1 + \mathfrak{q}O_{K,\mathfrak{q}}$ and is not a root of unity. Then the units $\equiv 1$ are $\pm$ even powers of the fundamental unit (if $\pm \varepsilon_0^j \equiv 1$, then $\varepsilon_0^{2j} \equiv 1$, and the order condition forces j to be even), so they have norm $+1$, $c_r \neq 1$ and $n = 1$. If S_f contains no prime above 2, then $n_{\mathfrak{m}}$ does not depend on the exponents: raising the exponent of an odd prime $\mathfrak{q}$ passes to a subgroup of odd index of the cyclic group of units $\equiv 1$, because the kernel of $(O_K/\mathfrak{q}^{k'})^\times \rightarrow (O_K/\mathfrak{q}^k)^\times$ has odd order; an odd power of a unit of signs $(-, +)$ has the same signs. This leaves the *open configuration*: every admissible modulus has $n = 2$, and S_f contains no prime above 2. There U_A is the square root of a value, so $\sqrt{U_A}$ is a fourth root of a value, and Theorem E gives nothing.

Proof of Theorem F. Reduce to $H_{\mathfrak{m}'\infty_u}$ by norms. Membership and the Artin law come from Section 3.6 ($n = 1$) or Theorem 4.5 ($n = 2$). (St2) follows from (V) and the normalization of Section 4.2; (St1) from Theorem D and the unit-circle statement at the start of Section 4.3. (St3) follows from Theorem E when some admissible $\mathfrak{m}'$ has $n = 1$, and from Theorem 4.6 in the open configuration. $\square$

4.4. The square-root clause in the open configuration. Assume the open configuration, with η , $t = |\text{tr } \eta| \geq 3$, $f_\eta = [O_K : \mathbb{Z}[\eta]]$ and $H = H_{\mathfrak{m}\infty_u}$ as in Section 1.1. Since A and B will denote subgroups of labels, in Sections 4.4 and 4.5 we write the Stark unit U_A of a class of $\mathcal{G}_{\mathfrak{m}}$ as $U_{\mathfrak{c}}$, $\mathfrak{c} \in \mathcal{G}_{\mathfrak{m}}$; the notation U_x , for a label x , is defined below. Put $e_0 := -\eta$. It is a unit of K with $e_0^2 = te_0 + 1$, positive at r . We write $T := -\eta_r > 1$; this real number should not be confused with the line T of Lemma 3.12. Since K^{ab}/K is abelian and $\text{Art}(\mathfrak{b})U_{\mathfrak{c}} = U_{\mathfrak{cb}}$, (St3) for H is equivalent to $\sqrt{U_{[1]}} \in K^{\text{ab}}$.

Reciprocity alone does not suffice. The Frobenius congruences determine the square roots of the factors $W(I, \varepsilon^{k_0}, \cdot)$ only up to sign, so they fix U but not its square class. What is needed is a multiplicative identity among U -values whose other factors are visibly squares. The identity (W) is one, and three arithmetic steps turn it into the result.

Theorem 4.6 (The square-root clause). *Assume identity (6) of [RW] and (I1). In the open configuration, for every t , every splitting of 2, every class number of H and every f_η , we have $\sqrt{U_{\mathfrak{c}}} \in K^{\text{ab}}$ for every $\mathfrak{c} \in \mathcal{G}_{\mathfrak{m}}$, and (St3) holds.*

The proof also uses (V). It has four steps: the identity (W) on the lattice $\mathbb{Z}[\eta]$, an additive square-class law, a change of order from $\mathbb{Z}[\eta]$ to O_K , and Artin transport.

Step 1: The model identity (W) on $\mathbb{Z}[\eta]$. Let $J := \mathbb{Z}[\eta] = \mathbb{Z}\theta + \mathbb{Z}$ with $\theta := e_0$, a fixed frame. In it η has matrix $g = \begin{pmatrix} -t & -1 \\ -1 & 0 \end{pmatrix}$, and $\varepsilon = \eta^2$ has matrix $\gamma = g^2 = \begin{pmatrix} t^2+1 & t \\ t & 1 \end{pmatrix}$, so $G = G_\gamma \cong (\mathbb{Z}/t)^2$.

Its subgroups $A := G[\eta - 1]$ (the labels of η -type data) and $B := G[\eta + 1]$ have order t . In the coordinates of Radchenko and Wheeler, $A = \{u_1 + u_2 \equiv 0\}$ and $B = \{u_1 \equiv u_2\}$. Put $D := 2A$, $U_x := U(J, \varepsilon, x)$ for $x \in A \setminus 0$, and $U_0 := e_0$. The *Weyl kernel* is the partial Fourier transform of E along B , read on A :

$$M(c, d) := \mu_\gamma^{-1} \sum_{\beta \in B} b(d, \beta) E(c + \beta) \quad (c, d \in A).$$

Lemma 4.7 (The frame of $\mathbb{Z}[\eta]$). *Let $t \geq 3$ and $\varsigma := -\eta^{-1}$, which acts as -1 on A and as $+1$ on B .*

- (a) $\mu_\gamma = 1$, by Rademacher's formula [RG]: $(t^2 + 2)/(12t) - s(1, t) - 1/4 = 0$, with the Dedekind sum $s(1, t) = (t-1)(t-2)/(12t)$.
- (b) $q \equiv 1$ on A and on B , $b \equiv 1$ on $A \times A$ and on $B \times B$, $b((m, -m), (n, n)) = e_t(-2mn)$, and $B^\perp = B$.
- (c) $E(\varsigma x)E(x) = 1$ whenever $x, \varsigma x \neq 0$. Hence $E(-c + \beta)E(c + \beta) = 1$ for $c \in A \setminus B$ and $\beta \in B$.
- (d) $\overline{E(x)} = E(\eta x) = q(x)E(x)$ for $x \neq 0$. Hence E is real on $A \cup B$, and $E(c - \beta) = b(c, \beta)E(c + \beta)$ for $c \in A$, $\beta \in B$.
- (e) $E(x) = U_x > 0$ for $x \in A \setminus 0$.
- (f) M is real.

Proof. Here $N = t^2$, $\varepsilon = T^2$ at r , since $T^2 = tT + 1$, and $\gamma - I = t(\begin{smallmatrix} t & 1 \\ 1 & 0 \end{smallmatrix})$, so $\Lambda_\gamma = t\mathbb{Z}^2$ and labels are read modulo t .

(a) For $c > 0$, Rademacher's formula [RG] gives $\mu_\gamma = \exp(\pi i(\frac{a+d}{12c} - s(d, c) - \frac{1}{4}))$. Here $a + d = t^2 + 2$, $c = t$, $d = 1$, and $s(1, t) = (t-1)(t-2)/(12t)$, so the exponent is $\pi i((t^2 + 2 - (t-1)(t-2))/(12t) - \frac{1}{4}) = \pi i(3t/(12t) - \frac{1}{4}) = 0$.

(b) For $\mathbf{u} = (m, -m)$ one finds $\mathbf{u}\gamma = (m(t^2 - t + 1), m(t-1))$ and $\omega(\mathbf{u}\gamma, \mathbf{u}) = -m^2t^2$. For $\mathbf{u} = (n, n)$, $\omega(\mathbf{u}\gamma, \mathbf{u}) = n^2t^2$. So $e_{2N}(\omega(\mathbf{u}\gamma, \mathbf{u}))$ is $(-1)^m$, resp. $(-1)^n$, which cancels the sign $(-1)^{u_1+u_2+u_1u_2}$. Hence $q = 1$ on A and on B . Further $(m, -m)(\gamma - I) = (mt^2 - mt, mt)$ and $(n, n)(\gamma - I) = (nt^2 + nt, nt)$, so

$$b((m, -m), (m', -m')) = e_N(-mm't^2) = 1, \quad b((n, n), (n', n')) = e_N(nn't^2) = 1,$$

$$b((m, -m), (n, n)) = e_N(mnt^2 - 2mnt) = e_t(-2mn).$$

Thus $B \subseteq B^\perp$, and equality holds because b is nondegenerate and $|B| = t$.

For (c)–(e) we express E through $\Phi := \Phi_S(\cdot; T)$. Let $f_{1/2} := \delta g = (\begin{smallmatrix} -t & -1 \\ 1 & 0 \end{smallmatrix})$ be the half step, so that $\gamma = (\delta f_{1/2} \delta) f_{1/2}$, $f_{1/2} T = -T$ and $j(f_{1/2}, T) = T$. For a representative $\mathbf{u} \in \mathbb{Z}^2$ of a label $x \neq 0$, the torsion point is

$$z_{\mathbf{u}} := \frac{u_1 T + u_2}{\varepsilon^{-1} - 1} = -\frac{T}{t}(u_1 T + u_2) = -u_1 T - \frac{u_1 + u_2 T}{t},$$

and [RW, Lemma 2] allows any representative. The cocycle relation with $(g, h) = (\delta f_{1/2} \delta, f_{1/2})$ and any $n \in \mathbb{Z}$ gives

$$\Phi_{\gamma, u_1, 0}(z; T) = \Phi_{\delta f_{1/2} \delta, n, 0}(z/T; -T) \Phi_{f_{1/2}, u_1, n}(z; T).$$

The reflection formula turns the first factor into $\Phi_{f_{1/2}, 1-n, 1}(z/T; T)^{-1}$, and [RW, (19)] at $c = 1$ gives $\Phi_{f_{1/2}, m, l}(z; T) = \Phi(z + mT - l)$. Hence

$$(4.1) \quad \frac{E(x)}{\mu_\gamma} = \frac{\Phi(P_x)}{\Phi(Q_x)}, \quad P_x := z_{\mathbf{u}} + u_1 T - n, \quad Q_x := \frac{z_{\mathbf{u}}}{T} + (1 - n)T - 1,$$

for every representative $\mathbf{u}$ and every $n \in \mathbb{Z}$. As $z_{\mathbf{u}}/T^2 = z_{\mathbf{u}} + u_1T + u_2$, this can also be written $Q_x = P_x/T + T - 1 - u_1 - nt$.

(c) We have $\eta^2 + t\eta - 1 = 0$, so $\varsigma = -\eta^{-1} = \theta - t$, $\varsigma\theta = 1$ and $\varsigma \cdot 1 = \theta - t$, and $\varsigma_r = T - t = 1/T$. Hence if $\mathbf{u}$ represents x , then $(u_2, u_1 - tu_2)$ represents ςx , with torsion point $z_{\mathbf{u}}/T$. Take $n = 1 - u_2$ for x , so that $P_x = z_{\mathbf{u}} + u_1T + u_2 - 1$ and $Q_x = z_{\mathbf{u}}/T + u_2T - 1$, and $n = 1$ for ςx . Then $P_{\varsigma x} = z_{\mathbf{u}}/T + u_2T - 1 = Q_x$ and $Q_{\varsigma x} = z_{\mathbf{u}}/T^2 - 1 = P_x$. By (4.1), $E(\varsigma x)/\mu_{\gamma} = \mu_{\gamma}/E(x)$, and $\mu_{\gamma} = 1$ by (a). For $x = c + \beta$ with $c \in A \setminus B$, $\beta \in B$, we have $\varsigma x = -c + \beta$, and both are nonzero.

(d) By the index rules, $\Phi(w) = (1 - e(w))\Phi(w + T)$ and $\Phi(w - 1) = (1 - e(w/T))\Phi(w)$. Faddeev's unitarity reads $\Phi(v) = 1/\Phi(-v + T - 1)$ for real v . Take a representative $\mathbf{u}$ of $x \neq 0$ with n for x , and $-\mathbf{u}$ with $1 - n$ for $-x$. Then

$$\frac{\overline{E(x)} E(-x)}{|\mu_{\gamma}|^2} = \frac{\Phi(-Q_x + T - 1)}{\Phi(-P_x + T - 1)} \cdot \frac{\Phi(P_{-x})}{\Phi(Q_{-x})} = \frac{\Phi(w_1 + T)}{\Phi(w_1 + T - 1)} \cdot \frac{\Phi(w_2)}{\Phi(w_2 + T)} = \frac{1 - e(w_2)}{1 - e(w_1/T + 1)},$$

where $w_1 := -z_{\mathbf{u}}/T + (n - 1)T$ and $w_2 := -z_{\mathbf{u}} - u_1T + n - 1$, since $-Q_x + T - 1 = w_1 + T$, $-P_x + T - 1 = w_2 + T$, $P_{-x} = w_2$ and $Q_{-x} = w_1 + T - 1$. Now $e(w_1/T) = e(-z_{\mathbf{u}}/T^2) = e(-z_{\mathbf{u}} - u_1T) = e(w_2)$, so $\overline{E(x)} E(-x) = 1$. Applying (c) to $\eta x = \varsigma^{-1}(-x)$ gives $E(-x)E(\eta x) = 1$, and [RW, (5)] gives $1/E(-x) = q(x)E(x)$. So $\overline{E(x)} = E(\eta x) = q(x)E(x)$. By (b), E is real on $A \cup B$, and $E(c - \beta) = E(\eta(c + \beta)) = q(c + \beta)E(c + \beta) = b(c, \beta)E(c + \beta)$.

(e) Every element of $A \setminus 0$ is $x_{\ell} = (\ell, -\ell)$ with $1 \leq \ell \leq t - 1$. For $\mathbf{u} = (\ell, -\ell)$, $z_{\mathbf{u}} = -Tz_{\ell}$ with $z_{\ell} := \ell T/(T + 1)$, and $n = 0$ in (4.1) gives $P_{x_{\ell}} = z_{\ell}$ and $Q_{x_{\ell}} = -z_{\ell} + T - 1$. By Faddeev's unitarity,

$$E(x_{\ell})/\mu_{\gamma} = \Phi(z_{\ell}) \overline{\Phi(z_{\ell})} = |\Phi(z_{\ell})|^2 > 0.$$

Here $z_{\ell} \notin \mathbb{Z}T + \mathbb{Z}$, which contains the zeros and poles of Φ . The same computation holds for every integer $\ell \not\equiv 0 \pmod{t}$. As $E(x_{\ell})^2 = q(x_{\ell})^{-1}W(J, \varepsilon, x_{\ell}) = U_{x_{\ell}}^2$ by (b) and (iv), and $\mu_{\gamma} = 1$, we get $E(x_{\ell}) = U_{x_{\ell}}$.

(f) By (d), $\overline{M(c, d)} = \sum_{\beta} b(d, \beta)^{-1} b(c, \beta) E(c + \beta) = M(c, c - d)$, and the substitution $\beta \mapsto -\beta$ together with (d) returns $M(c, d)$. $\square$

Since $A \cap B = A[2]$ is killed by b against A and against B , $M(c, d)$ depends on c and d only modulo $A[2]$, and $a + A[2] \mapsto 2a$ identifies $A/A[2]$ with D . With $\kappa := (e_0 - 1)/\sqrt{e_0}$ put $N(a, b) := M(c, d)/\kappa$ for $a, b \in D$, where $2c = a$ and $2d = a + b$.

Theorem W (Model identity on $\mathbb{Z}[\eta]$). *Assume (I1). On J , for every $t \geq 3$, with $\kappa^2 = (e_0 - 1)^2/e_0 \in K$,*

$$(W) \quad M(c, d)^2 = \kappa^2 \cdot \frac{U_{2c} U_{2(d-c)}}{U_{2d}} \quad \text{whenever } 2c, 2d, 2(d-c) \neq 0;$$

equivalently, $N(a, b)^2 = U_a U_b / U_{a+b}$ for $a, b \in D$ with $a, b, a + b \neq 0$.

The identity (W) is not asserted on the excluded lines, and no argument uses them. The rest of Step 1 proves Theorem W.

For a matrix of $\text{SL}_2(\mathbb{Z})$ we write $\Phi_{m,n}$ for its functions of Section 2, with $q_{\tau} := e(\tau)$, $\tilde{q}_{\tau}$ and j as there. Put $\Phi := \Phi_S(\cdot; T)$, $q_T := e(T)$ and, in this subsection only, $s := T - 1$ and $z_{\ell} := \ell T/(T + 1)$. The point z_{ℓ} carries the label $x_{\ell} := (\ell, -\ell) \in A$, modulo the lattice; the representative used in [RW] is $-Tz_{\ell} = z_{\ell} - \ell T$. As $T^2 = tT + 1$, we have $z_{\ell}/s = \ell/t$ and $z_{\ell+t} = z_{\ell} + s$.

Radchenko and Wheeler prove their five-term relation [RW, (7)] by integrating against a kernel $1/D_{m,n}$, $D_{m,n} = \tilde{q}_{\tau}^n e(z/j) - q_{\tau}^m e(z)$, whose poles sit at the torsion points, and then shifting the contour by a lattice period. The difference of the shifted integrands is the integrand

of the Fourier identity [RW, (15)], and the difference of the integrals is a finite sum of residues. For γ at its fixed point, $\tilde{q} = q$; the zeros of $D_{m,n}$ then depend only on $m - n$ and survive the identity $\Phi_{m,n} - \Phi_{m+1,n+1} = D_{m,n} \Phi_{m+1,n}$. The η -type labels, however, belong to the *half step*

$$f_{1/2} := \delta g = \begin{pmatrix} -t & -1 \\ 1 & 0 \end{pmatrix},$$

with $\gamma = (\delta f_{1/2} \delta) f_{1/2}$, $f_{1/2} T = -T$ and automorphy factor $j(f_{1/2}, T) = T = E(0)$. At $\tau = T$ its function is $\Phi_{f_{1/2}, m, n}(z; T) = \Phi(z + mT - n)$, by [RW, (19)] at $c = 1$. There $\tilde{q}_T = q_T^{-1}$, so the zeros of $D_{m,n}$ depend on $m + n$, move under the index shift, and lie on B rather than on A . A different kernel is needed.

Lemma 4.8 (The A -kernel). *For any matrix of $\mathrm{SL}_2(\mathbb{Z})$ put $\mathcal{K}_{m,n} := 1 - q_\tau^m \tilde{q}_\tau^n e(z) e(z/j)$. Then*

$$\Phi_{m,n} + q_\tau^m e(z) \Phi_{m+1,n+1} = \mathcal{K}_{m,n} \Phi_{m+1,n},$$

and for $f_{1/2}$ at T , $\mathcal{K}_{m+1,n+1} = \mathcal{K}_{m,n}$. On J , $\mathcal{K} := \mathcal{K}_{0,0} = 1 - e(z) e(z/T)$ satisfies $\mathcal{K}(z + s) = \mathcal{K}(z)$. Its zeros are the z_ℓ ; they are all simple, they are the points with labels in A , and $\mathrm{Res}_{z_\ell}(1/\mathcal{K}) = -T/(2\pi i(T + 1))$.

Proof. For the first identity, add $\Phi_{m,n} = (1 - q^m e(z)) \Phi_{m+1,n}$ and $q^m e(z) \Phi_{m+1,n+1} = q^m e(z) (1 - \tilde{q}^n e(z/j)) \Phi_{m+1,n}$. For $f_{1/2}$ at T , $q_T \tilde{q}_T = 1$ gives $\mathcal{K}_{m+1,n+1} = \mathcal{K}_{m,n}$. On J , $\mathcal{K}(z) = 1 - e(z(T + 1)/T)$. Since $s(T + 1)/T = (T^2 - 1)/T = t$, we get $\mathcal{K}(z + s) = \mathcal{K}(z)$. The zeros are the z with $z(T + 1)/T \in \mathbb{Z}$, that is, the z_ℓ . There $\mathcal{K}' = -2\pi i(T + 1)/T$, which gives simplicity and the residue. Finally z_ℓ is the torsion point of label x_ℓ (proof of Lemma 4.7(e)), and these labels exhaust A . $\square$

Thus $1/\mathcal{K}$ has, for the half step, the diagonal invariance that the kernel of [RW] has for γ , and its poles lie on A .

The analytic input is (I1). It is the identity [RW, (15)]: for $v \in (0, T + 1)$ and $y \in (-1, T - v)$,

$$\int_{i\mathbb{R}-0} \frac{\Phi_{S,1,0}(z; T)}{\Phi(z + y)} e\left(\frac{zv}{T}\right) dz = e(1/8) \sqrt{T} e\left(-\frac{T + 1/T}{24}\right) \frac{\Phi(v + y)}{\Phi(y) \Phi(v - 1)},$$

together with its continuation, which Radchenko and Wheeler state (by adding residues). In continued form the identity holds for complex (v, y) with $\mathrm{Re} v > 0$, $\mathrm{Re}(v + y) < T$ and $y \notin \mathbb{Z}_{<0} + T\mathbb{Z}_{\leq 0}$, over any contour, vertical outside a compact set, that separates $R := \mathbb{Z}_{\geq 0} + T\mathbb{Z}_{\geq 0}$ (the poles of $\Phi_{S,1,0}$) from $L(y) := -y + \mathbb{Z}_{<0} + T\mathbb{Z}_{\leq 0}$ (the zeros of $\Phi(\cdot + y)$). The separating-contour form follows from the identity theorem. Contour independence is Cauchy's theorem. The integrand decays exponentially at $\pm i\infty$, locally uniformly in (v, y) ; at $-i\infty$ this follows from [RW, (13)]. The domain is connected, because the excluded set is a locally finite union of complex hyperplanes in a convex open set. The continuation is essential: the parameters below have $y < -1$.

Proposition 4.9 (Half-step five-term relation). *Let $1 \leq j \leq t - 1$ and $y_j := z_j - T$. For $k \in \{-1, 0, \dots, t - 2\}$ put $w'_k := (k - y_j)/s$, $v_k := w'_k T$ and $h_k(z) := e(w'_k z) \Phi(z) / \Phi(z + y_j + 1)$. Then*

$$\begin{aligned} \frac{T}{T + 1} \left[h_k(0) + h_k(z_{2t-j}) + \sum_{\substack{1 \leq \ell \leq t-1 \\ \ell \neq -j}} h_k(z_\ell) \right] \\ = (1 - e(-y_j)) e(1/8) \sqrt{T} e\left(-\frac{T + 1/T}{24}\right) \frac{\Phi(v_k + y_j)}{\Phi(y_j) \Phi(v_k - 1)}. \end{aligned}$$

Proof. Step 1: Combining g_k with its shift. Put $g_k := h_k/K$, $\Xi = e(z)$, $\Xi' = e(z/T)$ and $Y = e(y_j)$. We have $\mathcal{K}(z+s) = \mathcal{K}(z)$, $e(w'_k s) = Y^{-1}$ and $e(y_j)e(y_j/T) = q_T^{-1}$, and the index rules give

$\Phi(z) = (1-\Xi)\Phi_{S,1,0}(z)$, $\Phi(z+s) = (1-\Xi')\Phi_{S,1,0}(z)$, $\Phi(z+y_j+1) = (1-\Xi Y)\Phi_{S,1,-1}(z+y_j)$, and $\Phi(z+s+y_j+1) = \Phi_{S,1,0}(z+y_j) = (1-\Xi'/Y)\Phi_{S,1,-1}(z+y_j)$. These write $g_k(z) - g_k(z+s)$ as $e(w'_k z)\Phi_{S,1,0}(z)/(\mathcal{K}\Phi_{S,1,-1}(z+y_j))$ times

$$\frac{(1-\Xi)(1-\Xi'/Y) - Y^{-1}(1-\Xi')(1-\Xi Y)}{(1-\Xi Y)(1-\Xi'/Y)},$$

and the numerator is $(1-Y^{-1})(1-\Xi\Xi') = (1-e(-y_j))\mathcal{K}(z)$. The kernel cancels:

$$g_k(z) - g_k(z+s) = (1-e(-y_j))e(w'_k z)\Phi_{S,1,0}(z)/\Phi(z+y_j),$$

which is the integrand of (II) at (v_k, y_j) .

Step 2: The contour. All singularities are real. The sets R and $L(y_j)$ are disjoint since $y_j \notin \mathbb{Z}T + \mathbb{Z}$, so there is a separating contour $\mathcal{C}$ that avoids every z_ℓ , and the non-kernel singularities of g_k and of $g_k(\cdot+s)$ have winding number 0 about the strip between $\mathcal{C}$ and $\mathcal{C}+s$.

Step 3: Convergence. The function $|g_k|$ decays at $\pm i\infty$ if and only if $w'_k > 0$ and $w'_k + y_j/T < 1$, that is, $y_j < k < s + y_j/T$. As $\max_j y_j = -(2T+1)/(T+1) < -1$ and $\min_j (s + y_j/T) = T-2+1/(T+1) > t-2$, this holds for all listed k and j .

Step 4: Residues. The difference $\int_{\mathcal{C}} g_k - \int_{\mathcal{C}+s} g_k$, both contours oriented upward, is $-2\pi i$ times the sum of the kernel residues in the strip, one for each class ℓ modulo t . Along $z_\ell + s\mathbb{Z}$ the residue of g_k is constant, by Step 1, except where the orbit meets $R \cup L(y_j)$. This happens only for $\ell \equiv 0$, at $0 \in R$, and for $\ell \equiv -j$, at $-y_j - 1 \in L(y_j)$, whose representative in the strip is $T-2-y_j = z_{2t-j}$. These two terms are forced by the contour, as in the bookkeeping of [RW].

Step 5: Evaluation. The integral over $\mathcal{C}$ of the integrand in Step 1 is given by (II), continued. $\square$

Three translation facts turn both sides of Proposition 4.9 into values. Put $\psi(m) := \Phi(z_m)$.

- (1) Theta shift: $\Phi(z_m - s) = -e(z_m)q_T^{-1}\psi(m)$ for $m \neq 0$.
- (2) Right side: as k runs modulo t , the label of v_k runs once through a coset $c+B$ with $c \in A$, $2c = -x_j$, provided $j \in 2(\mathbb{Z}/t)$; this is automatic for t odd, while for t even $A+B$ has index 2. Moreover $E(c+\beta_k)\Phi(v_k+y_j)/\Phi(v_k-1) = \mu_\gamma$. With $c' := -c \notin B$, Lemma 4.7(c) turns $1/E(c+\beta_k)$ into $E(c'+\beta_k)$.
- (3) Moduli: $|\psi(\ell)|^2 = E(x_\ell) = U_{x_\ell}$ for $\ell \neq 0$; $|\Phi(z_{\ell+j} - s)| = \sqrt{U_{x_{\ell+j}}}$, by fact 1; and $|\mathfrak{E}(y_j)| = \sqrt{T}/\sqrt{U_{x_j}}$ for

$$\mathfrak{E}(y) := (1-e(-y))e(1/8)\sqrt{T}e(-(T+1/T)/24)\mu_\gamma/\Phi(y),$$

since $\Phi(y_j) = (1-e(y_j))\psi(j)$ by the index rule.

Proof of facts 1–3. We use the index rules in the form $\Phi(w) = (1-e(w))\Phi(w+T)$ and $\Phi(w-1) = (1-e(w/T))\Phi(w)$, and the identities $z_m + z_m/T = m$ and $1/T = T-t$.

Fact 1. The two rules give $\Phi(w-s) = (1-q_T^{-1}e(w))\Phi(w+1)$ and $\Phi(w) = (1-e((w+1)/T))\Phi(w+1)$. At $w = z_m$ we have $e((z_m+1)/T) = e(-z_m)e(T-t) = q_T e(-z_m)$. With $X := q_T^{-1}e(z_m)$, which is not 1 as $z_m - T \notin \mathbb{Z}$, this gives $\Phi(z_m-s)/\psi(m) = (1-X)/(1-X^{-1}) = -X$.

Fact 2. From $(T-1)(T+1) = tT$ and $y_j = jT/(T+1) - T$ one computes

$$v_k = \frac{(k-y_j)T}{T-1} = \frac{(k+1-j+t)T + (k+1)}{t}.$$

Let $\mathbf{u}^{(k)} := (t-1-k, j-1-k-t)$. By the formula for $z_{\mathbf{u}}$ in the proof of Lemma 4.7, $z_{\mathbf{u}^{(k)}} + u_1^{(k)}T = -(u_1^{(k)} + u_2^{(k)}T)/t = v_k - 1$. So v_k is, modulo $\mathbb{Z}T + \mathbb{Z}$, the torsion point of label $\mathbf{u}^{(k)} \equiv (-1-k, j-1-k)$. Consecutive labels differ by $(-1, -1) \in B$, so they run once through a coset of B ; writing $(-1-k, j-1-k) = (a, -a) + (b, b)$ forces $2a = -j$, so the coset meets A if and only if $j \in 2(\mathbb{Z}/t)$, and then $c = (-j/2, j/2)$ has $2c = (-j, j) = -x_j$. Now take $\mathbf{u}^{(k)}$ and $n = 0$ in (4.1). Then $P = v_k - 1$ and $Q = P/T + T - 1 - u_1^{(k)} = (v_k - 1)/T + T - t + k = w'_k + k$, using $1/T = T - t$. On the other hand $v_k + y_j = w'_k + w'_k(T - 1) + y_j = w'_k + k$. Hence $E(c + \beta_k)/\mu_\gamma = \Phi(v_k - 1)/\Phi(v_k + y_j)$.

Fact 3. For $\ell \neq 0$, the proof of Lemma 4.7(e) gives $|\psi(\ell)|^2 = E(x_\ell)/\mu_\gamma = U_{x_\ell}$. By fact 1, $|\Phi(z_{\ell+j} - s)| = |\psi(\ell + j)|$ when $\ell + j \neq 0$. Finally $\Phi(y_j) = \Phi(z_j - T) = (1 - e(y_j))\psi(j)$ by the first index rule, $|1 - e(-y)| = |1 - e(y)|$ for real y , and $e(1/8)$, $e(-(T+1/T)/24)$ and μ_γ have modulus 1; so $|\mathfrak{C}(y_j)| = \sqrt{T}/|\psi(j)|$. $\square$

Proof of Theorem W. Since $e(w'_k z_\ell) = e(k\ell/t)e(-y_j\ell/t)$, the left side of Proposition 4.9 is a Fourier transform over ℓ modulo t . Let $\tilde{\ell} \equiv \ell$ denote the representative used, which is $2t - j$ for the class $-j$. By fact 2 the relation reads

$$\frac{T}{T+1} \sum_{\ell \bmod t} e(k\ell/t) e(-y_j\tilde{\ell}/t) \mathcal{A}(\ell) = \mathfrak{C}(y_j) E(c' + \beta_k), \quad \mathcal{A}(\ell) := \frac{\Phi(z_{\tilde{\ell}})}{\Phi(z_{\tilde{\ell}} + y_j + 1)}.$$

We invert over k . Let $d = (m, -m) \in A$. As $\beta_{k+1} - \beta_k = (-1, -1)$ and $b((m, -m), (n, n)) = e_t(-2mn)$, we have $b(d, \beta_k) = b(d, \beta_0)e_t(2mk)$, and $\sum_k e_t((2m + \ell)k)$ over the t values of k is t if $\ell \equiv -2m$ and 0 otherwise. Since $\mu_\gamma = 1$, $M(c', d) = \sum_k b(d, \beta_k)E(c' + \beta_k)$. Multiplying the relation by $b(d, \beta_k)$ and summing over k gives

$$M(c', d) = \zeta \cdot t \frac{T}{T+1} e(-y_j\tilde{\ell}/t) \mathcal{A}(\ell) / \mathfrak{C}(y_j), \quad \zeta := b(d, \beta_0) \in \mu_t, \quad x_\ell = -2d.$$

The forced classes $\ell \equiv 0$ and $\ell \equiv -j$ are the cases $2d = 0$ and $2(d - c') = 0$.

For the other ℓ we compute moduli. Here $z_\ell + y_j + 1 = z_{\ell+j} - s$, so $\mathcal{A}(\ell) = \psi(\ell)/\Phi(z_{\ell+j} - s)$. By fact 3 and $t^2T^2/(T+1)^2 = T\kappa^2$,

$$|M(c', d)|^2 = \kappa^2 \frac{U_{x_j} U_{x_\ell}}{U_{x_{\ell+j}}} = \kappa^2 \frac{U_{2c'} U_{2(d-c')}}{U_{2d}} \quad (U_{-x} = U_x^{-1}).$$

The function M is real (Lemma 4.7(f)) and the right side is positive, so $M^2 = |M|^2$. Finally, as j and ℓ run through $2(\mathbb{Z}/t) \setminus 0$ and $2(\mathbb{Z}/t) \setminus \{0, -j\}$ respectively, (c', d) runs through the whole domain of (W) modulo $A[2] \times A[2]$, and both sides depend only on $2c'$ and $2d$. $\square$

Step 2: The additive square-class law. Put $F := K^{\text{ab}}$ and $\mathcal{V} := F^\times / F^{\times 2}$, written additively.

Lemma 4.10 (Square classes). *Assume identity (6) of [RW] and that (W) holds on J .*

- (a) Every U_x , every value $E(x)$ on J , every $M(c, d)$ and $N(a, b)$, $\sqrt{e_0}$ and κ lie in F .
- (b) The map $\chi : D \rightarrow \mathcal{V}$, $\chi(x) := [U_x]$, is a homomorphism.
- (c) χ vanishes on the odd part D_{odd} , on $2D$ and, when $4 \mid t$, at the element of order 2 of D .

Proof. (a) For $x \in G \setminus 0$, (J, ε, x) is a datum, so $\sqrt{W(J, \varepsilon, x)} \in F$ by Theorem E (for every order). This gives U_x , and $E(x) = \pm\sqrt{W}/\nu$ with $\nu^2 = q(x)$ a root of unity. Moreover $K(\sqrt{e_0})/K$ is abelian, μ_γ and b are roots of unity, and $e_0 \neq 1$.

(b) If $a, b, a+b \neq 0$, then (W) says that $U_a U_b / U_{a+b}$ is the square $N(a, b)^2$ in F . If $a = 0$, then $\chi(0) = [e_0] = 0$ by (a). If $b = -a$, reflection gives $U_a U_{-a} = 1$.

(c) The group $\mathcal{V}$ has exponent 2, so $\chi(2w) = 2\chi(w) = 0$, and every element of D_{odd} is twice an element of D_{odd} . If $h = -h$ has order 2, reflection gives $W(J, \varepsilon, h)^2 = 1$, so $U_h = 1$. $\square$

Step 3: The orbit-product order change. The identity (W) is proved on J , while the Stark units live on O_K . Lemma 3.11(b), applied to $J \subseteq O_K$ with the unit η itself, relates the two. The action of η on O_K/J makes every term except the η -fixed ones a visible square.

Lemma 4.11 ($\mathbb{Z}[\eta]$ inside O_K). (a) $J = \mathbb{Z} + f_\eta O_K$, and $\eta \equiv n \pmod{f_\eta O_K}$ with $n \in \mathbb{Z}$ and $n^2 \equiv -1 \pmod{f_\eta}$. Thus η acts on $O_K/J \cong \mathbb{Z}/f_\eta$ as multiplication by n .
 (b) $4 \nmid f_\eta$, $\gcd(f_\eta, t) \mid 2$, and if f_η is even then $4 \mid t$.
 (c) Every $\langle \eta \rangle$ -orbit on O_K/J has 1 or 4 elements. The fixed set C_η is $\{0\}$ if f_η is odd, and $\{0, y_2\}$, with y_2 of order 2, if f_η is even. When $4 \mid t$, $C_\eta \subseteq D$.
 (d) $A_J := (\eta - 1)^{-1}J/J$ is cyclic of order t , and the map $A_J \rightarrow A_O := (\eta - 1)^{-1}O_K/O_K$ is an isomorphism on odd parts.

Proof. (a) $-1 = \eta\eta' \equiv n^2$ modulo $f_\eta O_K \cap \mathbb{Z} = f_\eta \mathbb{Z}$.

(b) The congruence $n^2 \equiv -1$ excludes $4 \mid f_\eta$. Next, $-t = \text{tr } \eta \equiv 2n$ with n a unit modulo f_η . If f_η is even, then $(\eta + 1)/2 \in O_K$ has norm $-t/4$.

(c) We have $n^4 \equiv 1$, and an orbit of length ≤ 2 requires $(n^2 - 1)y = -2y \equiv 0$. For $y \in C_\eta$, $(\eta - 1)y \in J$, so $C_\eta \subseteq A_J = A$. For $4 \mid t$, y_2 is the element of order 2 of the cyclic group A_J , which lies in $2A_J = D$.

(d) Both groups have order $|\text{Nm}(\eta - 1)| = t$, $A_J \cong \mathbb{Z}[x]/(x^2 + tx - 1, x - 1)$, and the kernel is killed by $\gcd(f_\eta, t)$. $\square$

Proposition 4.12 (Order change). For $z \in K$ with $(\eta - 1)z \in J$ and $z \notin O_K$, as positive reals,

$$U(O_K, \varepsilon, z) = \prod_{h \in C_\eta} U(J, \varepsilon, z + h) \cdot \prod_P W(J, \varepsilon^2, z + y_P),$$

where the second product runs over the $\langle \eta \rangle$ -orbits P of length 4 on O_K/J , and $y_P \in P$. The W -factors are squares in F by Theorem E.

Proof. Apply Lemma 3.11(b) with $u = \eta$, $I = J$ and $I' = O_K$; here $\eta J = J$, $\eta O_K = O_K$, $\eta_r < -1$ and $\eta_u > 0$. As $m(\eta) = 2$, $\mathcal{W}(O_K, \eta, z) = U(O_K, \varepsilon, z)$. A fixed point contributes $\mathcal{W}(J, \eta, z + h) = U(J, \varepsilon, z + h)$, and an orbit of length 4 contributes $\mathcal{W}(J, \eta^4, \cdot) = W(J, \varepsilon^2, \cdot)$, since η^4 is totally positive. $\square$

Here Lemma 3.11(b) uses (ii) along $J \subset O_K$, (iii), and homothety at $\beta = \eta$, a case of Theorem DC. It decomposes $z + O_K$ into J -cosets, and no Euler factor appears.

Step 4: Conclusion and Artin transport.

Proof of Theorem 4.6. Since $\mathfrak{m}$ is odd and $\eta \equiv 1 \pmod{\mathfrak{m}}$, the group A_O contains an element of exact annihilator $\mathfrak{m}$, of odd order. By Lemma 4.11(d) it lifts to $z \in D_{\text{odd}} \setminus 0$, represented by some $z \in (\eta - 1)^{-1}J \setminus O_K$, and $U(O_K, \varepsilon, z) = U_{\mathfrak{c}}$ for its class $\mathfrak{c}$ (Section 3.6 with (V)). By Proposition 4.12, in $\mathcal{V}$,

$$[U_{\mathfrak{c}}] = \sum_{h \in C_\eta} [U(J, \varepsilon, z + h)] = \sum_{h \in C_\eta} (\chi(z) + \chi(h)) = 0,$$

by Lemma 4.10, since $C_\eta \subseteq D$ (if f_η is even then $4 \mid t$), $z \in D_{\text{odd}}$, and h is 0 or of order 2. So $\sqrt{U_{\mathfrak{c}}} \in F$ for this $\mathfrak{c}$. For $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/K)$, Theorem 4.5 gives $\sigma U_{\mathfrak{c}} = U_{\mathfrak{c}'}$, where $\mathfrak{c}'$ is the product of $\mathfrak{c}$ and the class of $\sigma|_H$, and $\sigma \sqrt{U_{\mathfrak{c}}} \in F$ squares to $U_{\mathfrak{c}'}$. So $\sqrt{U_{\mathfrak{c}'}} \in K^{\text{ab}}$ for every $\mathfrak{c}'$, in particular for $\mathfrak{c}' = [1]$, which is (St3) for H . $\square$

Only the lattice $\mathbb{Z}[\eta]$ enters, whatever f_η and the class number, and only two instances of (W) are used: $\chi(z) = 2\chi(w)$ for $2w = z$ is (W) at $(c, 2c)$ with $4c = z$, and $\chi(z + y_2) = \chi(z) + \chi(y_2)$ is (W) at one further pair when f_η is even. Nothing here places the roots in H rather than in K^{ab} .

4.5. The root law. The root law $-\eta U_{\mathfrak{c}} \in H^{\times 2}$ asks for more than Theorem 4.6: a root in H itself, hence control of signs and not only of square classes, and on maximal-order lattices rather than on $\mathbb{Z}[\eta]$.

Proposition 4.13 (The target). *If 2 is unramified in K , then (St3) holds if and only if $U_{\mathfrak{c}} \in H^{\times 2} \cup (-\eta)H^{\times 2}$ for one, equivalently every, $\mathfrak{c}$. If 2 ramifies in K , then $-\eta U_{\mathfrak{c}} \in H^{\times 2}$ still implies (St3).*

Proof. If $-\eta U_{\mathfrak{c}} \in H^{\times 2}$, then $\sqrt{U_{\mathfrak{c}}} \in H(\sqrt{-\eta})$, which is abelian over K ; this gives the second sentence and one direction of the first. For the converse let 2 be unramified in K , and suppose $L = H(\sqrt{U_{\mathfrak{c}}})$ is abelian over K and $U_{\mathfrak{c}} \notin H^{\times 2}$. We first show that $L \subseteq H_{4m\infty_u}$. The extension L/H is unramified at the odd primes, as $U_{\mathfrak{c}}$ is a unit there. So at a prime dividing $\mathfrak{m}$ the inertia group of L/K meets $\text{Gal}(L/H)$ trivially and has the same upper filtration as that of H/K . Above 2, where H/K is unramified, the conductor exponent is that of a quadratic extension generated by the square root of a unit, which is at most 2. Finally, L is real above r , as $U_{\mathfrak{c}} > 0$ there.

Since $\mathfrak{m}$ is odd and the units $\equiv 1 \pmod{\mathfrak{m}}$ and positive at u are the powers of η , we have

$$\text{Gal}(H_{4m\infty_u}/H) \cong (O_K/4)^{\times} / \langle \bar{\eta} \rangle.$$

The 2-part of $(O_K/4)^{\times}$ is C_2^2 , and $\bar{\eta}$ has nontrivial 2-part, since otherwise η would be congruent modulo 4 to a root of unity of odd order, and $\text{Nm } \eta \equiv 1 \pmod{4}$. So the 2-part of the quotient has order 2, and $H_{4m\infty_u}$ contains exactly one quadratic extension of H . The field $H(\sqrt{-\eta})$ is one. Indeed, $K(\sqrt{-\eta})$ has conductor dividing $4\infty_u$, and $\sqrt{-\eta} \notin H$. Otherwise $K(\sqrt{-\eta})/K$ would be unramified above 2, $-\eta$ would be a square times an element of $1 + 4O_{K,2}$, and, taking norms, -1 would be a square modulo 4. Hence $L = H(\sqrt{-\eta})$. $\square$

So where 2 is unramified, the root law is (St3) together with $U_{\mathfrak{c}} \notin H^{\times 2}$; where 2 ramifies, it is strictly stronger.

The case t odd. As $t^2 + 4 \equiv 5 \pmod{8}$, 2 is inert in K , f_η is odd, and H is unramified above 2. On an invertible O_K -ideal the ideals $(\eta - 1)$ and $(\eta + 1)$ are coprime, since their sum contains 2 and their norms $\pm t$ are odd. Hence $G = A \oplus B$. Corollary 4.4 gives $E/\mu_\gamma = +U$ on $A \setminus 0$ and $E/\mu_\gamma = +1$ on $B \setminus 0$. Theorem DC, applied to the matrix of η (which commutes with γ , has determinant -1 and $j = \eta_r < 0$), gives $E(\eta x) = \mu_\gamma^2 \overline{E(x)}$.

For $\beta \in B \setminus 0$ this gives $E(-\beta) = \mu_\gamma$, and reflection gives $\mu_\gamma^2 = q(\beta)^{-1}$. Here $q(\beta)$ has odd order, as $q(n\beta) = q(\beta)^{n^2}$, and lies in μ_{2t} , hence in μ_t , while $\mu_\gamma^2 \in \mu_{12}$. So $\mu_\gamma^2 \in \mu_3$, and $\mu_\gamma^2 = 1$ unless $3 \mid t$. If $3 \mid t \neq 3$, choose β of order t . This is possible because the group $B \cong O_K/(\eta + 1)$ is cyclic, as no rational integer $n > 1$ divides $\eta + 1$: if a rational prime ℓ divides $\eta + 1$ in O_K , it divides $\eta' + 1$, hence $(\eta + 1)(\eta' + 1) = \text{tr } \eta$ and $(\eta + 1) + (\eta' + 1) = \text{tr } \eta + 2$, so $\ell \mid 2$ and $\ell \mid \text{tr } \eta$, which is impossible as $t = |\text{tr } \eta|$ is odd. Then $q(3\beta) = q(\beta)^9$ gives $\mu_\gamma^{16} = 1$, so again $\mu_\gamma^2 = 1$. If $t = 3$, then $K = \mathbb{Q}(\sqrt{13})$, $O_K = \mathbb{Z}[\eta]$ has class number 1, every invertible ideal I is a totally positive multiple $\alpha\mathbb{Z}[\eta]$, and in the fixed frame $\alpha(\theta, 1)$, which we use, $\mu_\gamma = 1$ by Rademacher's formula (Lemma 4.7(a)).

So $q \equiv 1$ on B . On A , $E(x) = \overline{E(x)} = q(x)E(x)$ gives $q \equiv 1$. Thus $b \equiv 1$ on $A \times A$ and on $B \times B$, and $b : A \times B \rightarrow \mu_t$ is perfect, since b is nondegenerate. Finally, (vi) at $y \in B$, inverted

over this pairing as in Step 0 of the proof of Theorem 4.14, gives $\sum_{\beta \in B} E(\beta) = \lambda(e_0 + (t-1)\mu_\gamma)$, where $e_0 = -\eta = E(0)$. The left side is $e_0 + (t-1)\mu_\gamma$, which is not 0, since otherwise $\mu_\gamma = \pm 1$ and e_0 would be rational. So $\lambda = 1$, and $\mu_\gamma = \lambda = 1$ by [RW, (6)].

The *digit* of a unit x at a dyadic place w of H is its class in $O_w^\times/O_w^{\times 2}(1+4O_w)$; it is trivial if and only if $H_w(\sqrt{x})/H_w$ is unramified.

Theorem 4.14 (Dyadic digit). *Assume identity (6) of [RW], and identity (7) of [RW] at the three families of pairs of labels listed in the proof. In the open configuration with t odd, on an invertible O_K -ideal, $\text{digit}_w(U_c) = \text{digit}_w(-\eta)$ for every class c and every dyadic place w of H . Hence $H(\sqrt{-\eta U_c})/H$ is unramified everywhere.*

Proof. Let I be an invertible O_K -ideal, γ the matrix of $\varepsilon = \eta^2$ in a fixed frame (for $t = 3$, the frame chosen above), $G = G_\gamma = A \oplus B$ as above, $e_0 := -\eta = E(0)$, so that $e_0^2 = te_0 + 1$, and $U_c := E(c)$ for $c \in A$, with $U_0 := e_0$. By the discussion before the theorem, $\mu_\gamma = \lambda = 1$, $q \equiv 1$ on A and on B , $b \equiv 1$ on $A \times A$ and on $B \times B$, $b : A \times B \rightarrow \mu_t$ is perfect, $E = U > 0$ on $A \setminus 0$ and $E = 1$ on $B \setminus 0$.

We use three families of instances of the identities of [RW], transported to the attracting point as in Section 2. In them $F_\gamma^+ = E$, and F_γ^- agrees with E off the origin while $F_\gamma^-(0) = e_0^{-1}$ [RW, (2), (4)]. Since $|G| = t^2$, the normalization $\sqrt{N}$ of [RW] is t , and $\langle x; y \rangle = b(x, y)$.

- (i) The five-term relation [RW, (7)] at the pairs $(y, -v)$, $(y, v) \in B \times B \setminus \{(0, 0)\}$:

$$\sum_{x \in G} b(x, y) \frac{F^+(x)}{F^-(x-v)} = t \frac{F^+(y-v)}{F^-(y)F^-(v)}.$$

- (ii) The form [RW, (36)] of the pentagon relation, which Radchenko and Wheeler deduce from their identities (4)–(7), at pairs $(a, a') \in A \times A$ with $a, a', a + a' \neq 0$ (it is reduced modulo 2 in Step 4):

$$\frac{1}{t} \sum_{s \in G} E(s)q(s)E(a-s)b(s, a') = q(a+a')E(-a-a')E(a)E(a').$$

- (iii) The identity [RW, (36)] at the pairs $(2u, -u)$, $u \neq 0$, which is the identity in the proof of [RW, Thm. 10]:

$$tE(2u) = E(u)^4 + E(u)^2 \sum_{s \neq 0} q(s)E(u+s)E(u-s).$$

By Theorem D every value is a unit at every prime above 2. Let L be a number field containing H , the values on I and μ_t , and let a bar denote reduction modulo a prime $\mathfrak{P} \mid 2$ of L . Since t is odd, $\bar{e}_0$ is a root of $X^2 + X + 1$, so $\bar{e}_0 \in \mathbb{F}_4 \setminus \mathbb{F}_2$, $\bar{e}_0^3 = 1$ and $1 + \bar{e}_0 = \bar{e}_0^2$.

Step 0: Square roots and symmetries. Since $|G|$ is odd, $q(x)^{n^2} = q(nx) = 1$ for n the order of x , so $q(x)$ has odd order and a unique square root $q(x)^{1/2}$ of odd order. These roots are multiplicative. Put $\Psi := q^{1/2}E$. Then $\Psi = E$ on $A \cup B$. Theorem DC, applied to the matrix of η (which commutes with γ , has determinant -1 and $j = \eta_r < 0$), together with real type (iv), gives $E(\eta x) = \mu_\gamma^2 \overline{E(x)} = q(x)E(x)$. With reflection, $q(\eta x) = q(x)^{-1}$. Hence $\Psi(\eta x) = \Psi(x)$, and $\Psi(x)\Psi(-x) = 1$ for $x \neq 0$. As η is 1 on A and -1 on B , for $c \in A$ and $\beta \in B$

$$\Psi(c - \beta) = \Psi(c + \beta), \quad \Psi(c + \beta) = b(c, \beta)^{1/2}E(c + \beta) = b(c/2, \beta)E(c + \beta),$$

using $q(c + \beta) = q(c)q(\beta)b(c, \beta) = b(c, \beta)$. Here $c/2$ is the half in A , which has odd order.

We use the following sums over the coset $c + B$. Put

$$T(c) := \sum_{\beta \in (B \setminus 0)/\pm} \Psi(c + \beta),$$

$$\Sigma^+(c) := \sum_{\beta \in B} \Psi(c + \beta) = U_c + 2T(c), \quad \Sigma^-(c) := \sum_{\beta \in B} \Psi(c + \beta)^{-1} = U_c^{-1} + 2T(-c),$$

the last because $\Psi(c + \beta)^{-1} = \Psi(-c - \beta)$. In terms of the Weyl kernel $M(c, d) = \sum_{\beta} b(d, \beta)E(c + \beta) = \sum_{\beta} b(d - c/2, \beta)\Psi(c + \beta)$ we have $\Sigma^+(c) = M(c, c/2)$, $\Sigma^-(c) = M(-c, -c/2)$, and, replacing β by $-\beta$, $M(c, d) = M(c, c - d)$.

Finally, as $b(c + \beta, \beta') = b(c, \beta')$ and $q(\beta') = 1$ for $\beta, \beta' \in B$, identity (vi) at $y = \beta' \in B$ reads $\sum_{c \in A} b(c, \beta')M(c, 0) = t\lambda E(-\beta')$. Inverting over the perfect pairing $A \times B \rightarrow \mu_t$ gives, for $c \in A \setminus 0$,

$$M(c, 0) = \sum_{\beta \in B} E(c + \beta) = \lambda \left(e_0 + \sum_{\beta' \neq 0} b(c, \beta')^{-1} \right) = e_0 - 1.$$

Step 1: The digit criterion. Let w be a dyadic place of H ; it is unramified over 2. Every unit of O_w can be written as $\rho(1 + 2a)$ with ρ a root of unity of odd order, which is unique, and $a \in O_w$, unique modulo 2. The map $\rho(1 + 2a) \mapsto \bar{a} \in k_w$ is a homomorphism onto the additive group k_w , whose kernel $\rho(1 + 4O_w)$ is $O_w^{\times 2}(1 + 4O_w)$, since squares $\rho^2(1 + 4a + 4a^2)$ lie in the kernel and ρ is a square. So it identifies the digit with $\bar{a}$, and the digit is additive. Write $t = 1 + 2D$ and $e_0 = \rho_0(1 + 2y)$, where ρ_0 is a primitive cube root of unity; the residue field contains $O_K/2 = \mathbb{F}_4$. Modulo 4, $e_0^2 \equiv \rho_0^2 = -1 - \rho_0$ and $te_0 + 1 \equiv \rho_0 + 1 + 2\rho_0(D + y)$, so $-2(1 + \rho_0) \equiv 2\rho_0(D + y)$ and $\bar{D} + \bar{y} = \bar{e}_0^{-1}(1 + \bar{e}_0) = \bar{e}_0$. Hence

$$\text{digit}(te_0) = \bar{D} + \bar{y} = \bar{e}_0.$$

Now let $u \in A$ with $\text{Ann}(u) = \mathfrak{m}$. In (iii), $q(u) = 1$ and $q(u + s)q(u - s) = q(s)^2$, so $q(s)E(u + s)E(u - s) = \Psi(u + s)\Psi(u - s)$. Pairing s with $-s$ gives

$$tU_{2u} = U_u^4 + 2U_u^2 S(u), \quad S(u) := \sum_{s \in (G \setminus 0)/\pm} \Psi(u + s)\Psi(u - s).$$

So $S(u) = (tU_{2u} - U_u^4)/(2U_u^2)$ lies in H and is integral at w , and $\text{digit}(t) + \text{digit}(U_{2u}) = \bar{S}(u)/\bar{U}_u^2$. With $\text{digit}(t) + \text{digit}(e_0) = \bar{e}_0$, this gives

$$\text{digit}_w(U_{2u}) = \text{digit}_w(e_0) \iff \bar{S}(u) = \bar{e}_0 \bar{U}_u^2.$$

Reducing (iii) modulo 2 also gives $\bar{U}_{2y} = \bar{U}_y^4$ for $y \in A \setminus 0$, and this holds at $y = 0$ since $\bar{e}_0^4 = \bar{e}_0$. We prove $\bar{S}(u) = \bar{e}_0 \bar{U}_u^2$ in Steps 2–6; write $x := \bar{U}_u$ and $h := (t - 1)/2 \bmod 2$.

Step 2: A product identity in characteristic 0. We show that for $c \in A \setminus 0$

$$\Sigma^+(c)\Sigma^-(c) = t + 2 - 2e_0.$$

By Step 0, $\Sigma^-(c) = \sum_{\beta} b(c/2, \beta)^{-1}E(c + \beta)^{-1}$, so with $x' = c + \beta'$, $v = \beta - \beta'$,

$$\Sigma^+\Sigma^- = \sum_{\beta, \beta'} b(c/2, \beta - \beta') \frac{E(c + \beta)}{E(c + \beta')} = \sum_{v \in B} \psi(v) \sum_{x' \in c + B} g_v(x'),$$

where $\psi := b(c/2, \cdot)$ and $g_v(x') := E(x' + v)/E(x')$. As $c \notin B$, the coset $c + B$ does not contain 0, and we put $g_v(0) := e_0 E(v)$. Since $B^\perp = B$, $\sum_{y \in B} b(y, x' - c)$ is t on $c + B$ and 0 elsewhere, so

$$\sum_{x' \in c + B} g_v(x') = \frac{1}{t} \sum_{y \in B} b(y, c)^{-1} \sum_{x' \in G} b(y, x') g_v(x').$$

Now $g_v(x') = F^+(x' + v)/F^-(x')$ for all x' , including $x' = 0$ and $x' = -v$. Substituting $x' = x - v$ in (i) and using $b(v, y) = 1$, the inner sum is $tW(y, v)$ with $W(y, v) := F^+(y - v)/(F^-(y)F^-(-v))$ for $(y, v) \neq (0, 0)$, and $t^2 - 1 + e_0^2$ for $(y, v) = (0, 0)$. On B , F^+ is 1 off the origin and e_0 at it, and F^- is 1 off the origin and e_0^{-1} at it. So $W(y, v) = e_0$ if exactly one of $y = 0, v = 0, y = v$ holds, and $W(y, v) = 1$ if none does. As $b(y, c)^{-1} = \psi(y)^{-2}$,

$$\Sigma^+ \Sigma^- = \frac{t^2 - 1 + e_0^2}{t} + e_0 \sum_{v \neq 0} \psi(v) + e_0 \sum_{y \neq 0} \psi(y)^{-2} + e_0 \sum_{y \neq 0} \psi(y)^{-1} + \sum_{y \neq 0} \psi(y)^{-2} \sum_{v \notin \{0, y\}} \psi(v).$$

The character ψ is nontrivial, as $b : A \times B \rightarrow \mu_t$ is perfect and $c/2 \neq 0$, and it has odd order, so ψ^{-1} and ψ^{-2} are nontrivial too. The first term is $t + e_0$, as $e_0^2 = te_0 + 1$. The three middle terms are $-e_0$ each, and the last is $\sum_{y \neq 0} \psi(y)^{-2}(-1 - \psi(y)) = 2$. The total is $t + 2 - 2e_0$.

Step 3: Reduction modulo 4. Expanding, $\Sigma^+ \Sigma^- = 1 + 2(T(c)/U_c + U_c T(-c)) + 4T(c)T(-c)$, and all terms are integral at $\mathfrak{P}$. Comparing with Step 2 and dividing by 2 gives $T(c)/U_c + U_c T(-c) \equiv (t + 1)/2 - e_0 \pmod{2}$. At $c = u$, multiplying by x :

$$(\alpha) \quad \bar{T}(u) + x^2 \bar{T}(-u) = x(h + 1 + \bar{e}_0).$$

Step 4: The Weyl kernel modulo 2. We show that for $c, d \in A$ with $c, d, d - c \neq 0$

$$\bar{M}(c, d) = (\bar{U}_c \bar{U}_{d-c} / \bar{U}_d)^2,$$

while $\bar{M}(c, 0) = \bar{M}(c, c) = \bar{e}_0 + 1 = \bar{e}_0^2$ for $c \neq 0$ by Step 0.

Let $a, a' \in A$ with $a, a', a + a' \neq 0$. The right side of (ii) is $U_a U_{a'} / U_{a+a'}$, by $q \equiv 1$ on A and reflection. On the left write $s = k + \beta$ with $k \in A, \beta \in B$. Then $q(s) = b(k, \beta)$, $q(a - s) = b(a - k, \beta)^{-1}$, $b(s, a') = b(\beta, a')$ and $\Psi(a - k - \beta) = \Psi(a - k + \beta)$, so

$$q(s)E(s)E(a - s) = q(s)^{1/2}q(a - s)^{-1/2}\Psi(s)\Psi(a - s) = b(a/2, \beta)\Psi(k + \beta)\Psi(a - k + \beta),$$

and the left side of (ii), times t , is $\sum_{k \in A} \sum_{\beta \in B} b(a/2 + a', \beta)\Psi(k + \beta)\Psi(a - k + \beta)$. The involution $k \mapsto a - k$ of A preserves each summand and has the single fixed point $k = a/2$. Modulo 2 the exchanged pairs cancel and $t \equiv 1$, so

$$\begin{aligned} \frac{\bar{U}_a \bar{U}_{a'}}{\bar{U}_{a+a'}} &= \sum_{\beta \in B} b(a/2 + a', \beta) \bar{\Psi}(a/2 + \beta)^2 \\ &= \left(\sum_{\beta \in B} b(a/4 + a'/2, \beta) \bar{\Psi}(a/2 + \beta) \right)^2 = \bar{M}(a/2, (a + a')/2)^2. \end{aligned}$$

Put $a = 2c$ and $a' = 2(d - c)$. With $\bar{U}_{2y} = \bar{U}_y^4$ this reads $\bar{M}(c, d)^2 = (\bar{U}_c \bar{U}_{d-c} / \bar{U}_d)^4$, and the Frobenius of the residue field is injective.

Step 5: The A-pairs. Put $Y_0(u) := \sum_{a \in (A \setminus 0)/\pm} \Psi(u + a)\Psi(u - a)$, where $\Psi(0) = e_0$. We show

$$(\beta) \quad \bar{Y}_0(u) = x^4 \bar{T}(-u)^2 + h x^2.$$

First, for an even function f on B with values in the residue field, put $\hat{f}(s) := \sum_{\beta} b(s, \beta)f(\beta)$ for $s \in A$. Since $b : A \times B \rightarrow \mu_t$ is perfect, $f(\beta) = t^{-1} \sum_s b(s, \beta)^{-1} \hat{f}(s)$, and $t \equiv 1$. Summing over a set of representatives of $(B \setminus 0)/\pm$, the coefficient of $\hat{f}(0)$ is $(t - 1)/2 \equiv h$. For $s \neq 0$ the coefficients of $\hat{f}(s)$ and $\hat{f}(-s) = \hat{f}(s)$ add up to $\sum_{\beta \neq 0} b(s, \beta) = -1 \equiv 1$. So

$$\sum_{\beta \in (B \setminus 0)/\pm} f(\beta) = h \hat{f}(0) + \sum_{s \in (A \setminus 0)/\pm} \hat{f}(s).$$

We apply this to $f(\beta) = \bar{\Psi}(-u + \beta)$, which is even by Step 0, with $\hat{f}(s) = \bar{M}(-u, s - u/2)$ and $\hat{f}(0) = \bar{\Sigma}^-(u) = x^{-1}$. The class $s = \pm u/2$ gives $\bar{M}(-u, 0) = \bar{M}(-u, -u) = \bar{e}_0^2$. For the other classes, Step 4 applies at $(-u, s - u/2)$, and with $\bar{U}_{-y} = \bar{U}_y^{-1}$ it gives $\bar{M}(-u, s - u/2) = x^{-2}(\bar{U}_{u/2+s}\bar{U}_{u/2-s})^2$. Hence

$$x^2\bar{T}(-u) = hx + \bar{e}_0^2x^2 + \sum'_s (\bar{U}_{u/2+s}\bar{U}_{u/2-s})^2,$$

where $\sum'$ runs over $(A \setminus 0)/\pm$ without the class of $u/2$. On the other hand, the square root of $\bar{U}_y$ in the residue field is $\bar{U}_{y/2}^2$, also at $y = 0$, since $\bar{e}_0^{1/2} = \bar{e}_0^2$. So, with $a = 2s$,

$$\bar{Y}_0(u)^{1/2} = \sum_{s \in (A \setminus 0)/\pm} (\bar{U}_{u/2+s}\bar{U}_{u/2-s})^2 = \bar{e}_0^2x^2 + \sum'_s (\bar{U}_{u/2+s}\bar{U}_{u/2-s})^2 = x^2\bar{T}(-u) + hx,$$

the class of $u/2$ contributing $(\bar{U}_u\bar{U}_0)^2 = \bar{e}_0^2x^2$. Squaring gives (β) .

Step 6: Assembly. Split the classes $\{\pm s\}$ in $S(u)$ according to $s = k + \beta$, $k \in A$, $\beta \in B$. The classes with $\beta = 0$ give $Y_0(u)$. Those with $k = 0$ give $\sum_{\beta \in (B \setminus 0)/\pm} \Psi(u + \beta)^2$, whose residue is $\bar{T}(u)^2$. On the remaining classes $s \mapsto \eta s = k - \beta$ acts without fixed points, since $k - \beta = \pm(k + \beta)$ would force $k = 0$ or $\beta = 0$; and it preserves the summand, as $\Psi(u \pm \eta s) = \Psi(\eta(u \pm s)) = \Psi(u \pm s)$. So these classes contribute an element of $2O_L$. Hence, by (β) and (α) ,

$$\bar{S}(u) = \bar{Y}_0(u) + \bar{T}(u)^2 = (\bar{T}(u) + x^2\bar{T}(-u))^2 + hx^2 = x^2(h + 1 + \bar{e}_0^2) + hx^2 = x^2(1 + \bar{e}_0^2) = \bar{e}_0x^2.$$

By Step 1, $\text{digit}_w(U_v) = \text{digit}_w(e_0)$ for every $v = 2u \in A$ with $\text{Ann}(v) = \mathfrak{m}$.

Step 7: Conclusion. Every class $\mathfrak{c} \in \mathcal{G}_{\mathfrak{m}}$ is the class of a datum $(I, \varepsilon_{\mathfrak{m}}, z)$ on an invertible O_K -ideal with annihilator $\mathfrak{m}$ (Lemma 3.20). It is η -type, so its label lies in $A \setminus 0$, and $U_{\mathfrak{c}} = E(z/v_2)$ by Corollary 4.4 and (V). So $\text{digit}_w(U_{\mathfrak{c}}) = \text{digit}_w(e_0) = \text{digit}_w(-\eta)$, that is, $e_0U_{\mathfrak{c}} \in O_w^{\times 2}(1 + 4O_w)$, and $H_w(\sqrt{e_0U_{\mathfrak{c}}})/H_w$ is unramified at every dyadic w . It is unramified at the odd places, because $e_0U_{\mathfrak{c}}$ is a unit there (Theorem D). Every real place of H lies above r , where $e_0 > 0$ and every conjugate of $U_{\mathfrak{c}}$ is positive (Theorem 4.5). The places above u are complex. So $H(\sqrt{-\eta U_{\mathfrak{c}}})/H$ is unramified everywhere. $\square$

Proof of Theorem G for t odd. The field $L := H(\sqrt{-\eta U_{\mathfrak{c}}})$ lies in K^{ab} by Theorem 4.6, and L/H is unramified everywhere by Theorem 4.14. Every inertia group of L/K meets $\text{Gal}(L/H)$ trivially, so it maps isomorphically, with its upper filtration, onto that of H/K . Hence $\mathfrak{f}(L/K) = \mathfrak{f}(H/K)$, r stays unramified, and $L \subseteq H_{\mathfrak{m}\infty_u} = H$. $\square$

The case $f_{\eta} = 1$, for every t .

Theorem 4.15 (Root law from (W)). *Assume identity (6) of [RW], and identity (36) of [RW] at the pairs $(2c, y)$, $y \in B$, for the label c below. Suppose $f_{\eta} = 1$, and let $I := O_K = \mathbb{Z}[\eta]$ with the frame $(\theta, 1)$, $\theta = -\eta$, so that γ is the matrix of Lemma 4.7 and $\mu_{\gamma} = 1$; define A , B , M and U_x on I as in Section 4.4. Let $c \in A_{\text{odd}}$ with $\text{Ann}(4c) = \mathfrak{m}$ (so $2c, 4c \neq 0$), and put $S(c) := \sum_{\beta \in B} b(c, \beta)W(c + \beta)$ and $X(c) := S(c) + \eta + 1$.*

- (a) $U_{2c}M(-2c, -c) = X(c)$.
- (b) $X(c) \in H$, and $X(c)$ is real at r .
- (c) *If (W) holds at $(-2c, -c)$ on I , then $-\eta U_{4c} = ((\eta+1)U_{2c}^2/X(c))^2$. Hence $-\eta U_{[4c]} \in H^{\times 2}$ for the class $[4c] \in \mathcal{G}_{\mathfrak{m}}$ of the datum $(I, \varepsilon, 4c)$, and, by Theorem 4.5, $-\eta U_{\mathfrak{c}} \in H^{\times 2}$ for every $\mathfrak{c}$.*

Proof. We use $\mu_{\gamma} = 1$, which holds in this frame by Lemma 4.7(a). Then Corollary 4.4 gives $E = U$ on $A \setminus 0$ and $E = 1$ on $B \setminus 0$. Theorem DC, applied to the matrix of η , gives

$E(\eta x) = \overline{E(x)} = q(x)E(x)$, hence $q \equiv 1$ on A and on B , $b \equiv 1$ on $B \times B$, and $B^\perp = B$ since $|B| = t$.

(a) We use [RW, (36)] at the pairs $(2c, y)$, $y \in B$ (the *slice* $\{2c\} \times B$), in characteristic 0:

$$\frac{1}{t} \sum_{s \in G} E(s)q(s)E(2c-s)b(s, y) = q(2c+y)E(-2c-y)E(2c)E(y).$$

The term with $\delta(x)\delta(y)$ vanishes as $2c \neq 0$. Multiply by $b(c, y)^{-1}$ and sum over $y \in B$. On the left, $\sum_{y \in B} b(s-c, y)$ is t for $s \in c + B^\perp = c + B$ and 0 otherwise, and for $s = c + \beta$ we have $E(2c-s) = E(\eta s) = q(s)E(s)$ and $q(s) = b(c, \beta)$. So the left side becomes $\sum_{\beta} b(c, \beta) q(c + \beta) E(c + \beta)^2 = \sum_{\beta} b(c, \beta) W(c + \beta) = S(c)$. On the right, $q(2c+y) = b(c, y)^2$, $E(2c) = U_{2c}$, $E(-2c) = U_{2c}^{-1}$, $E(0) = e_0$ and $E(y) = 1$ for $y \neq 0$. So the right side becomes

$$U_{2c} \sum_{y \in B} b(c, y) E(-2c-y) E(y) = e_0 + U_{2c}(M(-2c, -c) - U_{2c}^{-1}),$$

since $M(-2c, -c) = \sum_{\beta} b(c, \beta) E(-2c - \beta)$. Hence $S(c) = e_0 - 1 + U_{2c}M(-2c, -c)$, which is (a), as $\eta = -e_0$.

(b) As $\eta \in K$, it suffices to treat $S(c)$. It is real: conjugation replaces $b(c, \beta)$ by $b(c, -\beta)$, and $W(c + \beta) = W(\eta(c + \beta)) = W(c - \beta)$. The group $\text{Gal}(K^{\text{ab}}/H)$ is generated by complex conjugation at r and by the $\sigma = \text{Art}(s)$ with s a finite idele, $s \equiv 1 \pmod{\mathfrak{m}}$ and a unit everywhere. For such s , the label group G is a free $O_K/(\varepsilon-1)$ -module of rank one, and s acts on it through an element $\sigma_0 \in O_K$ with $\sigma_0 \equiv s \pmod{(\varepsilon-1)\widehat{O}_K}$; as $\text{Ann}(c) = \mathfrak{m}$, $\sigma_0 c = c$. By $(\star)$, $\sigma W(c + \beta) = W(c + \sigma_0^{-1}\beta)$, and σ acts on μ_t by the exponent $\text{Nm}(s)^{-1} \equiv (\sigma_0 \bar{\sigma}_0)^{-1}$ modulo t . The bicharacter satisfies $b(\alpha x, y) = b(x, \bar{\alpha} y)$ for $\alpha \in O_K$: in a frame, multiplication by α is $\mathbf{u} \mapsto \mathbf{u} M_\alpha$ with M_α commuting with γ , and $\omega(\mathbf{v} M_\alpha, \mathbf{w}) = \omega(\mathbf{v}, \mathbf{w} \text{adj } M_\alpha)$ with $\text{adj } M_\alpha = M_{\bar{\alpha}}$. Substituting $\beta = \sigma_0 \beta'$,

$$\sigma S(c) = \sum_{\beta'} b(c, \sigma_0 \beta')^{(\sigma_0 \bar{\sigma}_0)^{-1}} W(c + \beta') = \sum_{\beta'} b(\sigma_0^{-1} c, \beta') W(c + \beta') = S(c).$$

(c) The identity (W) at $(-2c, -c)$ reads $M^2 = \kappa^2 U_{2c}^2 / U_{4c}$, with $\kappa^2 = (\eta+1)^2 / (-\eta)$; substitute (a). As $\text{Ann}(2c) = \text{Ann}(4c) = \mathfrak{m}$, U_{2c} and $U_{4c} = U_{[4c]}$ are Stark units (Section 4.2), so the root $\rho := (\eta+1)U_{2c}^2 / X(c)$ lies in H by (b) and Theorem 4.5. For $\sigma \in \text{Gal}(\mathbb{Q}/K)$, Theorem 4.5 gives $\sigma U_{[4c]} = U_{c'}$, where c' is the product of $[4c]$ and the class of $\sigma|_H$. Hence $-\eta U_{c'} = (\sigma \rho)^2$ with $\sigma \rho \in H$. As in the proof of Theorem 4.6, c' runs through all of $\mathcal{G}_{\mathfrak{m}}$. $\square$

Proof of Theorem G for $f_\eta = 1$. Let $O_K = \mathbb{Z}[\eta]$. Since $\mathfrak{m}$ is odd and $\eta \equiv 1 \pmod{\mathfrak{m}}$, the group A on $I = O_K$ contains a label of exact annihilator $\mathfrak{m}$ and odd order, as in the proof of Theorem 4.6; it is $4c$ for some $c \in A_{\text{odd}}$. Theorem W supplies (W) at $(-2c, -c)$ on $I = O_K = J$, for every t , and Theorem 4.15(c) applies. $\square$

The remaining case: t even and $f_\eta > 1$. Here (St3) holds by Theorem 4.6, but the root law is open. The identity (W) is proved only on $\mathbb{Z}[\eta]$, which is not an O_K -lattice, and the dyadic method has no analogue: $(\eta-1)$ and $(\eta+1)$ share the dyadic primes, the reduction of M modulo 2 collapses to 0, and Fourier inversion on B modulo 2 fails because $|B|$ is even.

5. VERIFICATION

Review. No specialist has yet read the argument; all review so far has been by AI agents (Anthropic's Claude and OpenAI's GPT-6). The reviews consisted of re-derivations of statements from their stated hypotheses, carried out without access to the proofs, and adversarial

reviews of specific claims, with independently written code for the numerical parts. The depth of review varies across the argument.

Imported proofs. We re-checked Radchenko and Wheeler’s proof of [RW, (6)] step by step at both fixed points and for all $c \geq 1$, filling gaps in its written form (the representative window, the tilted contours, the closing of contours and the continuation to the real point), up to classical facts on Faddeev’s function. We also re-checked their proof of [RW, (7)] for every admissible γ , given [RW, Thm. 3, (23)], whose proof we re-checked as well, and their proof of [RW, Thm. 7], given their identities (4), (5), (7) and (38). We import (I1) as stated in [RW].

Yamamoto’s boundary case $x_k = 1$ (in his notation) of [Ya08, Thm. 5.1.1] is covered by [Ya10, Thms. 3.5, 4.10]; we also carried out the omitted computation in the framework of [Ya08], confirming his theorem. Numerically, 135 boundary pairs and 100 full products agree with PARI’s values of $L'(0, \chi)$ to 10^{-45} .

Numerical and exact checks. Values were computed in ball arithmetic. For values, “exact” means checked exactly with PARI/GP [PARI] after recognizing the minimal polynomial in $O_K[X]$; the identification of its roots with the analytic values is numerical. For lattices and codes, “exact” means integer arithmetic.

- Identities (6) and (7) of [RW] at both fixed points for nine matrices, to $5 \cdot 10^{-29}$; Theorem 3.1 on all eight sign classes; Lemma 3.12 to $8 \cdot 10^{-32}$, with negative controls (deliberately altered inputs) failing as expected.
- For Theorem 3.2: the tube lemma exactly, in an integer model of the tree, in 20 configurations, split and inert, with a negative control; the tree geometry used in the proof of (U0) (inward and outward lines, levels of neighbors, the stabilizer of ε_0 , the image of $X[p]$) exactly in a lattice model at $p = 5, 7$, split and inert, $R \leq 3$; the defect identity on actual values, split and inert, $R = 1, 2$; and, for Proposition 3.5, that the reduction of $\mathfrak{L}$ is the span of the $\binom{x}{i}$ with $i < m$, exactly at $(p, R) = (5, 1), (7, 1), (11, 1), (11, 2), (13, 2), (17, 2)$.
- On non-maximal orders, Frobenius follows the line predicted by $(\star)$ at all 12 discriminating primes tested, and the Artin law passes 22 of 22 exact tests; $K(W_m(1)) = H_{m\infty_u}$ exactly in three fields with class numbers 3 and 4.
- Stark units computed from L -values, at moduli chosen to test bad primes, have integral unit minimal polynomials in 28 of 28 cases over 12 fields; Theorem 4.5 holds at 660 pairs of a prime and a class. Through Kopp’s formula, (V) holds at every class of 19 moduli in nine fields to 10^{-19} , with negative controls.
- The identity (W) at about 950 points on O_K -lattices and in eight further configurations; the half-step five-term relation to $1.3 \cdot 10^{-23}$; the order change in five fields; the root law in about 130 cases of the open configuration.

Formal verification. Part of the argument is formalized in Lean 4 with Mathlib, in a library of 151 files and about 39,000 lines. It contains no unproved steps and uses only the standard axioms. Radchenko and Wheeler’s identities, Chebotarev’s theorem and the statements of class field theory that we use enter the library as hypotheses, as do the identifications of our frames and labels with theirs. Within that framework the library proves:

- the local theorem: the conclusions (U0)–(U2) of Theorem 3.2 for $p \geq 5$, at split primes for every R and at inert primes, with identities (3), (5) and (6) of [RW] in their own form, and the line product formula (ii) and the other identities of Section 2 as hypotheses; the tube lemma at split and inert primes; Proposition 3.5 and Lemma 3.6;

and Proposition 3.10, with the extension of a valuation to the Gauss valuation taken as a classical hypothesis;

- Lemma FX, Theorem DC (except its clause at the origin), Theorem 3.1 and the homothety law (viii), from functional equations of [RW] taken as hypotheses;
- the reciprocity law ($\star$), including the selection step, from the local theorem in that form at split primes $p \geq 7$, Chebotarev's theorem and class field theory, with parts (c) and (d) of Lemma 3.15 as hypotheses;
- Theorem E, with one consequence of Theorem A (that the relevant value lies in a given abelian extension) taken as a hypothesis;
- for Theorem D, the lattice distribution of Lemma 3.11(a), the anchor step, and the assembly of the theorem from the cases of its proof; the case analysis itself and parts (c) and (d) of Lemma 3.15 are hypotheses.

Theorems B, C, F and G are not formalized. In Section 4 only three pieces are formalized, with (W) as a hypothesis: the square-class law (Lemma 4.10), part of Lemma 4.7 in an integer model, and the abstract orbit product behind Proposition 4.12. For the main formal theorems listed above no model of the hypotheses has been constructed; constant values cannot satisfy identity (6). Only earlier variants of the local theorem, with the Fourier identity in an encoded form, have been shown to have satisfiable hypotheses, in small models with constant values.

Status. Theorems A–F, and Theorem G in the cases stated, form a candidate proof under the hypotheses named in their statements. The main risks are the following. Radchenko and Wheeler's analytic theorems are in a preprint; identity (6) is used everywhere, and (7) and (23) for Theorems C and G. The identity (I1) is used in the open configuration. The limit formulas behind (N) and (V) are Shintani's and Yamamoto's (refereed), with Kopp's (a preprint) as an alternative route. Finally, some steps have had less review than others.

APPENDIX A. THE ROUTE THROUGH KOPP'S LIMIT FORMULA

In this appendix we derive (N) and (V) from Kopp's limit formula [Ko, Thm. 1.1], which is stated in a preprint. This gives an alternative to Proposition 3.24 and Theorem 4.2. We keep the notation of Section 3.6.

Kopp's Shintani–Faddeev modular cocycle is

$$\mathrm{Sh}_A^{\mathbf{r}}(w) := \frac{\varpi_{\mathbf{r}}(Aw)}{\varpi_{\mathbf{r}}(w)}, \quad \varpi_{\mathbf{r}}(w) := (e(r_2w - r_1); e(w))_{\infty},$$

for $\mathbf{r} \in \mathbb{Q}^2$, A in the stabilizer $\Gamma_{\mathbf{r}}$ of $\mathbf{r} \bmod \mathbb{Z}^2$ in $\mathrm{SL}_2(\mathbb{Z})$, and w in the upper half-plane. Kopp works with the modulus $\mathfrak{m}\infty_2$; his ∞_1 is our r , his class $\mathfrak{A}$ is our c_u , and his n is our $n_{\mathfrak{m}}$. To a class $\mathfrak{A} \in \mathcal{G}_{\mathfrak{m}}$ he attaches a real quadratic number $\beta > \beta'$ and a characteristic $\mathbf{r}$ with $r_2\beta' - r_1 > 0$. Suppose the stabilizer of β in $\Gamma_{\mathbf{r}}$ is generated, possibly together with $-I$, by a hyperbolic matrix A with eigenvalue $\lambda > 1$ on $(\beta, 1)^{\top}$. Then his Theorem 1.1 gives $\exp(n_{\mathfrak{m}}Z'_{\mathfrak{m}}(0, \mathfrak{A}))$ as a root of unity times

$$\mathrm{Sh}_A^{\mathbf{r}}(\beta)^2 := \lim_{y \rightarrow 0^+} \mathrm{Sh}_A^{\mathbf{r}}(\beta + iy)^2.$$

We use only the modulus of this identity.

To compare with Kopp's statement we translate our data into his. Let $(I, z) \in \mathcal{P}_{\mathfrak{f}}$ be written in a fixed frame, with $x := z/v_2 = s_1\theta + s_2$ and label $\mathbf{u}$, and put $\mathbf{r} := (-s_2, s_1)^{\top}$. The map $B \mapsto c_B\theta + d_B$ identifies the stabilizer of θ in $\Gamma_{\mathbf{r}}$ with the group of norm-one units $\equiv 1 \pmod{\mathfrak{f}}$. Indeed, if $B\theta = \theta$ then $B(\theta, 1)^{\top} = \eta(\theta, 1)^{\top}$ with $\eta := c_B\theta + d_B$. Multiplication by η preserves $\mathbb{Z}\theta + \mathbb{Z} = v_2^{-1}I$, so $\eta \in O_K^{\times}$, and $\mathrm{Nm}\eta = \det B = 1$. Conversely, every unit of norm 1

has such a matrix. It remains to identify the condition $B \in \Gamma_{\mathbf{r}}$. For $B \in \mathrm{SL}_2(\mathbb{Z})$ one has $B\mathbf{r} = J_0(\mathbf{s}B^{-1})^\top$, where $J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{s} = (s_1, s_2)$. So $B \in \Gamma_{\mathbf{r}}$ if and only if $\mathbf{s}B^{-1} - \mathbf{s} \in \mathbb{Z}^2$. As $\mathbf{s}B^{-1}(\theta, 1)^\top = \eta^{-1}x$, this says $(\eta^{-1} - 1)z \in I$, that is, $\eta \equiv 1 \pmod{\mathfrak{f}}$.

We say that $\mathfrak{f}$ is of *negative type* if this group contains a totally negative unit. If $\mathfrak{f}$ is not of negative type, every unit in it is totally positive, so the group is generated by $\varepsilon_{\mathfrak{f}}$, $-I$ is not in the stabilizer, and Kopp's A is the matrix γ of $\varepsilon_{\mathfrak{f}}$, with eigenvalue $(\varepsilon_{\mathfrak{f}})_r > 1$ on $(\theta, 1)^\top$. His identification of the cocycle with a q -Pochhammer quotient [Ko, Lemma 4.14, Props. 4.19, 4.35, 4.39], written in Radchenko and Wheeler's normalization, then gives

$$(A.1) \quad W(\mathbf{c}) = |\mathrm{Sh}_{\gamma}^{\mathbf{r}}(\theta)|^{-2}$$

for the class $\mathbf{c}$ of (I, z) .

To see this, write $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\mathbf{u} = \mathbf{s}(\gamma^{-1} - I)$, $z_{\mathbf{s}}(T) := s_1T + s_2$ and $\mathbf{r}^b := \mathbf{r} + (0, u_1)^\top$. For T in the upper half-plane, $(T, 1)^\top = (cT + d)\gamma^{-1}(\gamma T, 1)^\top$ gives $z_{\mathbf{s}}(T)/(cT + d) = (s_1 + u_1)\gamma T + s_2 + u_2$, and $u_1T + z_{\mathbf{s}}(T) = r_2^bT - r_1^b$. So the definition of Φ in Section 2 gives, exactly,

$$\Phi_{\gamma, u_1, 0}(z_{\mathbf{s}}(T); T) = \varpi_{\mathbf{r}^b}(T)/\varpi_{\mathbf{r}^b}(\gamma T) = \mathrm{Sh}_{\gamma}^{\mathbf{r}^b}(T)^{-1},$$

since $u_2 \in \mathbb{Z}$ disappears under $e(\cdot)$. As $T \rightarrow \theta$ the left side tends to $E_{\gamma}(\mathbf{u})/\mu_{\gamma}$. Passing to the limit at the real point requires the joint holomorphy of $\Phi_{\gamma, u_1, 0}$ in both variables near (x, θ) . This follows from [RW, Prop. 2(i)] and Faddeev's analyticity of Φ_S [Fa95].

It remains to pass from $\mathbf{r}^b$ back to $\mathbf{r}$. We have $\varpi_{\mathbf{r}+(0,1)^\top}(T) = \varpi_{\mathbf{r}}(T)/(1 - e(r_2T - r_1))$, so

$$\mathrm{Sh}_{\gamma}^{\mathbf{r}+(0,1)^\top} = \mathrm{Sh}_{\gamma}^{\mathbf{r}} \cdot \frac{1 - e(r_2T - r_1)}{1 - e(r_2\gamma T - r_1)},$$

and the last factor tends to 1 because $\gamma\theta = \theta$ and $r_2\theta - r_1 \notin \mathbb{Z}$. The same holds at every intermediate vector, as $z \notin I$. These steps are [Ko, Prop. 4.35]. After $|u_1|$ of them, and with $|\mu_{\gamma}| = 1$, we get $W(\mathbf{c}) = |E_{\gamma}(\mathbf{u})|^2 = |\mathrm{Sh}_{\gamma}^{\mathbf{r}}(\theta)|^{-2}$.

Kopp's class of these data is $\mathrm{pr} \mathbf{c}$, provided the representative is chosen with x positive at u . For instance, take $z = 1$ and $I = \mathfrak{a}^{-1}\mathfrak{f}$ with $\mathfrak{a} \in \mathbf{c}$ integral and prime to $\mathfrak{f}$. Then Kopp's condition $r_2\theta' - r_1 = x_u > 0$ holds, and positivity at u fixes the c_u -coset by Lemma 3.20(d).

Proposition A.1 (Kopp's route to (N) and (V)). *Assume [Ko, Thm. 1.1], including the steps its written proof leaves out (listed in Remark A.2). Then:*

- (a) *for every proper $\mathfrak{f}$ and every $\mathbf{c} \in C_{\mathfrak{f}}$, $\log W(\mathbf{c}) = -n_{\mathfrak{f}} Z'_{\mathfrak{f}}(0, \mathrm{pr} \mathbf{c})$;*
- (b) *for every primitive character χ of $\mathcal{G}_{\mathfrak{f}}$ with $\chi(c_u) = -1$,*

$$\hat{\ell}_{\mathfrak{f}}(\chi) = -2n_{\mathfrak{f}} L'(0, \bar{\chi}) = -\frac{n_{\mathfrak{f}}}{\pi} \mathbf{w}(\bar{\chi}) \sqrt{d_K N_{\mathfrak{f}}} L(1, \chi) \neq 0,$$

which is the same value as in Proposition 3.24.

Proof. If $\mathfrak{f}$ is of negative type, then $c_u = 1$ in $\mathcal{G}_{\mathfrak{f}}$, so $Z_{\mathfrak{f}} \equiv 0$, while positive homothety and reflection alone give $W \equiv 1$. Hence (a) holds and (b) is vacuous. In this case Kopp's A need not exist, and if $2 \notin \mathfrak{f}$ it does not; an example is $K = \mathbb{Q}(\sqrt{3})$, $\mathfrak{f} = (\sqrt{3})$. If $\mathfrak{f}$ is not of negative type, then (a) is Kopp's theorem by the identity (A.1), and (b) follows from (a), the relation $\sum_B \bar{\chi}(B) Z_{\mathfrak{f}}(s, B) = 2L(s, \bar{\chi})$ and Hecke's functional equation. $\square$

Our contribution in this route is the translation into Radchenko and Wheeler's labels and our class indexing, the stabilizer case analysis and the deduction of (N). The q -Pochhammer identity and the periodicity step are Kopp's, as is the formula. The route uses none of [Ya08] and, in our steps, does not use Theorem DC. Kopp's proof rests on Shintani's formula and on

a relation for the class c_r (see Remark A.2), so the two routes share only Shintani's work and Hecke theory.

Remark A.2 (Gaps in the written proof of [Ko, Thm. 1.1]). Kopp's proof rests on Shintani's formula [Sh, Props. 3, 5], the quasi-periodicity of the double sine [KK], continued-fraction reduction, and a relation for the class c_r whose proof he omits, with a pointer to Arakawa [Ar] and Tangedal [Tan]. For the maximal order that relation is [Ya10, Thm. 5.1], stated for Yamamoto's invariants; the identification of these invariants with Kopp's, taken over his half-open cones, is not written out. Kopp also leaves to the reader one case of his Lemma 7.19 and a mirrored cone partition in one pairing step. These are gaps in written detail, not known errors. Our numerical checks of his formula, including at boundary label points, agree with it.

APPENDIX B. MISPRINTS IN THE SOURCES

The following misprints occur in the sources we use. None of them affects a statement used here, once that statement is read in its corrected form.

- In [Ya08] (arXiv:math/0602615v1), in the auxiliary display in the proof of Theorem 5.1.1, the term $+B_1(y_k) \log \omega_{k+1}$ should read $-B_1(y_k) \log \omega_{k+1}$. Yamamoto's theorem is unaffected. We have not checked whether the journal version corrects the display.
- In [RW], the origin clause $F^\pm(0) := \varepsilon^{\pm 1/2}$ of their definition (2) is right only for $\varepsilon > 1$. At the repelling point only their (4) holds, and their identities (6) and (7) fail with the literal (2). In [RW, Thm. 3], " $\tau \in \mathbb{R}_{>0}$ " should read $c\tau + d > 0$. Theorem 7 of [RW] leaves the sign convention of the Heisenberg operators unspecified, and their identity (39) fails with $q \mapsto q^{-1}$; Section 3.7 fixes the convention through the choice of D_{ij} and θ .
- In [Ko], the sentence after display (8.6) in the proof of Theorem 8.2 has the $t = 1$ sign pattern reversed. Kopp's Theorem 1.1 is unaffected.

APPENDIX C. INPUTS AND DEPENDENCIES

Results of other authors are cited where they are used. Where we re-checked a proof or found a misprint, this is recorded in Section 5 and Appendix B.

Imported results.

- *Radchenko and Wheeler* [RW]:
 - definitions: (1)–(4) and (37);
 - the four hypotheses of Section 1.1: (6) rests on [RW, Thm. 3, (22)] and (7) on [RW, Thm. 3, (23)], while (39) is proved in [RW] from (36), (38) and (5);
 - for the root law (Theorem G): (7) at three families of pairs, (36) modulo 2 and on the slice $\{2c\} \times B$, the identity in the proof of [RW, Thm. 10], and (6) with $\lambda = \mu_\gamma$;
 - for Section 2: the results listed there, with (9), (16), (19) and (20), for identities (i)–(v), (vii), Lemma FX and Theorem DC (all of which we have re-derived from their definitions);
 - for Theorem 4.6: (13), [RW, Lemma 2] and [RW, Prop. 1];
 - for Section 3.6 and Theorem 4.3: [RW, Props. 3, 4, 5].
- *Shintani and Yamamoto*: [Sh, Props. 3, 5], in Yamamoto's form [Ya08, Thm. 5.1.1] (boundary case [Ya10, Thms. 3.5, 4.10]), with [Ya08, Thm. 5.2.3, Cor. 5.2.5(1)], for Proposition 3.24 and Theorem 4.2; [Sh76] for Theorem 4.3 and, through it, Theorem G.

- *Classical*: Rademacher’s formula [RG], which gives $\mu_\gamma = 1$ for the matrix of η^2 on $\mathbb{Z}[\eta]$ and is used by Theorem 4.6 for every t ; Faddeev’s unitarity of Φ_S at real arguments [Fa95]; class field theory; Chebotarev’s theorem [Ch]; Hecke L -functions ($L(1, \chi) \neq 0$, the functional equation, root numbers, orders at $s = 0$); [Ta, Ch. IV] and [Ru]; the conductor–discriminant formula, Gauss sums, Milgram’s formula and Lucas’s theorem; valuation rings; lattices over $\mathbb{Z}_p$, their multiplier rings and the Bruhat–Tits tree [Se].
- *Appendix A only*: [Ko, Thm. 1.1] (a preprint), modulus only.

Dependencies. Each main statement uses the following inputs, directly or through earlier statements. Here (6) is identity (6) of [RW], “Shintani’s formula” means Shintani’s product formula in Yamamoto’s form, and “Chebotarev” means Chebotarev’s theorem together with class field theory.

- Theorem 3.2: (6), only through Lemma 3.4.
- Theorem A: (6) and Chebotarev, and Theorem DC once, in the selection step. The restriction to rational ideles, the Galois form and the positivity above r do not use Theorem DC.
- Theorem B: (6), Theorem DC, Shintani’s formula and Chebotarev.
- Theorem C: (6), the quadratic relation [RW, (39)], which rests on [RW, (7)], and Chebotarev.
- Theorems D and E: (6), Theorem DC and Chebotarev.
- Theorem 4.2: Theorem DC and Shintani’s formula.
- Theorem 4.3: Shintani’s formula and Shintani’s evaluation at $s = 0$ [Sh76].
- Theorem 4.5: (6), Theorem DC and Chebotarev; its clause on U_A also uses (V), hence Shintani’s formula.
- Theorem 4.6: (6), (I1), Theorem DC, Shintani’s formula, Rademacher’s formula and Chebotarev.
- Theorem F: (6), Theorem DC, Shintani’s formula and Chebotarev; in the open configuration also (I1) and Rademacher’s formula.
- Theorem G: (6), (I1), Theorem DC, Shintani’s formula, [Sh76], Rademacher’s formula and Chebotarev; for t odd also [RW, (7)] at three families of pairs of labels, and for $f_\eta = 1$ also [RW, (36)] on the slice $\{2c\} \times B$.

What is new. The following are new.

- Part A: Lemma FX (a transport of the results of [RW]); Theorems DC and 3.1; the local theory of Section 3.2, beyond the classical facts on lattices and the tree; Lemmas 3.11–3.17 and Theorem RL; Theorems A, D and E; the indexing of Section 3.6, which turns [Ya08, Cor. 5.2.5(1)] into Proposition 3.24, and the reading of Shintani’s two factors as the two place values of one datum; the assembly of Theorem C.
- Part B: Theorems 4.2, 4.3 and 4.5 and the assembly of Theorem F; for (St3) in the open configuration, the application of Radchenko and Wheeler’s contour method to the half step with the A -kernel, the square-class law and the order change; the root-law arguments of Section 4.5.
- Appendix A: the translation and case analysis.

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REFERENCES

- [AFK] M. Appleby, S. T. Flammia and G. S. Kopp, A constructive approach to Zauner’s conjecture via the Stark conjectures, preprint, arXiv:2501.03970.
- [Ar] T. Arakawa, Generalized eta-functions and certain ray class invariants of real quadratic fields, *Math. Ann.* **260** (1982), 475–494.
- [Ch] N. Tschebotareff, Die Bestimmung der Dichtigkeit einer Menge von Primzahlen, welche zu einer gegebenen Substitutionsklasse gehören, *Math. Ann.* **95** (1926), 191–228.
- [Cox] D. A. Cox, *Primes of the Form $x^2 + ny^2$* , 2nd ed., Wiley, 2013.
- [DK1] S. Dasgupta and M. Kakde, On the Brumer–Stark conjecture, *Ann. of Math. (2)* **197** (2023), 289–388.
- [DK2] S. Dasgupta and M. Kakde, Brumer–Stark units and explicit class field theory, *Duke Math. J.* **173** (2024), 1477–1555.
- [DPV] H. Darmon, A. Pozzi and J. Vonk, The values of the Dedekind–Rademacher cocycle at real multiplication points, *J. Eur. Math. Soc.* **26** (2024), 3987–4032.
- [DV] H. Darmon and J. Vonk, Singular moduli for real quadratic fields: a rigid analytic approach, *Duke Math. J.* **170** (2021), 23–93.
- [Fa95] L. D. Faddeev, Discrete Heisenberg–Weyl group and modular group, *Lett. Math. Phys.* **34** (1995), 249–254.
- [FK94] L. D. Faddeev and R. M. Kashaev, Quantum dilogarithm, *Modern Phys. Lett. A* **9** (1994), 427–434.
- [FKV] L. D. Faddeev, R. M. Kashaev and A. Yu. Volkov, Strongly coupled quantum discrete Liouville theory. I. Algebraic approach and duality, *Comm. Math. Phys.* **219** (2001), 199–219.
- [FM] M. Fouquet and F. Morain, Isogeny volcanoes and the SEA algorithm, in: *Algorithmic Number Theory (ANTS-V)*, Lecture Notes in Comput. Sci. **2369**, Springer, 2002, 276–291.
- [GKZ] S. Garoufalidis, R. Kashaev and D. Zagier, An $SL_2(\mathbb{Z})$ -extension of Faddeev’s quantum dilogarithm, manuscript, 2018 (cited in [RW]).
- [Koh] D. Kohel, *Endomorphism rings of elliptic curves over finite fields*, PhD thesis, University of California, Berkeley, 1996.
- [Ko] G. S. Kopp, The Shintani–Faddeev modular cocycle: Stark units from q -Pochhammer ratios, preprint, arXiv:2411.06763v3.
- [KK] N. Kurokawa and S. Koyama, Multiple sine functions, *Forum Math.* **15** (2003), 839–876.
- [MH] J. Milnor and D. Husemoller, *Symmetric Bilinear Forms*, *Ergeb. Math. Grenzgeb.* **73**, Springer, 1973.
- [PARI] The PARI Group, PARI/GP 2.17.1, Univ. Bordeaux.
- [RG] H. Rademacher and E. Grosswald, *Dedekind Sums*, *Carus Math. Monographs* **16**, Math. Assoc. America, 1972.
- [Ro] X.-F. Roblot, Stark’s conjectures and Hilbert’s twelfth problem, *Experiment. Math.* **9** (2000), no. 2, 251–260.
- [Ru] K. Rubin, A Stark conjecture “over $\mathbb{Z}$ ” for abelian L -functions with multiple zeros, *Ann. Inst. Fourier (Grenoble)* **46** (1996), 33–62.
- [RW] D. Radchenko and C. Wheeler, Real quadratic fields and finite quantum dilogarithms I, preprint, arXiv:2609.21892v2 (29 September 2026). Equation and statement numbers refer to this version.
- [RW2] D. Radchenko and C. Wheeler, Stark units for real quadratic fields and reciprocity laws, preprint, 4 October 2026, <https://www.danrad.net/papers/stark-reciprocity.pdf>.
- [Se] J.-P. Serre, *Trees*, Springer, 1980.
- [Sh] T. Shintani, On a Kronecker limit formula for real quadratic fields, *J. Fac. Sci. Univ. Tokyo Sect. IA Math.* **24** (1977), 167–199.
- [Sh76] T. Shintani, On evaluation of zeta functions of totally real algebraic number fields at non-positive integers, *J. Fac. Sci. Univ. Tokyo Sect. IA Math.* **23** (1976), 393–417.
- [Shi] G. Shimura, *Introduction to the Arithmetic Theory of Automorphic Functions*, Publ. Math. Soc. Japan **11**, Iwanami Shoten and Princeton University Press, 1971.
- [St] H. M. Stark, L -functions at $s = 1$. III. Totally real fields and Hilbert’s twelfth problem, *Adv. Math.* **22** (1976), 64–84.
- [Ta] J. Tate, *Les conjectures de Stark sur les fonctions L d’Artin en $s = 0$* , *Progr. Math.* **47**, Birkhäuser, Boston, 1984.
- [Tan] B. A. Tangedal, Continued fractions, special values of the double sine function, and Stark units over real quadratic fields, *J. Number Theory* **124** (2007), 291–313.
- [Ya08] S. Yamamoto, On Kronecker limit formulas for real quadratic fields, *J. Number Theory* **128** (2008), 426–450 (theorem numbers as in arXiv:math/0602615).

- [Ya10] S. Yamamoto, On Shintani's ray class invariant for totally real number fields, *Math. Ann.* **346** (2010), 449–476.
- [Za] G. Zauner, Quantum designs: foundations of a noncommutative design theory, *Int. J. Quantum Inf.* **9** (2011), 445–507.
- [Zag75] D. Zagier, A Kronecker limit formula for real quadratic fields, *Math. Ann.* **213** (1975), 153–184.

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